Seriously, Models? A Source-Verified Audit of Frontier LLM Benchmark Reporting (2021-2026)

Every number in this repository is a verbatim quote on a specific page of a SHA-256-pinned primary source (190 documents: arXiv papers by version, vendor pages and system cards by Wayback Machine capture, GitHub files by commit). 1,083 claims are checked mechanically on every build; a number that cannot be found on its cited page is not published.

Findings

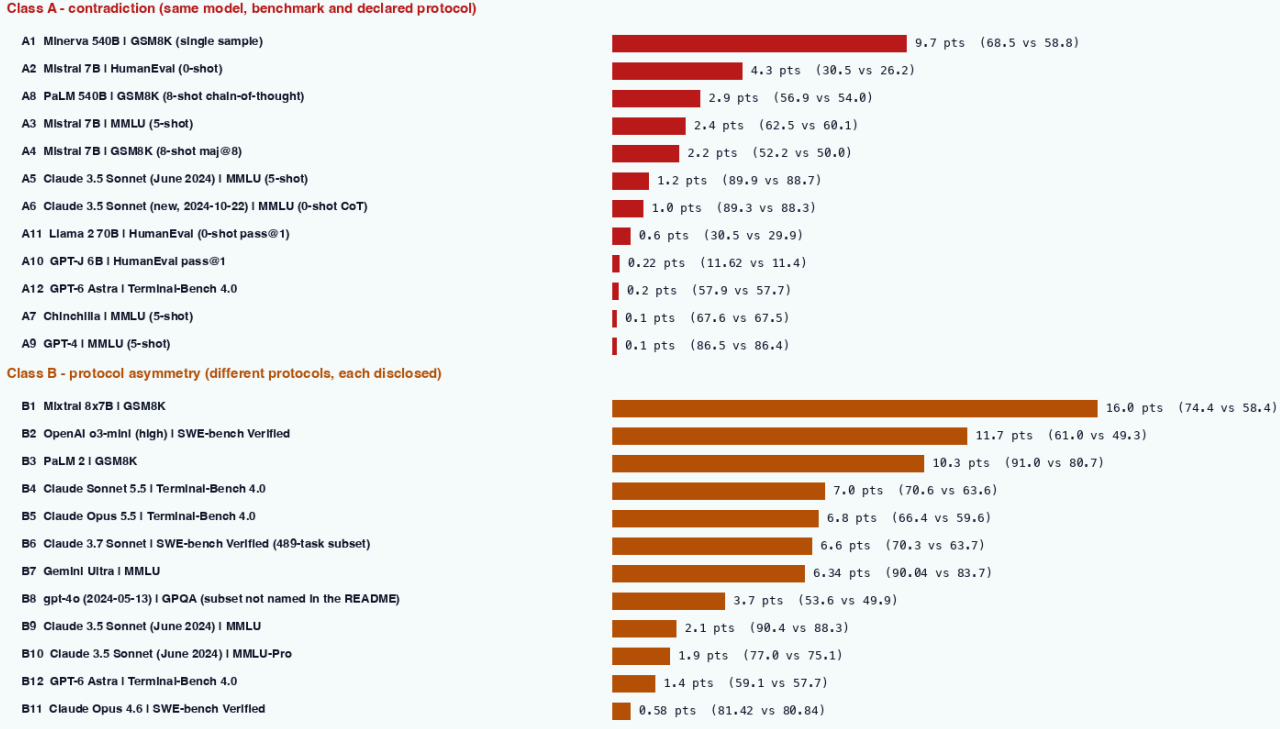
1. **12 contradictions (class A):** the same model, benchmark and declared protocol carry different published numbers. Median difference 1.1 points, range 0.1 to 9.7. They include a transcription error in a competitor's table, a vendor publishing different numbers for its own model in two papers under identical protocol text, competitors reporting a rival higher or lower than the rival itself, and text that disagrees with the table next to it.
2. **12 protocol asymmetries (class B):** the same model and benchmark under different disclosed protocols. Median difference 6.47 points, range 0.58 to 16.0. The protocol a headline chooses (consensus voting, parallel sampling, a modified prompt, a different scaffold or checkpoint under the same label) moves the reported number by several points.
3. **12 pairs are not comparable (class C)** and are shown only as context: different benchmarks or test sets, unestablished checkpoint identity, or an unstated protocol. No statistic pools classes.

Case	Class	Model	Benchmark	Published values	Difference (pts)
A1	A	Minerva 540B	GSM8K (single sample)	68.5 (Meta (LLaMA-1 TR, Table 7)) vs 58.8 (Google (Minerva TR, Table 3))	9.7
A2	A	Mistral 7B	HumanEval (0-shot)	30.5 (Mistral AI (Mistral 7B TR, Table 2)) vs 26.2 (Mistral AI (Mixtral TR, Table 2))	4.3
A3	A	Mistral 7B	MMLU (5-shot)	62.5 (Mistral AI (Mixtral TR, Table 2)) vs 60.1 (Mistral AI (Mistral 7B TR, Table 2))	2.4
A4	A	Mistral 7B	GSM8K (8-shot maj@8)	52.2 (Mistral AI (Mistral 7B TR, Table 2)) vs 50.0 (Mistral AI (Mixtral TR, Table 2))	2.2
A5	A	Claude 3.5 Sonnet (June 2024)	MMLU (5-shot)	89.9 (Meta (Llama 3 Herd TR, Table 2)) vs 88.7 (Anthropic (Claude 3.5 Sonnet model card addendum, Table 1))	1.2
A6	A	Claude 3.5 Sonnet (new, 2024-10-22)	MMLU (0-shot CoT)	89.3 (Anthropic (October 2024 addendum, Table 7)) vs 88.3 (DeepSeek (DeepSeek-R1 TR v1, Table 4))	1.0
A7	A	Chinchilla	MMLU (5-shot)	67.6 (DeepMind (Chinchilla TR, Table 6 and Section 4.2.2)) vs 67.5 (DeepMind (Chinchilla TR, abstract))	0.1
A8	A	PaLM 540B	GSM8K (8-shot chain-of-thought)	56.9 (Wei et al. (Google), arXiv:2201.11903v6 Table 1) vs 54.0 (Chowdhery et al. (Google), arXiv:2204.02311v5 Table 10)	2.9
A9	A	GPT-4	MMLU (5-shot)	86.5 (TII (Falcon series paper, Table 19), quoting GPT-4 paper) vs 86.4 (OpenAI (GPT-4 Technical Report))	0.1
A10	A	GPT-J 6B	HumanEval pass@1	11.62 (Chen et al. (Codex paper), Table 1) vs 11.4 (Chen et al. (Codex paper), abstract)	0.22
A11	A	Llama 2 70B	HumanEval (0-shot pass@1)	30.5 (Meta (Code Llama paper, v3)) vs 29.9 (Meta (Touvron et al. 2023b, Table 4))	0.6
A12	A	GPT-6 Astra	Terminal-Bench 4.0	57.9 (OpenAI, GPT-6 Astra launch post (Wayback capture 2026-09-03)) vs 57.7 (OpenAI, GPT-6 Astra launch post (Wayback capture 2026-09-03))	0.2
B1	B	Mixtral 8x7B	GSM8K	74.4 (Mistral AI (Mixtral TR, Table 2)) vs 58.4 (Mistral AI (Mixtral TR, Table 3))	16.0
B2	B	OpenAI o3-mini (high)	SWE-bench Verified	61.0 (OpenAI (GPT-4.5 launch post, Feb 27 2025)) vs 49.3 (OpenAI (GPT-4.1 launch post, Apr 14 2025, appendix))	11.7
B3	B	PaLM 2	GSM8K	91.0 (Google (PaLM 2 TR, Table 7)) vs 80.7 (Google (PaLM 2 TR, Table 7))	10.3
B4	B	Claude Sonnet 5.5	Terminal-Bench 4.0	70.6 (Anthropic (Claude Sonnet 5.5 page)) vs 63.6 (Artificial Analysis (model page, capture 2026-10-16))	7.0
B5	B	Claude Opus 5.5	Terminal-Bench 4.0	66.4 (Anthropic (Claude Opus 5.5 page)) vs 59.6 (Artificial Analysis (model page, capture 2026-10-18))	6.8
B6	B	Claude 3.7 Sonnet	SWE-bench Verified (489-task subset)	70.3 (Anthropic (Claude 3.7 Sonnet launch post)) vs 63.7 (Anthropic (Claude 3.7 Sonnet launch post))	6.6
B7	B	Gemini Ultra	MMLU	90.04 (Google (Gemini 1.0 TR, Table 2)) vs 83.7 (Google (Gemini 1.0 TR, Table 2))	6.34
B8	B	gpt-4o (2024-05-13)	GPQA (subset not named in the README)	53.6 (OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)) vs 49.9 (OpenAI (simple-evals README, commit 294cb1f, 2024-05-13))	3.7

B9	B	Claude 3.5 Sonnet (June 2024)	MMLU	90.4 (Anthropic (Claude 3.5 Sonnet model card addendum, Table 1)) vs 88.3 (Anthropic (Claude 3.5 Sonnet model card addendum, Table 1))	2.1
B10	B	Claude 3.5 Sonnet (June 2024)	MMLU-Pro	77.0 (Meta (Llama 3 Herd of Models, Table 2)) vs 75.1 (Anthropic (Claude 3.5 Sonnet (New) & Haiku addendum, Oct 2024, Table 7))	1.9
B11	B	Claude Opus 4.6	SWE-bench Verified	81.42 (Anthropic, Introducing Claude Opus 4.6) vs 80.84 (Anthropic, Claude Opus 4.6 System Card (Feb 2026))	0.58
B12	B	GPT-6 Astra	Terminal-Bench 4.0	59.1 (Artificial Analysis model page (Wayback)) vs 57.7 (OpenAI, GPT-6 Astra launch post (Wayback capture 2026-09-03))	1.4

FIG 1: VERIFIED CROSS-SOURCE DISCREPANCIES BY CLASS

Every bar is the difference between two published numbers, each a verbatim quote on a pinned source page (data/claims.json)



Bar length proportional to the difference in percentage points (0-20 scale). Class C rows are context only: no bar, excluded from every statistic.

FIG 2: THE SAME QUANTITY, TWO PUBLISHED NUMBERS

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Class A and B cases: open circle = lower published value, filled circle = higher; scale 0-100%

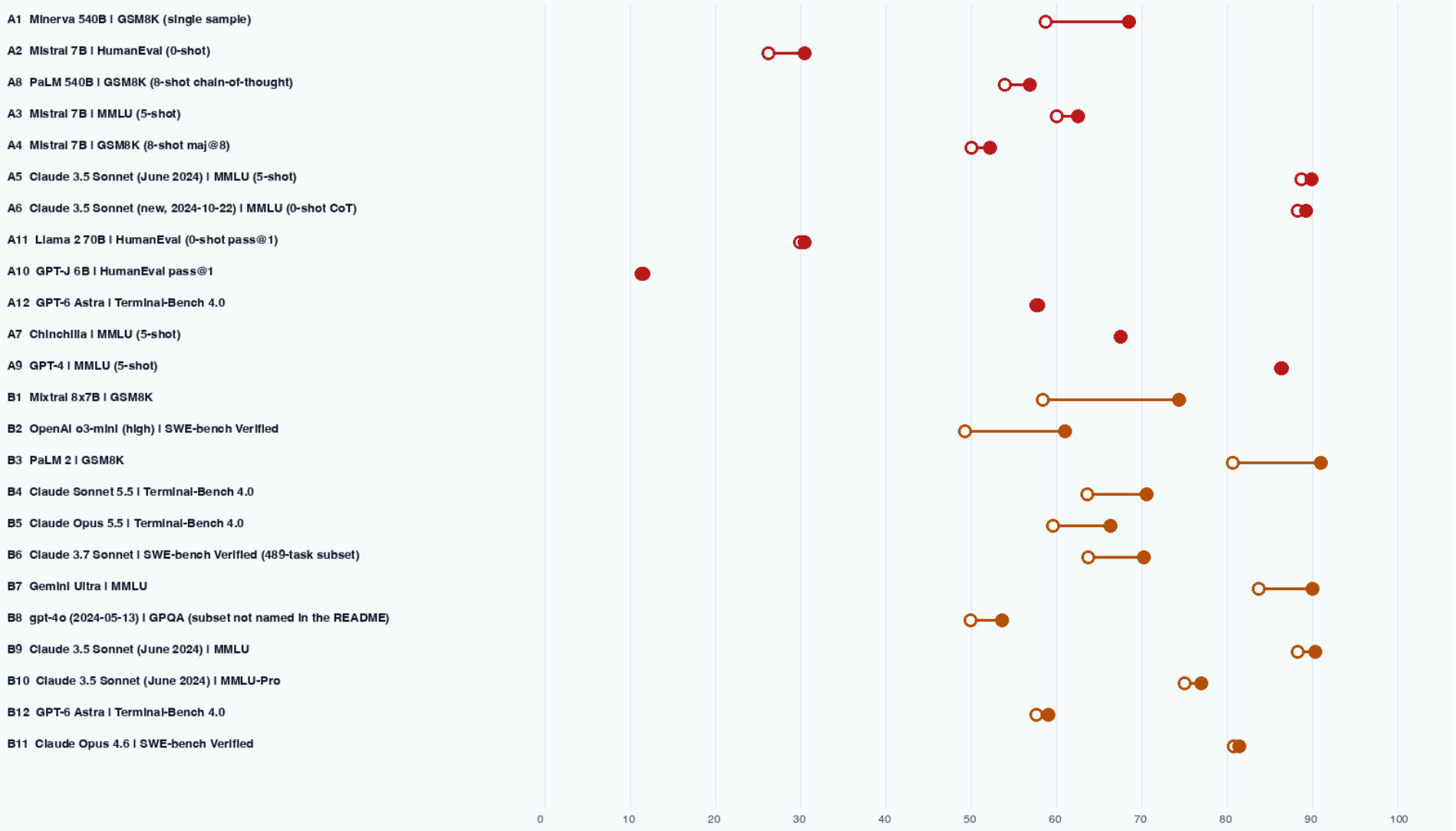


FIG 3: PRIMARY-SOURCE CROPS FOR CASES A1, A2-A4 AND B7

Rendered from the SHA-256-pinned PDFs at build time (scripts/build_forensic_analysis.py); physical PDF page numbers

A1 Meta LLaMA-1 TR, Table 7 (PDF p.6): the Minerva 540B row prints 68.5 in the GSM8k column

		MATH	MATH +maj1@k	GSM8k	GSM8k +maj1@k
PaLM	8B	1.5	-	4.1	-
	62B	4.4	-	33.0	-
	540B	8.8	-	56.5	-
Minerva	8B	14.1	25.4	16.2	28.4
	62B	27.6	43.4	52.4	68.5
	540B	**33.6**	**50.3**	**68.5**	**78.5**
LLaMA	7B	2.9	6.9	11.0	18.1
	13B	3.9	8.8	17.8	29.3
	33B	7.1	15.2	35.6	53.1
	65B	10.6	20.5	50.9	69.7

Table 7: **Model performance on quantitative reason-**

A1 Google Minerva TR, Table 3 (PDF p.8): Minerva 540B GSM8k 58.8; Minerva 62B maj1@k GSM8k 68.5

Table 3: **Model performance on several quantitative reasoning datasets.** For majority voting we use $k = 256$ (64 for 540B) samples for MATH, $k = 64$ for OCWCourses, $k = 100$ (40 for 540B) for GSM8k and $k = 16$ for MMLU-STEM. The PaLM GSM8k results do not use a calculator and were reported in (Chowdhery et al., 2022). We evaluated datasets that did not have published results on recent models on OpenAI davinci-002. Despite MMLU-STEM being a multiple choice task, we can apply majority vote by prompting the model to generate a rationale prior to the final answer, sampling multiple times, and then using majority vote on the final answers. Superscripts denote results that are quoted from previous work: [a] GPT-2 Hendrycks et al. (2021), [b] PaLM 540B maj1@40 Wang et al. (2022), and [c] Chinchilla Hoffmann et al. (2022).

	MATH	OCWCourses	GSM8k	MMLU-STEM
PaLM 8B	1.5%	1.5%	4.1%	22.0%
Minerva 8B	14.1%	7.7%	16.2%	35.6%
Minerva 8B, maj1@k	25.4%	12.5%	28.4%	43.4%
PaLM 62B	4.4%	5.9%	33.0%	39.1%
Minerva 62B	27.6%	12.9%	52.4%	53.9%
Minerva 62B, maj1@k	43.4%	23.5%	68.5%	63.5%
PaLM 540B	8.8%	7.1%	56.5%	58.7%
Minerva 540B	33.6%	17.6%	58.8%	63.9%
Minerva 540B, maj1@k	**50.3%**	**30.8%**	**78.5%**	**75.0%**
OpenAI davinci-002	19.1%	14.8%	-	-
Published SOTA	6.9%[a]	-	74.4%[b]	54.9%[c]

A2-A4 Mistral AI, Mistral 7B TR, Table 2 (PDF p.4)

Model	Modality	MMLU	HellaSwag	WinoG	PIQA	Arc-e	Arc-c	NQ	TriviaQA	HumanEval	MBPP	MATH	GSM8K
LLaMA 2 7B	Pretrained	44.4%	77.1%	69.5%	77.9%	68.7%	43.2%	24.7%	63.8%	11.6%	26.1%	3.9%	16.0%
LLaMA 2 13B	Pretrained	55.6%	**80.7%**	72.9%	80.8%	75.2%	48.8%	**29.0%**	**69.6%**	18.9%	35.4%	6.0%	34.3%
Code-Llama 7B	Finetuned	36.9%	62.9%	62.3%	72.8%	59.4%	34.5%	11.0%	34.9%	**31.1%**	**52.5%**	5.2%	20.8%
Mistral 7B	Pretrained	**60.1%**	**81.3%**	**75.3%**	**83.0%**	**80.0%**	**55.5%**	**28.8%**	**69.9%**	**30.5%**	47.5%	**13.1%**	**52.2%**

Table 2: Comparison of Mistral 7B with Llama. Mistral 7B outperforms Llama 2 13B on all metrics, and

A2-A4 Mistral AI, Mixtral TR, Table 2 (PDF p.4): same models, same declared protocols

Model	Active Params	MMLU	HellaS	WinoG	PIQA	Arc-e	Arc-c	NQ	TriQA	HumanE	MBPP	Math	GSM8K
LLaMA 2 7B	7B	44.4%	77.1%	69.5%	77.9%	68.7%	43.2%	17.5%	56.6%	11.6%	26.1%	3.9%	16.0%
LLaMA 2 13B	13B	55.6%	80.7%	72.9%	80.8%	75.2%	48.8%	16.7%	64.0%	18.9%	35.4%	6.0%	34.3%
LLaMA 1 33B	33B	56.8%	83.7%	76.2%	82.2%	79.6%	54.4%	24.1%	68.5%	25.0%	40.9%	8.4%	44.1%
LLaMA 2 70B	70B	69.9%	**85.4%**	**80.4%**	82.6%	79.9%	56.5%	25.4%	**73.0%**	29.3%	49.8%	13.8%	69.6%
Mistral 7B	7B	62.5%	81.0%	74.2%	82.2%	80.5%	54.9%	23.2%	62.5%	26.2%	50.2%	12.7%	50.0%
Mixtral 8x7B	13B	**70.6%**	84.4%	77.2%	**83.6%**	**83.1%**	**59.7%**	**30.6%**	71.5%	**40.2%**	**60.7%**	**28.4%**	**74.4%**

Table 2: Comparison of Mixtral with Llama. Mixtral outperforms or matches Llama 2 70B performance on

B7 Google Gemini 1.0 TR, Table 2 (PDF p.8): MMLU and GSM8K rows

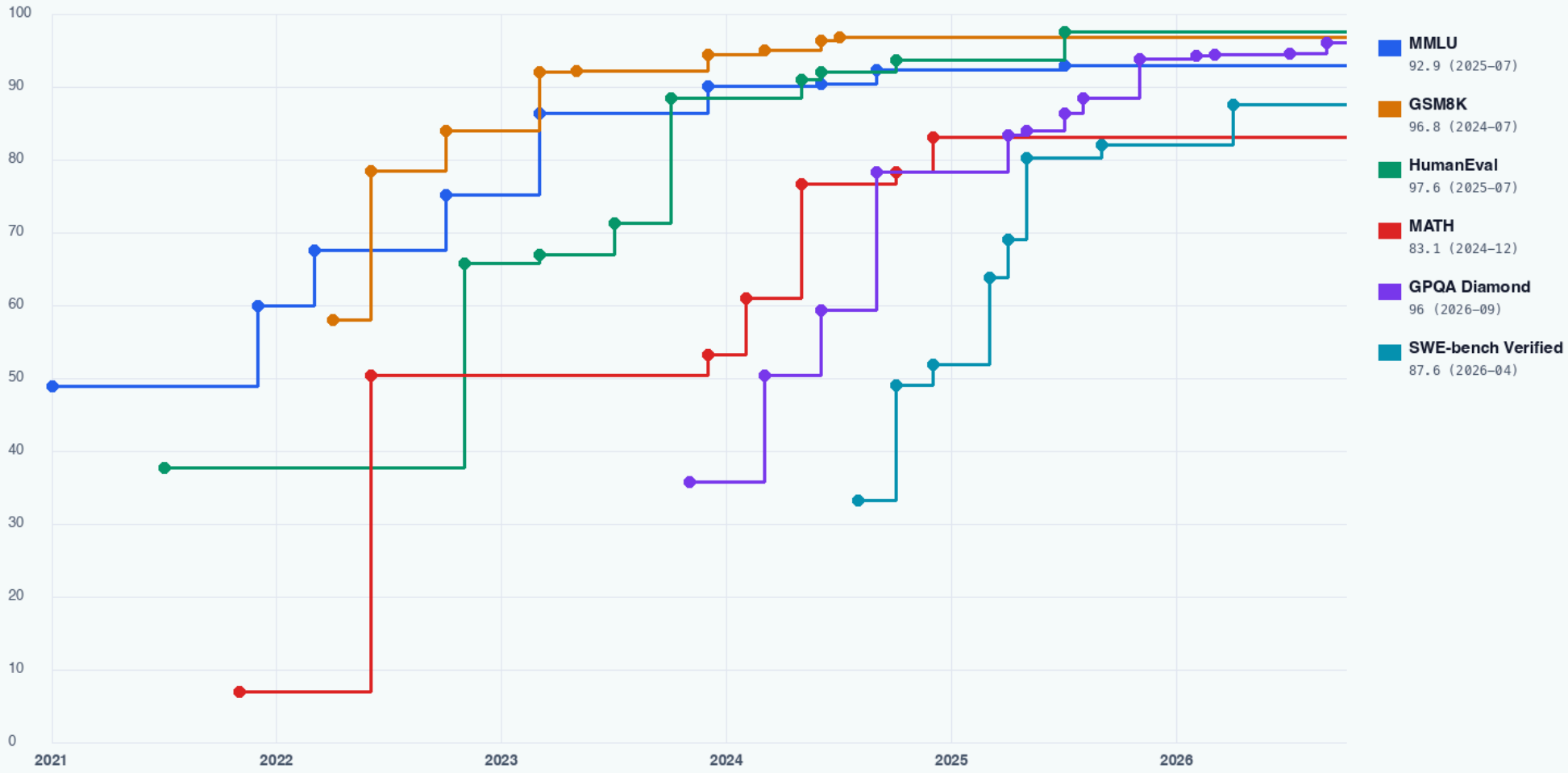
	Gemini Ultra	Gemini Pro	GPT-4	GPT-3.5	PaLM 2-L	Claude 2	Inflect-ion-2	Grok 1	LLAMA-2
MMLU Multiple-choice questions in 57 subjects (professional & academic) (Hendrycks et al., 2021a)	**90.04%** CoT@32*	79.13% CoT@8*	87.29% CoT@32 (via API**)	70% 5-shot	78.4% 5-shot	78.5% 5-shot CoT	79.6% 5-shot	73.0% 5-shot	68.0%***
	83.7% 5-shot	71.8% 5-shot	86.4% 5-shot (reported)						
GSM8K	**94.4%**	86.5%	92.0%	57.1%	80.0%	88.0%	81.4%	62.9%	56.8%

Month-by-month dataset

718 verified scores for 229 model names across 70 months (January 2021 to October 2026; 64 months contain at least one verified score). Each score cites its claim. Figure 4 tracks, per benchmark, the highest value published by each month, dated by when the reporting document was public (the first arXiv version when the number is already in it). Protocols differ between points; oracle metrics such as pass@100 are excluded.

FIG 4: HIGHEST PUBLISHED VALUE TO DATE, BY BENCHMARK (2021-01 TO 2026-10)

Verified records only, dated by when the reporting document was public. Protocols differ between points; each cites a claim.



Scores in %. A step is drawn when a verified record exceeds the previous highest value; the protocol of each point is in data/monthly_benchmarks_2021_2026.json.

Month	Verified scores	Models	MMLU	MMLU-Pro	HumanEval	GSM8K	MATH	MATH-500
2021-01	4	GPT-3 X-Large (few-shot) [davinci, 175B], UnifiedQA (11B, T5 backbone), davinci (175B)	48.9					
2021-02	0		48.9					
2021-03	5	GPT-2 1.5B (AMPS-pretrained, MATH fine-tuned), GPT-3 175B* (few-shot), GPT-Neo 1.3B, GPT-Neo 2.7B	48.9					
2021-04	0		48.9					

2021-05	0		48.9			
2021-06	3	GPT-J (6B), GPT-J 6B	48.9			
2021-07	5	Codex-12B, Codex-S, GPT-3 (size not stated)	48.9	37.7		
2021-08	2	J1-Jumbo v1 (178B)	48.9	37.7		
2021-09	0		48.9	37.7		
2021-10	4	GPT-3 175B finetuned plus an additional 175B verifier, GPT-3 175B finetuned with 7.5k examples, TNLG v2 (530B)	48.9	37.7		
2021-11	0		48.9	37.7		6.9
2021-12	2	Gopher (280B), Gopher (5-shot) 280B	60.0	37.7		6.9
2022-01	15	code-davinci-002, text-davinci-002	60.0	37.7		6.9
2022-02	3	GPT-NeoX-20B	60.0	37.7		6.9
2022-03	5	Chinchilla 70B, code-davinci-002	67.6	37.7		6.9
2022-04	15	Anthropic-LM v4-s3 (52B), PaLM 540B, PaLM-Coder 540B, code-davinci-001 (Davinci Codex)	67.6	37.7	58.0	6.9
2022-05	4	OPT-175B	67.6	37.7	58.0	6.9
2022-06	8	Minerva 540B, Minerva 62B, OpenAI davinci-002 (variant unstated)	67.6	37.7	78.5	50.3
2022-07	4	BLOOM 176B	67.6	37.7	78.5	50.3
2022-08	3	GLM-130B	67.6	37.7	78.5	50.3
2022-09	1	code-davinci-002 + CodeT	67.6	37.7	78.5	50.3
2022-10	4	Flan-PaLM 540B	75.2	37.7	83.9	50.3
2022-11	11	GPT-3.5, GPT-3.5 (snapshot not named), Galactica 120B, gpt-3.5-turbo and 1 more	75.2	65.8	83.9	50.3
2022-12	2	OPT-IML-Max 175B	75.2	65.8	83.9	50.3
2023-01	1	SantaCoder 1.1B (final)	75.2	65.8	83.9	50.3
2023-02	8	LLaMA 65B, LLaMA-I 65B	75.2	65.8	83.9	50.3
2023-03	3	GPT-4	86.4	67.0	92.0	50.3
2023-04	0		86.4	67.0	92.0	50.3
2023-05	12	Flan-PaLM 2, Flan-PaLM 2 (L), GPT-4, PaLM 2 and 3 more	86.4	67.0	92.2	50.3
2023-06	4	StarCoder 15B, Vicuna 13B, WizardCoder 15B	86.4	67.0	92.2	50.3
2023-07	19	Claude 1.3, Claude 2, Claude Instant, Llama 2 13B and 2 more	86.4	71.2	92.2	50.3
2023-08	4	Code Llama - Instruct 34B, Code Llama - Python 34B, Code Llama 34B, Llama 2 70B	86.4	71.2	92.2	50.3
2023-09	6	LLaMA 2 13B, Mistral 7B	86.4	71.2	92.2	50.3
2023-10	6	LLEMMA 34B	86.4	88.4	92.2	50.3
2023-11	12	Falcon-180B, GPT-4, Grok-1, Qwen-72B (base)	86.4	88.4	92.2	50.3
2023-12	38	GPT-4, GPT-4 (gpt-4-0613), Gemini Ultra, Gemini Ultra (post-trained API model) and 7 more	90.04	88.4	94.4	53.2
2024-01	19	Code Llama - Instruct 70B, DeepSeek 67B (Base), DeepSeek 67B Chat, DeepSeek-Coder-Instruct 33B and 2 more	90.04	88.4	94.4	53.2
2024-02	13	DeepSeekMath 7B (variant not named in the sentence), DeepSeekMath-RL 7B, Gemini 1.5 Pro, Gemini 1.5 Pro (Feb 2024) and 1 more	90.04	88.4	94.4	60.9
2024-03	13	Claude 3 Opus, GPT-4, Grok-1.5	90.04	88.4	95.0	60.9

2024-04	19	Llama 3 70B (instruction tuned), Mixtral 8x22B (instructed), Mixtral 8x22B Inst., Phi-3-small 7b and 1 more	90.04		88.4	95.0	60.9	
2024-05	36	DeepSeek-V2 Chat (RL), GPT-4o, GPT-4o 0513, Gemini 1.5 Pro (May 2024) and 2 more	90.04		91.0	95.0	76.6	
2024-06	25	Claude 3.5 Sonnet, DS-Coder-V2-Instruct, Qwen2-72B (base), Qwen2-72B-Instruct	90.4		92.0	96.4	76.6	
2024-07	18	GPT4o-mini, LLaMA-3.1 405B-Inst., Llama 3 405B (Llama 3.1 405B Instruct), Llama 3 405B (pre-trained) and 3 more	90.4	77.0	92.0	96.8	76.6	
2024-08	11	GPT-4o (checkpoint not stated), Grok-2, Phi-3.5-MoE 16x3.8B, gpt-4o-2024-08-06	90.4	77.0	92.0	96.8	76.6	
2024-09	33	DeepSeek V2.5-0905, Gemini 1.5 Pro (September 2024 release), OpenAI-o1-mini, Qwen2.5-72B (base) and 4 more	92.3	77.0	92.0	96.8	76.6	94.8
2024-10	23	Claude 3.5 Haiku, Claude 3.5 Sonnet (New), Claude-3.5-Sonnet-1022	92.3	78.0	93.7	96.8	78.3	94.8
2024-11	9	Marco-o1-MCTS (step), QwQ-32B-Preview	92.3	78.0	93.7	96.8	78.3	94.8
2024-12	26	DeepSeek-V3, DeepSeek-V3 Base, Gemini 2.0 Flash (research coding agent), Llama-3.3 70B Instruct and 4 more	92.3	78.0	93.7	96.8	83.1	94.8
2025-01	19	DeepSeek-R1, Kimi k1.5 (long-CoT), OpenAI o3-mini (high), o3-mini (tools), non-final checkpoint and 2 more	92.3	84.0	93.7	96.8	83.1	97.3
2025-02	18	Claude 3.7 Sonnet, GPT-4.5, Grok 3 (Think), Grok 3 (Think) [chart code GROK_3_R] and 2 more	92.3	84.0	93.7	96.8	83.1	97.3
2025-03	13	DeepSeek-V3-0324, Gemini 2.5 Pro (Experimental 03-25), Gemini2.5-Pro, QwQ-32B	92.3	84.0	93.7	96.8	83.1	97.3
2025-04	25	GPT-4.1, Llama 4 Maverick (instruction tuned), OpenAI o3 (high), Qwen3-235B-A22B (Non-thinking) and 8 more	92.3	84.0	93.7	96.8	83.1	97.3
2025-05	19	Claude Opus 4, Claude Sonnet 4, DeepSeek-R1-0528, Vicuna-13b and 1 more	92.3	85.0	93.7	96.8	83.1	98.0
2025-06	12	Gemini 2.5 Pro, Magistral Medium, MiniMax-M1-80k	92.3	85.0	93.7	96.8	83.1	98.0
2025-07	17	GLM-4.5, Grok 4, Grok 4 (no tools) [chart code GROK_4_NT], Grok 4 Heavy [chart code GROK_4_HEAVY] and 2 more	92.9	85.0	97.6	96.8	83.1	98.0
2025-08	13	Claude Opus 4.1, DeepSeek V3.1-NonThinking, DeepSeek V3.1-Thinking, GPT-5 (thinking) and 4 more	92.9	86.6	97.6	96.8	83.1	98.2
2025-09	12	Claude Sonnet 4.5, Claude Sonnet 4.5 (Thinking), DeepSeek-V3.2-Exp, GLM-4.6	92.9	86.6	97.6	96.8	83.1	98.2
2025-10	1	Claude Haiku 4.5	92.9	86.6	97.6	96.8	83.1	98.2
2025-11	19	Claude Opus 4.5, GPT-5.1 Thinking, Gemini 3 Deep Think, Gemini 3 Pro and 2 more	92.9	86.6	97.6	96.8	83.1	98.2
2025-12	21	DeepSeek-V3.2 Thinking, DeepSeek-V3.2-Speciale, GLM-4.7, GPT-5.2 Pro and 2 more	92.9	90.1	97.6	96.8	83.1	98.2
2026-01	6	Kimi K2.5	92.9	90.1	97.6	96.8	83.1	98.2
2026-02	23	Claude Opus 4.6, Claude Sonnet 4.6, GLM-5, GPT-5.3-Codex (xhigh) and 2 more	92.9	90.1	97.6	96.8	83.1	98.2
2026-03	5	GPT-5.4, GPT-5.4 Pro	92.9	90.1	97.6	96.8	83.1	98.2
2026-04	12	Claude Opus 4.7, DeepSeek-V4-Pro (Think Max), GPT-5.5	92.9	90.1	97.6	96.8	83.1	98.2
2026-05	2	Gemini 3.5 Flash	92.9	90.1	97.6	96.8	83.1	98.2
2026-06	5	Claude Sonnet 5, GLM-5.2	92.9	90.1	97.6	96.8	83.1	98.2
2026-07	5	GPT-5.6 Sol, Grok 4.5	92.9	90.1	97.6	96.8	83.1	98.2
2026-08	3	Qwen3.8-Max	92.9	90.1	97.6	96.8	83.1	98.2
2026-09	8	Claude Opus 5.5, Claude Opus 5.5 (Artificial Analysis run: Claude Opus 5.5 (Max, Default Fallback)), Claude Sonnet 5.5, Claude Sonnet 5.5 (Artificial Analysis run: Claude Sonnet 5.5 (max with fallback)) and 1 more	92.9	90.1	97.6	96.8	83.1	98.2
2026-10	2	Claude Haiku 5.5, Gemini 4 Argon	92.9	90.1	97.6	96.8	83.1	98.2

Method and verification

```
python3 scripts/download_sources.py        # fetch and pin every source; refuses changed bytes
```

```
python3 scripts/evidence.py                 # verify every claim against its pinned page
python3 scripts/build_forensic_analysis.py  # build the case catalog and figures 1-3
python3 scripts/build_dataset.py            # build the month-by-month dataset
python3 scripts/generate_charts.py          # figure 4
python3 scripts/extract_original_figures.py # references to the cited source pages
python3 scripts/build_docs_and_pdf.py       # this README, the audit, the dossier and the PDF
python3 scripts/adversarial_verify.py       # all gates plus break tests
python3 -m unittest discover -s tests       # verifier unit and break tests
```

Requirements: Python 3, Pillow, poppler (pdftotext, pdftoppm, pdfinfo), curl; pandoc and mutool for the PDF.

Limitations

1. Mechanical verification proves that a quoted string occurs on the cited page and that the value sits in the stated column. It does not prove that the model label in the claim is the one the source means; that attribution was made by reading the source, and column order was checked on rendered pages where text extraction is ambiguous.
2. A source can be internally inconsistent. Example: the o3-mini launch post's image alt text gives 48.9 for o3-mini (high), while the chart itself shows 49.3 for o3-mini (high) and 48.9 for o1. Such cases are documented when found; the method does not detect them automatically.
3. Numbers that appear only inside images are not used, except where stated as visual checks.
4. Vendor web pages are cited through Wayback Machine captures; a capture shows the page as archived, which may differ from what was first published.
5. Most claims were extracted by AI agents working in parallel, each required to pass the same mechanical verifier. Every claim was then reviewed a second time by independent reviewers who re-derived model, benchmark variant, protocol and reporting party from the source page (rendered where text extraction is ambiguous). 995 of 1,083 claims were confirmed as written; the others were corrected, mostly protocol details, benchmark variants and re-quotations. Each claim records the outcome in its 'review' field.
6. The highest-value-to-date series mixes protocols by construction; it is a record of what was published, not a like-for-like leaderboard.

Repository layout

Path	Content
`data/claims.json`	Evidence ledger: one entry per number, with source, page, verbatim quotes and column
`data/source_specs.json`, `data/downloaded_sources_manifest.json`	Source addresses and SHA-256 pins
`data/inconsistencies_catalog.json/.csv`	Case catalog (classes A, B, C), concordant cross-reports, sampling resolution
`data/monthly_records.json`	Verified per-model scores, each citing a claim
`data/monthly_benchmarks_2021_2026.json/.csv`, `data/model_releases_master.csv`	Month-by-month dataset
`data/source_pages.json`	Links to the cited primary-source pages, with the captions found on each page
`assets/synthesis/`	Figures 1-4
`scripts/`, `tests/`	Build, verification and tests

License

Code: MIT (`LICENSE`). Data compilation, documents and figures: CC BY 4.0 (`LICENSE-DATA.md`). The source documents belong to their authors and are not redistributed; the repository stores their pinned addresses and hashes. Short quoted excerpts and table crops remain the property of their owners.

Discrepancy Audit: Verified Cross-Source Benchmark Discrepancies (2021-2026)

This document is generated by `scripts/build_docs_and_pdf.py` from `data/inconsistencies_catalog.json`, which `scripts/build_forensic_analysis.py` builds from the evidence ledger `data/claims.json`. Do not edit it by hand.

1. Method

1. Every source document is pinned: arXiv papers by explicit version, vendor pages and system cards by a timestamped Wayback Machine capture, GitHub files by commit, HELM results by release. The SHA-256 of each file is recorded in `data/downloaded_sources_manifest.json`; `scripts/download_sources.py` refuses to re-pin a source whose bytes changed.
2. Every number is a claim in `data/claims.json`: source, physical PDF page, one or more verbatim quotes, the value, and for table cells the column index. `scripts/evidence.py` checks mechanically that each quote occurs on that page of the pinned file, that the value occurs in the first quote, and that the value sits in the stated column. A claim that fails is not used.
3. Cases are pairs of claims about the same model and benchmark. Deltas are computed, never typed. The build fails if a case's text contains a number that is not a value or quoted number of the claims it cites, or a statistic recomputed from the pinned sources.
4. Each case is assigned to exactly one class:

Class	Definition	Counted as a discrepancy
A	Contradiction: same model, benchmark and declared protocol; different published numbers	yes
B	Protocol asymmetry: same model and benchmark; different disclosed protocols	yes (protocol effect)
C	Not comparable: different benchmark/test set, unestablished checkpoint identity, or unstated protocol	no (context only)

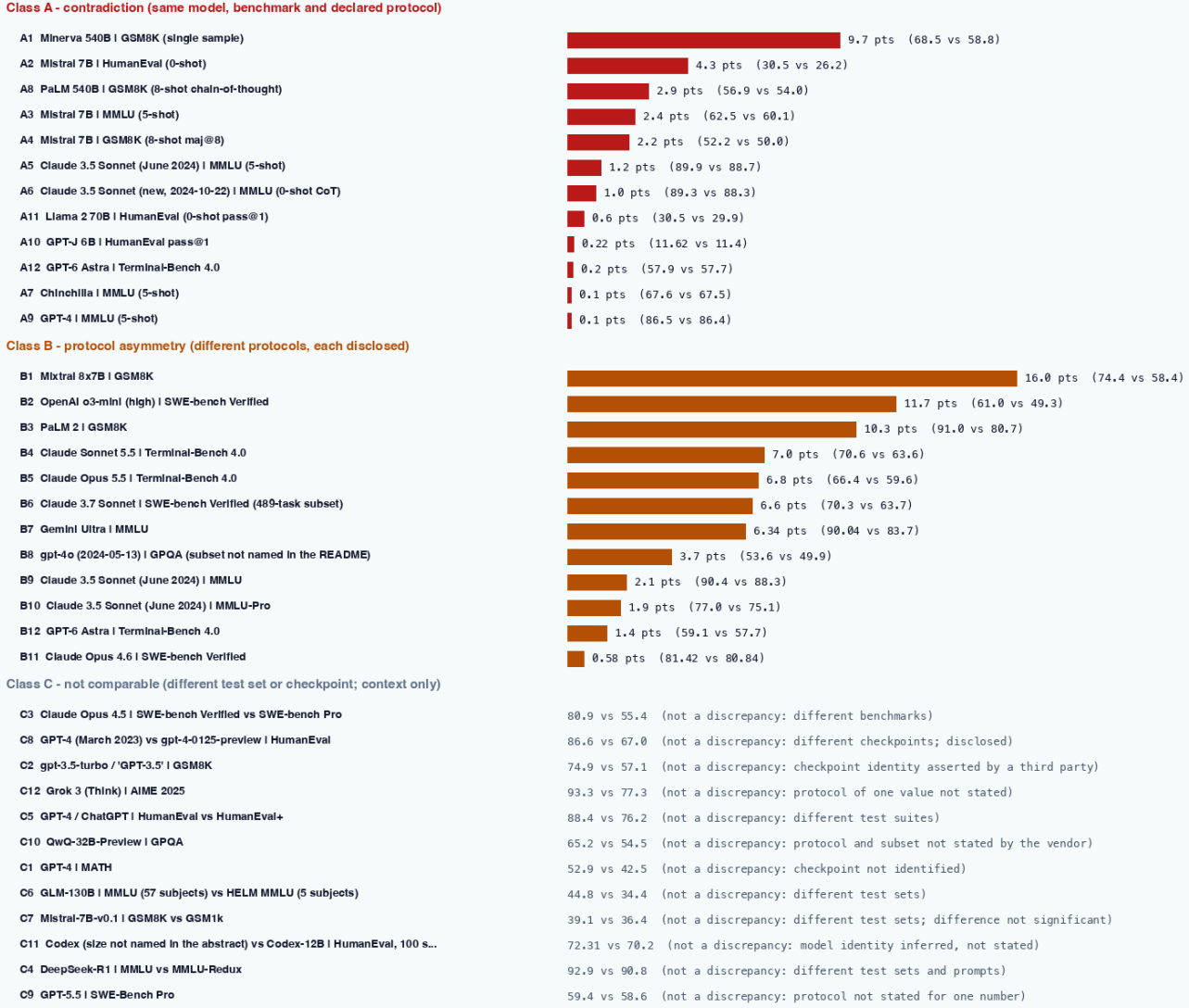
2. Summary

Class	Cases	Median difference (pts)	Range (pts)
A	12	1.1	0.1 to 9.7
B	12	6.47	0.58 to 16.0
C	12	not computed	not computed

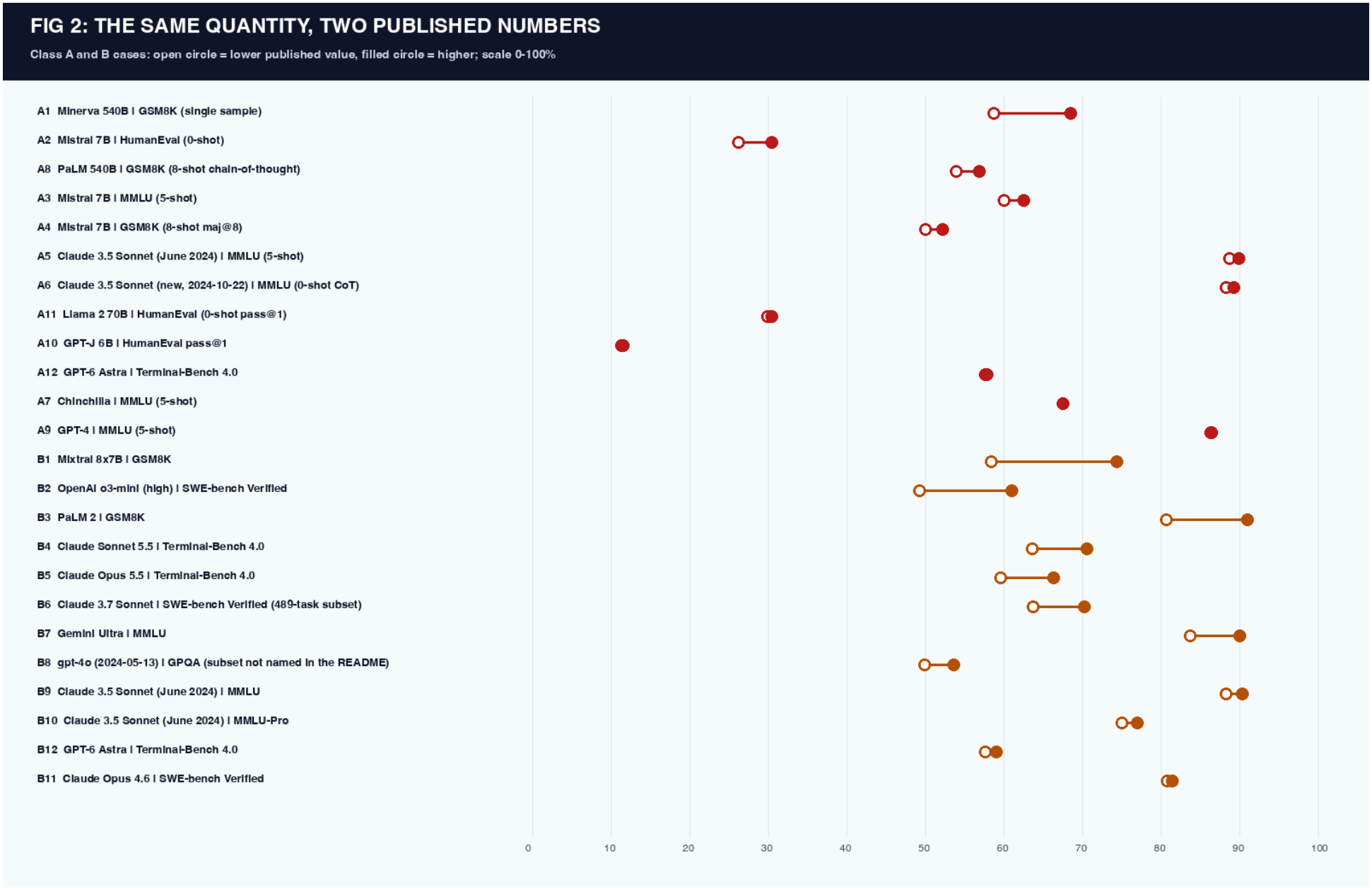
10 of 12 class A differences are below 3 points, but each is an error or an unexplained change in a published number. Class B differences are larger: the protocol a headline chooses moves the reported number by several points while every protocol is disclosed somewhere. No statistic in this repository pools A, B and C.

FIG 1: VERIFIED CROSS-SOURCE DISCREPANCIES BY CLASS

Every bar is the difference between two published numbers, each a verbatim quote on a pinned source page (data/claims.json)



Bar length proportional to the difference in percentage points (0-20 scale). Class C rows are context only: no bar, excluded from every statistic.



3. Cases

Case	Class	Model	Benchmark	Higher (source)	Lower (source)	Difference
A1	A	Minerva 540B	GSM8K (single sample)	68.5 (Meta (LLaMA-1 TR, Table 7))	58.8 (Google (Minerva TR, Table 3))	9.7

A2	A	Mistral 7B	HumanEval (0-shot)	30.5 (Mistral AI (Mistral 7B TR, Table 2))	26.2 (Mistral AI (Mixtral TR, Table 2))	4.3
A3	A	Mistral 7B	MMLU (5-shot)	62.5 (Mistral AI (Mixtral TR, Table 2))	60.1 (Mistral AI (Mistral 7B TR, Table 2))	2.4
A4	A	Mistral 7B	GSM8K (8-shot maj@8)	52.2 (Mistral AI (Mistral 7B TR, Table 2))	50.0 (Mistral AI (Mixtral TR, Table 2))	2.2
A5	A	Claude 3.5 Sonnet (June 2024)	MMLU (5-shot)	89.9 (Meta (Llama 3 Herd TR, Table 2))	88.7 (Anthropic (Claude 3.5 Sonnet model card addendum, Table 1))	1.2
A6	A	Claude 3.5 Sonnet (new, 2024-10-22)	MMLU (0-shot CoT)	89.3 (Anthropic (October 2024 addendum, Table 7))	88.3 (DeepSeek (DeepSeek-R1 TR v1, Table 4))	1.0
A7	A	Chinchilla	MMLU (5-shot)	67.6 (DeepMind (Chinchilla TR, Table 6 and Section 4.2.2))	67.5 (DeepMind (Chinchilla TR, abstract))	0.1
A8	A	PaLM 540B	GSM8K (8-shot chain-of-thought)	56.9 (Wei et al. (Google), arXiv:2201.11903v6 Table 1)	54.0 (Chowdhery et al. (Google), arXiv:2204.02311v5 Table 10)	2.9
A9	A	GPT-4	MMLU (5-shot)	86.5 (TII (Falcon series paper, Table 19), quoting GPT-4 paper)	86.4 (OpenAI (GPT-4 Technical Report))	0.1
A10	A	GPT-J 6B	HumanEval pass@1	11.62 (Chen et al. (Codex paper), Table 1)	11.4 (Chen et al. (Codex paper), abstract)	0.22
A11	A	Llama 2 70B	HumanEval (0-shot pass@1)	30.5 (Meta (Code Llama paper, v3))	29.9 (Meta (Touvron et al. 2023b, Table 4))	0.6
A12	A	GPT-6 Astra	Terminal-Bench 4.0	57.9 (OpenAI, GPT-6 Astra launch post (Wayback capture 2026-09-03))	57.7 (OpenAI, GPT-6 Astra launch post (Wayback capture 2026-09-03))	0.2
B1	B	Mixtral 8x7B	GSM8K	74.4 (Mistral AI (Mixtral TR, Table 2))	58.4 (Mistral AI (Mixtral TR, Table 3))	16.0
B2	B	OpenAI o3-mini (high)	SWE-bench Verified	61.0 (OpenAI (GPT-4.5 launch post, Feb 27 2025))	49.3 (OpenAI (GPT-4.1 launch post, Apr 14 2025, appendix))	11.7
B3	B	PaLM 2	GSM8K	91.0 (Google (PaLM 2 TR, Table 7))	80.7 (Google (PaLM 2 TR, Table 7))	10.3
B4	B	Claude Sonnet 5.5	Terminal-Bench 4.0	70.6 (Anthropic (Claude Sonnet 5.5 page))	63.6 (Artificial Analysis (model page, capture 2026-10-16))	7.0
B5	B	Claude Opus 5.5	Terminal-Bench 4.0	66.4 (Anthropic (Claude Opus 5.5 page))	59.6 (Artificial Analysis (model page, capture 2026-10-18))	6.8
B6	B	Claude 3.7 Sonnet	SWE-bench Verified (489-task subset)	70.3 (Anthropic (Claude 3.7 Sonnet launch post))	63.7 (Anthropic (Claude 3.7 Sonnet launch post))	6.6
B7	B	Gemini Ultra	MMLU	90.04 (Google (Gemini 1.0 TR, Table 2))	83.7 (Google (Gemini 1.0 TR, Table 2))	6.34
B8	B	gpt-4o (2024-05-13)	GPQA (subset not named in the README)	53.6 (OpenAI (simple-evals README, commit 294cb1f, 2024-05-13))	49.9 (OpenAI (simple-evals README, commit 294cb1f, 2024-05-13))	3.7
B9	B	Claude 3.5 Sonnet (June 2024)	MMLU	90.4 (Anthropic (Claude 3.5 Sonnet model card addendum, Table 1))	88.3 (Anthropic (Claude 3.5 Sonnet model card addendum, Table 1))	2.1
B10	B	Claude 3.5 Sonnet (June 2024)	MMLU-Pro	77.0 (Meta (Llama 3 Herd of Models, Table 2))	75.1 (Anthropic (Claude 3.5 Sonnet (New) & Haiku addendum, Oct 2024, Table 7))	1.9
B11	B	Claude Opus 4.6	SWE-bench Verified	81.42 (Anthropic, Introducing Claude Opus 4.6)	80.84 (Anthropic, Claude Opus 4.6 System Card (Feb 2026))	0.58
B12	B	GPT-6 Astra	Terminal-Bench 4.0	59.1 (Artificial Analysis model page (Wayback))	57.7 (OpenAI, GPT-6 Astra launch post (Wayback capture 2026-09-03))	1.4
C1	C	GPT-4	MATH	52.9 (Google (Gemini 1.0 TR, Table 2))	42.5 (Google (PaLM 2 TR, Table 7))	10.4
C2	C	gpt-3.5-turbo / 'GPT-3.5'	GSM8K	74.9 (Chain-of-Thought Hub (Fu et al., Table 1))	57.1 (OpenAI (GPT-4 TR, Table 2))	17.8
C3	C	Claude Opus 4.5	SWE-bench Verified vs SWE-bench Pro	80.9 (Moonshot AI (Kimi K2.5 TR, Table 4))	55.4 (Moonshot AI (Kimi K2.5 TR, Table 4))	25.5
C4	C	DeepSeek-R1	MMLU vs MMLU-Redux	92.9 (DeepSeek (DeepSeek-R1 TR v1, Table 4))	90.8 (DeepSeek (DeepSeek-R1 TR v1, Table 4))	2.1
C5	C	GPT-4 / ChatGPT	HumanEval vs HumanEval+	88.4 (EvalPlus (Liu et al., Table 3))	76.2 (EvalPlus (Liu et al., Table 3))	12.2
C6	C	GLM-130B	MMLU (57 subjects) vs HELM MMLU (5 subjects)	44.8 (Tsinghua/Zhipu (GLM-130B paper, Section 5.2))	34.4 (Stanford HELM classic v0.3.0)	10.4
C7	C	Mistral-7B-v0.1	GSM8K vs GSM1k	39.1 (Scale AI (GSM1k, Appendix F))	36.4 (Scale AI (GSM1k, Appendix F))	2.7

C8	C	GPT-4 (March 2023) vs gpt-4-0125-preview	HumanEval	86.6 (OpenAI (simple-evals README, commit 6ead091))	67.0 (Anthropic (Claude 3 model card, Table 1))	19.6
C9	C	GPT-5.5	SWE-Bench Pro	59.4 (OpenAI, GPT-5.6 launch post (Jul 9, 2026))	58.6 (OpenAI, Introducing GPT-5.5 (Apr 23, 2026))	0.8
C10	C	QwQ-32B-Preview	GPQA	65.2 (Alibaba Qwen (QwQ-32B-Preview post))	54.5 (DeepSeek (DeepSeek-R1 report v1, Table 5))	10.7
C11	C	Codex (size not named in the abstract) vs Codex-12B	HumanEval, 100 samples per problem	72.31 (OpenAI (Codex TR, Table 1))	70.2 (OpenAI (Codex TR, abstract))	2.11
C12	C	Grok 3 (Think)	AIME 2025	93.3 (xAI (Grok 3 launch post, chart data))	77.3 (xAI (Grok 3 launch post, chart data))	16.0

A1

Minerva 540B, GSM8K (single sample). Class A: Transcription error in a competitor's table. Difference: 9.7 points.

LLaMA-1 Table 7 prints 68.5 in the GSM8k column of the Minerva 540B row. The Minerva paper (Table 3) gives 58.8 for that cell; 68.5 is the maj1@k value of Minerva 62B, the row directly above. The other three cells of the row (33.6, 50.3, 78.5) match the Minerva paper.

Evidence:

- **68.5** Minerva 540B, GSM8K, column 'GSM8k' (no majority voting). Reported by Meta (LLaMA-1 TR, Table 7). arxiv_2302.13971, PDF p.6: "540B 33.6 50.3 68.5 78.5" (claim `llama1.minerva-540b.gsm8k.greedy`)
- **58.8** Minerva 540B, GSM8K, single sample (no majority voting). Reported by Google (Minerva TR, Table 3). arxiv_2206.14858, PDF p.8: "Minerva 540B 33.6% 17.6% 58.8% 63.9%" (claim `minerva.minerva-540b.gsm8k.greedy`)
- **68.5** Minerva 62B, GSM8K, maj1@k. Reported by Google (Minerva TR, Table 3). arxiv_2206.14858, PDF p.8: "Minerva 62B, maj1@k 43.4% 23.5% 68.5% 63.5%" (claim `minerva.minerva-62b.gsm8k.maj1k`)
- **78.5** Minerva 540B, GSM8K, maj1@k. Reported by Google (Minerva TR, Table 3). arxiv_2206.14858, PDF p.8: "Minerva 540B, maj1@k 50.3% 30.8% 78.5% 75.0%" (claim `minerva.minerva-540b.gsm8k.maj1k`)

A2

Mistral 7B, HumanEval (0-shot). Class A: Same vendor, same declared protocol, two papers. Difference: 4.3 points.

Mistral AI reports Mistral 7B at 30.5 (Mistral 7B paper, Oct 2023) and 26.2 (Mixtral paper, Jan 2024) under word-identical protocol descriptions and an identical 'Evaluation Differences' note. Across the two Table 2s, Mistral 7B differs in 12 of 12 benchmark cells, while the LLaMA 2 7B control row is reproduced exactly in 10 of 12 (only NQ and TriviaQA change). Neither paper explains the change for Mistral 7B.

Evidence:

- **30.5** Mistral 7B, HumanEval, 0-shot. Reported by Mistral AI (Mistral 7B TR, Table 2). arxiv_2310.06825, PDF p.4: "Mistral 7B Pretrained 60.1% 81.3% 75.3% 83.0% 80.0% 55.5% 28.8% 69.9% 30.5% 47.5% 13.1% 52.2%" (claim `mistral7b-tr.mistral-7b.humaneval`)
- **26.2** Mistral 7B, HumanEval, 0-shot. Reported by Mistral AI (Mixtral TR, Table 2). arxiv_2401.04088, PDF p.4: "Mistral 7B 7B 62.5% 81.0% 74.2% 82.2% 80.5% 54.9% 23.2% 62.5% 26.2% 50.2% 12.7% 50.0%" (claim `mixtral-tr.mistral-7b.humaneval`)
- **13.1** Mistral 7B, MATH, 4-shot maj@4. Reported by Mistral AI (Mistral 7B TR, Table 2). arxiv_2310.06825, PDF p.4: "Mistral 7B Pretrained 60.1% 81.3% 75.3% 83.0% 80.0% 55.5% 28.8% 69.9% 30.5% 47.5% 13.1% 52.2%" (claim `mistral7b-tr.mistral-7b.math`)
- **12.7** Mistral 7B, MATH, 4-shot maj@4. Reported by Mistral AI (Mixtral TR, Table 2). arxiv_2401.04088, PDF p.4: "Mistral 7B 7B 62.5% 81.0% 74.2% 82.2% 80.5% 54.9% 23.2% 62.5% 26.2% 50.2% 12.7% 50.0%" (claim `mixtral-tr.mistral-7b.math`)
- **44.4** LLaMA 2 7B, MMLU, 5-shot. Reported by Mistral AI (Mistral 7B TR, Table 2). arxiv_2310.06825, PDF p.4: "LLaMA 2 7B Pretrained 44.4% 77.1% 69.5% 77.9% 68.7% 43.2% 24.7% 63.8% 11.6% 26.1% 3.9% 16.0%" (claim `mistral7b-tr.llama2-7b.mmlu`)
- **44.4** LLaMA 2 7B, MMLU, 5-shot. Reported by Mistral AI (Mixtral TR, Table 2). arxiv_2401.04088, PDF p.4: "LLaMA 2 7B 7B 44.4% 77.1% 69.5% 77.9% 68.7% 43.2% 17.5% 56.6% 11.6% 26.1% 3.9% 16.0%" (claim `mixtral-tr.llama2-7b.mmlu`)

A3

Mistral 7B, MMLU (5-shot). Class A: Same vendor, same declared protocol, two papers. Difference: 2.4 points.

Same pair of papers as A2: 60.1 in the Mistral 7B paper, 62.5 in the Mixtral paper, both 5-shot.

Evidence:

- **62.5** Mistral 7B, MMLU, 5-shot. Reported by Mistral AI (Mixtral TR, Table 2). arxiv_2401.04088, PDF p.4: "Mistral 7B 7B 62.5% 81.0% 74.2% 82.2% 80.5% 54.9% 23.2% 62.5% 26.2% 50.2% 12.7% 50.0%" (claim `mixtral-tr.mistral-7b.mmlu`)

- **60.1** Mistral 7B, MMLU, 5-shot. Reported by Mistral AI (Mistral 7B TR, Table 2). arxiv_2310.06825, PDF p.4: "Mistral 7B Pretrained 60.1% 81.3% 75.3% 83.0% 80.0% 55.5% 28.8% 69.9% 30.5% 47.5% 13.1% 52.2%" (claim mistral7b-tr.mistral-7b.mmlu)

A4

Mistral 7B, GSM8K (8-shot maj@8). Class A: Same vendor, same declared protocol, two papers. Difference: 2.2 points.

Same pair of papers as A2: 52.2 in the Mistral 7B paper, 50.0 in the Mixtral paper, both declared as GSM8K (8-shot) with maj@8.

Evidence:

- **52.2** Mistral 7B, GSM8K, 8-shot maj@8. Reported by Mistral AI (Mistral 7B TR, Table 2). arxiv_2310.06825, PDF p.4: "Mistral 7B Pretrained 60.1% 81.3% 75.3% 83.0% 80.0% 55.5% 28.8% 69.9% 30.5% 47.5% 13.1% 52.2%" (claim mistral7b-tr.mistral-7b.gsm8k)
- **50.0** Mistral 7B, GSM8K, 8-shot maj@8. Reported by Mistral AI (Mixtral TR, Table 2). arxiv_2401.04088, PDF p.4: "Mistral 7B 7B 62.5% 81.0% 74.2% 82.2% 80.5% 54.9% 23.2% 62.5% 26.2% 50.2% 12.7% 50.0%" (claim mixtral-tr.mistral-7b.gsm8k)

A5

Claude 3.5 Sonnet (June 2024), MMLU (5-shot). Class A: Competitor reports a higher number than the vendor. Difference: 1.2 points.

Meta's Llama 3 paper lists Claude 3.5 Sonnet at 89.9 on 5-shot MMLU; Anthropic's model card addendum lists 88.7 for the same protocol. For 0-shot CoT both documents give 88.3.

Evidence:

- **89.9** Claude 3.5 Sonnet, MMLU, 5-shot. Reported by Meta (Llama 3 Herd TR, Table 2). arxiv_2407.21783, PDF p.3: "MMLU (5-shot) 69.4 72.3 61.1 83.6 76.9 70.7 87.3 82.6 85.1 89.1 89.9" (claim llama31-tr.claude-35-sonnet.mmlu.5shot)
- **88.7** Claude 3.5 Sonnet, MMLU, 5-shot. Reported by Anthropic (Claude 3.5 Sonnet model card addendum, Table 1). anthropic_claude35_addendum, PDF p.2: "General reasoning 5-shot 88.7% 86.8% 78.3%" (claim anthropic-35-addendum.claude-35-sonnet.mmlu.5shot)
- **88.3** Claude 3.5 Sonnet, MMLU, 0-shot CoT. Reported by Meta (Llama 3 Herd TR, Table 2). arxiv_2407.21783, PDF p.3: "MMLU (0-shot, CoT) 73.0 72.3△ 60.5 86.0 79.9 69.8 88.6 78.7◁ 85.4 88.7 88.3" (claim llama31-tr.claude-35-sonnet.mmlu.0shot-cot)
- **88.3** Claude 3.5 Sonnet, MMLU, 0-shot CoT. Reported by Anthropic (Claude 3.5 Sonnet model card addendum, Table 1). anthropic_claude35_addendum, PDF p.2: "0-shot CoT2 88.3% 85.7% 77.1% 88.7% 86.5%" (claim anthropic-35-addendum.claude-35-sonnet.mmlu.0shot-cot)

A6

Claude 3.5 Sonnet (new, 2024-10-22), MMLU (0-shot CoT). Class A: Competitor reports a lower number than the vendor. Difference: 1.0 points.

Anthropic's October 2024 addendum gives 89.3 (0-shot CoT). DeepSeek-R1 v1 Table 4 gives 88.3 for Claude-3.5-Sonnet-1022 with simple-evals prompts; Anthropic's own addenda state that the simple-evals MMLU implementation uses 0-shot CoT. 88.3 is the value Anthropic published for the June model. On MMLU-Pro (78.0) and GPQA Diamond (65.0) the two documents agree.

Evidence:

- **89.3** Claude 3.5 Sonnet (New), MMLU, 0-shot CoT. Reported by Anthropic (October 2024 addendum, Table 7). anthropic_claude35_oct_addendum, PDF p.6: "0-shot CoT 89.3% 88.3% 85.7% 77.1% 88.7% - 88.6%" (claim anthropic-35-oct.claude-35-sonnet-new.mmlu.0shot-cot)
- **88.3** Claude-3.5-Sonnet-1022, MMLU, simple-evals prompt, Pass@1 (shot count / CoT not stated in this source). Reported by DeepSeek (DeepSeek-R1 TR v1, Table 4). arxiv_2501.12948v1, PDF p.13: "MMLU (Pass@1) 88.3 87.2 88.5 85.2 91.8 90.8" (claim deepseek-r1-v1.claude-35-sonnet-1022.mmlu)
- **78.0** Claude 3.5 Sonnet (New), MMLU-Pro, 0-shot CoT. Reported by Anthropic (October 2024 addendum, Table 7). anthropic_claude35_oct_addendum, PDF p.6: "MMLU Pro 0-shot CoT 78.0% 75.1% 67.9%" (claim anthropic-35-oct.claude-35-sonnet-new.mmlu-pro.0shot-cot)
- **78.0** Claude-3.5-Sonnet-1022, MMLU-Pro, EM. Reported by DeepSeek (DeepSeek-R1 TR v1, Table 4). arxiv_2501.12948v1, PDF p.13: "MMLU-Pro (EM) 78.0 72.6 75.9 80.3 - 84.0" (claim deepseek-r1-v1.claude-35-sonnet-1022.mmlu-pro)
- **65.0** Claude 3.5 Sonnet (New), GPQA Diamond, 0-shot CoT. Reported by Anthropic (October 2024 addendum, Table 7). anthropic_claude35_oct_addendum, PDF p.6: "GPQA (Diamond) 0-shot CoT 65.0% 59.4% 50.4%" (claim anthropic-35-oct.claude-35-sonnet-new.gpqa.0shot-cot)
- **65.0** Claude-3.5-Sonnet-1022, GPQA Diamond, pass@1. Reported by DeepSeek (DeepSeek-R1 TR v1, Table 4). arxiv_2501.12948v1, PDF p.13: "GPQA Diamond (Pass@1) 65.0 49.9 59.1 60.0 75.7 71.5" (claim deepseek-r1-v1.claude-35-sonnet-1022.gpqa)
- **88.3** Claude 3.5 Sonnet, MMLU, 0-shot CoT. Reported by Anthropic (Claude 3.5 Sonnet model card addendum, Table 1). anthropic_claude35_addendum, PDF p.2: "0-shot CoT2 88.3% 85.7% 77.1% 88.7% 86.5%" (claim anthropic-35-addendum.claude-35-sonnet.mmlu.0shot-cot)

- (negative evidence) -, MMLU (simple-evals), protocol identity. Reported by Anthropic (October 2024 addendum, Table 7 footnote i). anthropic_claude35_oct_addendum, PDF p.6: "The simple-evals MMLU implementation in OpenAI's simple-evals suite [26] uses 0-shot CoT" (claim anthropic-35-oct.simple-evals-mmlu-is-0shot-cot)
- (negative evidence) -, MMLU (simple-evals), protocol identity. Reported by Anthropic (June 2024 addendum, footnote 2). anthropic_claude35_addendum, PDF p.2: "The simple-evals MMLU implementation uses 0-shot CoT" (claim anthropic-35-addendum.simple-evals-mmlu-is-0shot-cot)

A7

Chinchilla, MMLU (5-shot). Class A: Abstract vs table; mixed averaging conventions. Difference: 0.1 points.

Abstract 67.5, Table 6 67.6. Recomputed from the 57 per-task accuracies in Table A6: macro average 67.64, question-weighted average 67.56. Gopher's 60.0 in the same table matches Gopher's question-weighted average (60.01), not its macro average (60.61). With one convention for both models the gap is 7.0 (macro) or 7.6 (weighted) points; the text states 7.6. The difference is small; it is listed because it is exactly reproducible.

Evidence:

- **67.6** Chinchilla, MMLU, 5-shot, average over 57 tasks. Reported by DeepMind (Chinchilla TR, Table 6 and Section 4.2.2). arxiv_2203.15556, PDF p.11: "Chinchilla 5-shot 67.6%" (claim chinchilla-tr.table6.chinchilla.mmlu)
- **67.5** Chinchilla, MMLU, 5-shot. Reported by DeepMind (Chinchilla TR, abstract). arxiv_2203.15556, PDF p.1: "Chinchilla reaches a state-of-the-art average accuracy of 67.5% on the MMLU benchmark" (claim chinchilla-tr.abstract.chinchilla.mmlu)
- **60.0** Gopher, MMLU, 5-shot. Reported by DeepMind (Chinchilla TR, Table 6). arxiv_2203.15556, PDF p.11: "Gopher 5-shot 60.0%" (claim chinchilla-tr.table6.gopher.mmlu)
- **14042.0** -, MMLU, test split size. Reported by Hugging Face datasets-server (cais/mmlu). hf_cais_mmlu_size, JSON: ""dataset":"cais/mmlu","config":"all","split":"test","num_bytes_parquet_files":3504718,"num_bytes_memory":4055905,"num_rows":14042" (claim n.mmlu)

A8

PaLM 540B, GSM8K (8-shot chain-of-thought). Class A: Three Google papers, one model, one protocol. Difference: 2.9 points.

The chain-of-thought paper reports 56.9, the PaLM paper 54 (using the same eight exemplars), and the self-consistency paper 56.5 with greedy chain-of-thought. The Minerva paper prints 56.5 and attributes it to the PaLM paper, which prints 54.

Evidence:

- **56.9** PaLM 540B, GSM8K, 8-shot chain-of-thought, greedy. Reported by Wei et al. (Google), arXiv:2201.11903v6 Table 1. arxiv_2201.11903, PDF p.20: "PaLM 540B Standard 17.9 69.4 72.1 25.2 79.2 Chain of thought 56.9 (+39.0)" (claim m2022.wei.palm-540b.gsm8k.8shot-cot)
- **54.0** PaLM 540B, GSM8K, 8-shot chain-of-thought, no calculator. Reported by Chowdhery et al. (Google), arXiv:2204.02311v5 Table 10. arxiv_2204.02311, PDF p.21: "PaLM 540B+chain-of-thought 54%" (claim m2022.palm.palm-540b.gsm8k.8shot-cot)
- **56.5** PaLM-540B, GSM8K, few-shot chain-of-thought, greedy decoding. Reported by Wang et al. (Google), arXiv:2203.11171v4 Table 2. arxiv_2203.11171, PDF p.5: "CoT-prompting 91.9 94.7 74.0 35.8 79.0 56.5 PaLM-540B" (claim m2022.wang.palm-540b.gsm8k.cot-greedy)
- **56.5** PaLM 540B, GSM8K, chain-of-thought, no calculator. Reported by Lewkowycz et al. (Google), arXiv:2206.14858v2 Table 3. arxiv_2206.14858, PDF p.8: "PaLM 540B 8.8% 7.1% 56.5% 58.7%" (claim m2022.minerva.palm-540b.gsm8k.cot)

A9

GPT-4, MMLU (5-shot). Class A: Transcription error when quoting another report. Difference: 0.1 points.

The Falcon paper lists GPT-4 at 86.5 'as reported in GPT-4 paper'; the GPT-4 report gives 86.4.

Evidence:

- **86.5** GPT-4, MMLU, 5-shot. Reported by TII (Falcon series paper, Table 19), quoting GPT-4 paper. arxiv_2311.16867, PDF p.34: "MMLU (5-shot) 70.0 86.5 70.6" (claim m2023.almazrouei2023.gpt-4.mmlu.5shot-as-quoted)
- **86.4** GPT-4, MMLU, 5-shot. Reported by OpenAI (GPT-4 Technical Report). arxiv_2303.08774, PDF p.7: "MMLU [49] 86.4% 70.0% 70.7% 75.2%" (claim m2023.openai2023.gpt-4.mmlu.5shot)

A10

GPT-J 6B, HumanEval pass@1. Class A: Abstract vs table in the same paper. Difference: 0.22 points.

The Codex paper's abstract says GPT-J solves 11.4% of the problems; Table 1 gives pass@1 = 11.62.

Evidence:

- **11.62** GPT-J 6B, HumanEval, pass@1 (0-shot docstring prompt, unbiased estimator over sampled completions, temperature chosen per k). Reported by Chen et al. (Codex paper), Table 1. arxiv_2107.03374, PDF p.7: "GPT-J 6B 11.62% 15.74% 27.74%" (claim m2021.chen-codex.gpt-j-6b.humaneval.pass-at-1)
- **11.4** GPT-J, HumanEval, single sample, abstract. Reported by Chen et al. (Codex paper), abstract. arxiv_2107.03374, PDF p.1: "our model solves 28.8% of the problems, while GPT-3 solves 0% and GPT-J solves 11.4%" (claim m2021.chen-codex.gpt-j.humaneval.abstract)

A11

Llama 2 70B, HumanEval (0-shot pass@1). Class A: Same vendor, two papers. Difference: 0.6 points.

Meta's Code Llama paper lists Llama 2 70B at 30.5 pass@1; the Llama 2 paper lists 29.9 (0-shot).

Evidence:

- **30.5** Llama 2 70B, HumanEval, 0-shot pass@1. Reported by Meta (Code Llama paper, v3). arxiv_2308.12950, PDF p.6: "Llama 2 7B 12.2% 25.2% 44.4% 20.8% 41.8% 65.5% 13B 20.1% 34.8% 61.2% 27.6% 48.1% 69.5% 34B 22.6% 47.0% 79.5% 33.8% 56.9% 77.6% 70B 30.5% 59.4% 87.0%" (claim m2023.roziere2023.llama-2-70b.humaneval.pass1)
- **29.9** Llama 2 70B, HumanEval, 0-shot pass@1. Reported by Meta (Touvron et al. 2023b, Table 4). arxiv_2307.09288, PDF p.8: "HumanEval (0-shot) 48.1 67.0 26.2 - 29.9" (claim m2023.touvron2023b.llama-2-70b.humaneval.0shot)

A12

GPT-6 Astra, Terminal-Bench 4.0. Class A: Text vs table in the same launch post. Difference: 0.2 points.

OpenAI's GPT-6 Astra post states 57.9 in the text and 57.7 in its evaluation table.

Evidence:

- **57.9** GPT-6 Astra, Terminal-Bench 4.0, prose figure in the same post. Reported by OpenAI, GPT-6 Astra launch post (Wayback capture 2026-09-03). openai_gpt6_astra_blog, HTML: "Terminal-Bench 4.0 tests agents on complex terminal-based tasks, including software engineering, system configuration, and data analysis. GPT-6 Astra reaches a new high at 57.9%" (claim m2026.astra.gpt-6-astra.terminal-bench-4-0.prose)
- **57.7** GPT-6 Astra, Terminal-Bench 4.0, maximum at any effort (eval table). Reported by OpenAI, GPT-6 Astra launch post (Wayback capture 2026-09-03). openai_gpt6_astra_blog, HTML: "Terminal-Bench 4.0 57.7% 37.3% 55.8% 42.0% 52.3% 19.1%" (claim m2026.astra.gpt-6-astra.terminal-bench-4-0.table)

B1

Mixtral 8x7B, GSM8K. Class B: Two protocols in one paper. Difference: 16.0 points.

Table 2 reports 74.4 (8-shot maj@8); Table 3, the comparison with GPT-3.5, reports 58.4 (5-shot).

Evidence:

- **74.4** Mixtral 8x7B, GSM8K, 8-shot maj@8. Reported by Mistral AI (Mixtral TR, Table 2). arxiv_2401.04088, PDF p.4: "Mixtral 8x7B 13B 70.6% 84.4% 77.2% 83.6% 83.1% 59.7% 30.6% 71.5% 40.2% 60.7% 28.4% 74.4%" (claim mixtral-tr.mixtral-8x7b.gsm8k.maj8)
- **58.4** Mixtral 8x7B, GSM8K, 5-shot. Reported by Mistral AI (Mixtral TR, Table 3). arxiv_2401.04088, PDF p.5: "GSM-8K (5-shot) 53.6% 57.1% 58.4%" (claim mixtral-tr.mixtral-8x7b.gsm8k.5shot)

B2

OpenAI o3-mini (high), SWE-bench Verified. Class B: Same product label, different checkpoint and scaffold. Difference: 11.7 points.

OpenAI publishes two SWE-bench Verified numbers under the label 'OpenAI o3-mini (high)': 61.0 in the GPT-4.5 launch table (footnote: 'best internal performance') and 49.3 in the GPT-4.1 launch appendix and the o3 launch chart. The o3-mini system card explains 61: a non-final checkpoint with an internal tools scaffold on a 477-task subset; the launch candidate with Agentless scores 39. The o3-mini launch post's image alt text says 48.9, which in the chart image is o1's bar.

Evidence:

- **61.0** OpenAI o3-mini (high), SWE-bench Verified, 'best internal performance'. Reported by OpenAI (GPT-4.5 launch post, Feb 27 2025). openai_gpt45_blog, HTML: "SWE-Bench Verified (coding)* 38.0% 30.7% 61.0% *Numbers shown represent best internal performance." (claim gpt45-blog.o3-mini-high.swebench-verified)
- **49.3** OpenAI o3-mini (high), SWE-bench Verified (477 of 500 tasks), 23/500 tasks omitted (footnote 2). Reported by OpenAI (GPT-4.1 launch post, Apr 14 2025, appendix). openai_gpt41_blog, HTML: "SWE-

bench Verified 2 54.6% 23.6% - 33.2% 8.7% 41.0% 49.3% 38.0%" (claim `m2025.gpt41-blog.o3-mini-high.swebench`)

- **61.0** o3-mini (tools), non-final checkpoint, SWE-bench Verified (n=477 subset), internal tools scaffold, pass@1 over 4 tries. Reported by OpenAI (o3-mini system card, Section 5.7.2). openai_o3mini_system_card, PDF p.28: "o3-mini (tools) performs the best on SWE-bench Verified at 61%" (claim `o3mini-card.o3-mini-tools.swebench-verified`)
- **49.3** o3-mini (high reasoning effort per page footnote), SWE-bench Verified, high reasoning effort ('All models are evaluated at high reasoning effort settings'). Reported by OpenAI (o3 and o4-mini launch post, Apr 16 2025, chart data). openai_o3_o4mini_blog, HTML: "{"model":"o3-mini","value":49.3,"labelValue":49.3" (claim `m2025.o3-blog.o3-mini.swebench`)
- **48.9** o3-mini (high), SWE-bench Verified, image alt text of the launch chart (the chart itself prints 48.9 for o1 and 49.3 for o3-mini (high)). Reported by OpenAI (o3-mini launch post, Jan 31 2025). openai_o3mini_blog, HTML: "while "o3-mini (high)" (yellow) achieves the highest accuracy at 48.9%" (claim `o3mini-blog.o3-mini-high.swebench-verified`)

B3

PaLM 2, GSM8K. Class B: Two protocols in one table cell. Difference: 10.3 points.

PaLM 2 Table 7 prints 80.7 / 91.0 (chain-of-thought / self-consistency) in the same cell.

Evidence:

- **91.0** PaLM 2, GSM8K, self-consistency. Reported by Google (PaLM 2 TR, Table 7). arxiv_2305.10403, PDF p.15: "GSM8K 92.0b 56.5 / 74.4 58.8 / 78.5 92.0 80.7 / 91.0 84.7 / 92.2" (claim `palm2-tr.palm2.gsm8k.sc`)
- **80.7** PaLM 2, GSM8K, chain-of-thought. Reported by Google (PaLM 2 TR, Table 7). arxiv_2305.10403, PDF p.15: "GSM8K 92.0b 56.5 / 74.4 58.8 / 78.5 92.0 80.7 / 91.0 84.7 / 92.2" (claim `palm2-tr.palm2.gsm8k.cot`)

B4

Claude Sonnet 5.5, Terminal-Bench 4.0. Class B: Vendor harness vs independent harness. Difference: 7.0 points.

Anthropic reports 70.6; Artificial Analysis reports 63.6 on its own harness, for the configuration it labels 'Max, Default Fallback'.

Evidence:

- **70.6** Claude Sonnet 5.5, Terminal-Bench 4.0, vendor harness; effort level not stated for Sonnet 5.5. Reported by Anthropic (Claude Sonnet 5.5 page). anthropic_claude_sonnet55_page, HTML: "scores 70.6% on Terminal-Bench 4.0, an agentic coding evaluation" (claim `anthropic-sonnet55.sonnet-55.terminal-bench-4`)
- **63.6** Claude Sonnet 5.5, Terminal-Bench 4.0, Artificial Analysis harness, max effort, Default Fallback variant ('Claude Sonnet 5.5 (Max, Default Fallback)'). Reported by Artificial Analysis (model page, capture 2026-10-16). aa_claude_sonnet55, HTML: ""terminalBench40":0.636363636363636" (claim `aa.sonnet-55.terminal-bench-4`)

B5

Claude Opus 5.5, Terminal-Bench 4.0. Class B: Vendor harness vs independent harness. Difference: 6.8 points.

Anthropic reports 66.4 at xhigh effort, with production safeguards enabled, and states that its setup reproduces the public leaderboard for Opus 5 (51.8 vs 52.3); Artificial Analysis reports 59.6 for the configuration it labels 'Max, Default Fallback'.

Evidence:

- **66.4** Claude Opus 5.5, Terminal-Bench 4.0, xhigh effort, adaptive thinking, production safeguards enabled (on intervention, cyber tasks completed by Claude Opus 4.8 and bio/frontier-LLM tasks by Claude Opus 5). Reported by Anthropic (Claude Opus 5.5 page). anthropic_claude_opus55_page, HTML: "Terminal-Bench 4.01 66.4% 55.8% 52.3% 57.9% 37.3%" (claim `anthropic-opus55.opus-55.terminal-bench-4`)
- **59.6** Claude Opus 5.5, Terminal-Bench 4.0, Artificial Analysis harness, max effort, Default Fallback variant ('Claude Opus 5.5 (Max, Default Fallback)'). Reported by Artificial Analysis (model page, capture 2026-10-18). aa_claude_opus55, HTML: ""terminalBench40":0.595959595959596" (claim `aa.opus-55.terminal-bench-4`)

B6

Claude 3.7 Sonnet, SWE-bench Verified (489-task subset). Class B: Parallel test-time compute shown next to the single-attempt score. Difference: 6.6 points.

70.3 with parallel attempts and a scoring model, 63.7 without, both on the same 489-task subset.

Evidence:

- **70.3** Claude 3.7 Sonnet, SWE-bench Verified (n=489 subset), parallel attempts, regression-test rejection, scoring-model selection. Reported by Anthropic (Claude 3.7 Sonnet launch post). anthropic_claude37_blog, HTML: "This results in a score of 70.3% on the subset of n=489 verified tasks which work on our infrastructure" (claim `claude37-blog.claude-37-sonnet.swebench-verified.parallel`)
- **63.7** Claude 3.7 Sonnet, SWE-bench Verified (n=489 subset), without the parallel scaffold. Reported by Anthropic (Claude 3.7 Sonnet launch post). anthropic_claude37_blog, HTML: "Without this scaffold,

Claude 3.7 Sonnet achieves 63.7% on SWE-bench Verified using this same subset" (claim `claude37-blog.claude-37-sonnet.swebench-verified.single`)

B7

Gemini Ultra, MMLU. Class B: Headline uses the protocol that crosses a threshold. Difference: 6.34 points.

The launch post states 90.0 and 'first model to outperform human experts on MMLU' (expert estimate 89.8). The technical report shows 90.04 with uncertainty-routed CoT@32 and 83.7 with 5-shot, below GPT-4's reported 86.4.

Evidence:

- **90.04** Gemini Ultra, MMLU, CoT@32 with uncertainty routing. Reported by Google (Gemini 1.0 TR, Table 2). arxiv_2312.11805, PDF p.8: "(Hendrycks et al., 2021a) 90.04% CoT@32* 83.7% 5-shot 79.13% CoT@8* 71.8% 5-shot 87.29% CoT@32 (via API**) 86.4% 5-shot (reported)" (claim `gemini1-tr.gemini-ultra.mmlu.cot32`)
- **83.7** Gemini Ultra, MMLU, 5-shot. Reported by Google (Gemini 1.0 TR, Table 2). arxiv_2312.11805, PDF p.8: "(Hendrycks et al., 2021a) 90.04% CoT@32* 83.7% 5-shot 79.13% CoT@8* 71.8% 5-shot 87.29% CoT@32 (via API**) 86.4% 5-shot (reported)" (claim `gemini1-tr.gemini-ultra.mmlu.5shot`)
- **90.0** Gemini Ultra, MMLU, not stated in the post. Reported by Google (launch post, Dec 6 2023). google_gemini1_blog, HTML: "With a score of 90.0%, Gemini Ultra is the first model to outperform human experts on MMLU" (claim `gemini-blog.gemini-ultra.mmlu`)
- **89.8** Human expert (estimate), MMLU, 95th percentile test-taker estimate. Reported by Hendrycks et al. (MMLU paper). arxiv_2009.03300, PDF p.3: "we then estimate that expert-level accuracy is approximately 89.8%" (claim `mmlu-paper.human-expert`)
- **86.4** GPT-4, MMLU, 5-shot (reported). Reported by Google (Gemini 1.0 TR, Table 2). arxiv_2312.11805, PDF p.8: "(Hendrycks et al., 2021a) 90.04% CoT@32* 83.7% 5-shot 79.13% CoT@8* 71.8% 5-shot 87.29% CoT@32 (via API**) 86.4% 5-shot (reported)" (claim `gemini1-tr.gpt4.mmlu.5shot-reported`)

B8

gpt-4o (2024-05-13), GPQA (subset not named in the README). Class B: Two system prompts; the higher row later removed. Difference: 3.7 points.

On GPT-4o's launch day OpenAI's simple-evals README listed two rows: 53.6 with the 'chatgpt' system prompt and 49.9 with the 'assistant' prompt. The July 2025 README lists only the 49.9 row.

Evidence:

- **53.6** gpt-4o, GPQA (subset not stated), 0-shot CoT, 'chatgpt' system prompt. Reported by OpenAI (simple-evals README, commit 294cb1f, 2024-05-13). openai_simple_evals_readme_2024-05-13, MD: "gpt-4o chatgpt[^1] 88.7 53.6 76.6 90.2 90.5 83.4" (claim `simple-evals-2024-05-13.gpt-4o.gpqa.chatgpt-prompt`)
- **49.9** gpt-4o, GPQA (subset not stated), 0-shot CoT, 'assistant' system prompt. Reported by OpenAI (simple-evals README, commit 294cb1f, 2024-05-13). openai_simple_evals_readme_2024-05-13, MD: "gpt-4o assistant[^2] 87.2 49.9 76.6 91.0 89.9 83.7" (claim `simple-evals-2024-05-13.gpt-4o.gpqa.assistant-prompt`)
- **49.9** gpt-4o-2024-05-13, GPQA (subset not stated), 0-shot CoT, 'assistant' system prompt. Reported by OpenAI (simple-evals README, commit 6ead091, 2025-07-09). openai_simple_evals_readme, MD: "gpt-4o-2024-05-13 assistant 87.2 49.9 76.6 91.0 89.9 83.7 39.0" (claim `simple-evals-2025-07-09.gpt-4o-0513.gpqa.assistant-prompt`)

B9

Claude 3.5 Sonnet (June 2024), MMLU. Class B: Three protocols in one table. Difference: 2.1 points.

The addendum gives 90.4 (5-shot CoT), 88.7 (5-shot) and 88.3 (0-shot CoT).

Evidence:

- **90.4** Claude 3.5 Sonnet, MMLU, 5-shot CoT. Reported by Anthropic (Claude 3.5 Sonnet model card addendum, Table 1). anthropic_claude35_addendum, PDF p.2: "5-shot CoT 90.4% 88.2% 81.5%" (claim `anthropic-35-addendum.claude-35-sonnet.mmlu.5shot-cot`)
- **88.3** Claude 3.5 Sonnet, MMLU, 0-shot CoT. Reported by Anthropic (Claude 3.5 Sonnet model card addendum, Table 1). anthropic_claude35_addendum, PDF p.2: "0-shot CoT2 88.3% 85.7% 77.1% 88.7% 86.5%" (claim `anthropic-35-addendum.claude-35-sonnet.mmlu.0shot-cot`)
- **88.7** Claude 3.5 Sonnet, MMLU, 5-shot. Reported by Anthropic (Claude 3.5 Sonnet model card addendum, Table 1). anthropic_claude35_addendum, PDF p.2: "General reasoning 5-shot 88.7% 86.8% 78.3%" (claim `anthropic-35-addendum.claude-35-sonnet.mmlu.5shot`)

B10

Claude 3.5 Sonnet (June 2024), MMLU-Pro. Class B: Competitor uses a different protocol. Difference: 1.9 points.

Meta reports 77.0 (5-shot CoT); Anthropic reports 75.1 (0-shot CoT).

Evidence:

- **77.0** Claude 3.5 Sonnet, MMLU-Pro, 5-shot CoT. Reported by Meta (Llama 3 Herd of Models, Table 2). arxiv_2407.21783, PDF p.3: "MMLU-Pro (5-shot, CoT) 48.3 - 36.9 66.4 56.3 49.2 73.3 62.7 64.8 74.0 77.0" (claim `m2024.llama3herd.claude-3.5-sonnet.mmlu-pro.5-shot-cot`)
- **75.1** Claude 3.5 Sonnet, MMLU-Pro, 0-shot CoT. Reported by Anthropic (Claude 3.5 Sonnet (New) & Haiku addendum, Oct 2024, Table 7). anthropic_claude35_oct_addendum, PDF p.6: "MMLU Pro General reasoning 0-shot CoT 78.0% 75.1% 67.9% 54.9% - 75.8% 73.3%" (claim `m2024.c35oct.claude-3.5-sonnet.mmlu-pro.0-shot-cot`)

B11

Claude Opus 4.6, SWE-bench Verified. Class B: Prompt modification reported next to the standard score. Difference: 0.58 points.

The system card reports 80.84 with the standard prompt and 81.4 with a prompt modification; the launch post gives 81.42 with the prompt modification.

Evidence:

- **81.42** Claude Opus 4.6, SWE-bench Verified, prompt-modified. Reported by Anthropic, Introducing Claude Opus 4.6. anthropic_claude_opus46_blog, HTML: "SWE-bench Verified: Our score was averaged over 25 trials. With a prompt modification, we saw a score of 81.42%." (claim `m2026.opus46blog.claude-opus-4-6.swe-bench-verified.prompt-modified`)
- **80.84** Claude Opus 4.6, SWE-bench Verified, averaged over 25 trials, adaptive thinking, max effort, standard prompt. Reported by Anthropic, Claude Opus 4.6 System Card (Feb 2026). anthropic_claude_opus46_system_card, PDF p.19: "Claude Opus 4.6 achieves 80.84% on SWE-bench Verified and 77.83% on SWE-bench" (claim `m2026.opus46card.claude-opus-4-6.swe-bench-verified.standard`)
- **81.4** Claude Opus 4.6, SWE-bench Verified, prompt-modified harness (non-standard). Reported by Anthropic, Claude Opus 4.6 System Card (Feb 2026). anthropic_claude_opus46_system_card, PDF p.19: "For SWE-bench Verified, we found that the following prompt modification resulted in a score of 81.4%:" (claim `m2026.opus46card.claude-opus-4-6.swe-bench-verified.prompt-modified`)

B12

GPT-6 Astra, Terminal-Bench 4.0. Class B: Independent harness reports a higher number than the vendor. Difference: 1.4 points.

Artificial Analysis reports 59.1 at max effort; OpenAI's table gives 57.7. Here the independent number is the higher one.

Evidence:

- **59.1** GPT-6 Astra (max), Terminal-Bench 4.0, Artificial Analysis harness, effort 'max'. Reported by Artificial Analysis model page (Wayback). aa_claude_opus55, HTML: ""terminalBench40":0.590909090909091" (claim `m2026.aa.gpt-6-astra.terminal-bench-4-0`)
- **57.7** GPT-6 Astra, Terminal-Bench 4.0, maximum at any effort (eval table). Reported by OpenAI, GPT-6 Astra launch post (Wayback capture 2026-09-03). openai_gpt6_astra_blog, HTML: "Terminal-Bench 4.0 57.7% 37.3% 55.8% 42.0% 52.3% 19.1%" (claim `m2026.astra.gpt-6-astra.terminal-bench-4-0.table`)

C1

GPT-4, MATH. Class C: Checkpoint not identified. Difference: 10.4 points.

52.9 (Gemini report, own API run in Nov 2023, snapshot not stated), 50.3 (Zheng et al., cited by Gemini) and 42.5 (PaLM 2 report). The documents do not establish that the same GPT-4 snapshot was measured, so the spread is not attributed to evaluation alone.

Evidence:

- **52.9** GPT-4 (API, Nov 2023; snapshot not stated), MATH, 4-shot. Reported by Google (Gemini 1.0 TR, Table 2). arxiv_2312.11805, PDF p.8: "MATH 53.2% 32.6% 52.9% 34.1%" (claim `gemini1-tr.gpt4.math.api`)
- **42.5** GPT-4, MATH, as cited from OpenAI (2023b). Reported by Google (PaLM 2 TR, Table 7). arxiv_2305.10403, PDF p.15: "MATH 50.3a 8.8 33.6 / 50.3 42.5 34.3 / 48.8 33.2 / 45.2" (claim `palm2-tr.gpt4.math`)
- **50.3** GPT-4, MATH, as cited from Zheng et al. (2023). Reported by Google (Gemini 1.0 TR, Table 2). arxiv_2312.11805, PDF p.8: "50.3% (Zheng et al., 2023)" (claim `gemini1-tr.gpt4.math.zheng`)

C2

gpt-3.5-turbo / **'GPT-3.5', GSM8K.** Class C: Checkpoint identity asserted by a third party. Difference: 17.8 points.

Chain-of-Thought Hub measures gpt-3.5-turbo at 74.9 with its own scripts; the GPT-4 report lists 'GPT-3.5' at 57.1 (5-shot). CoT Hub treats the GPT-4 report's GPT-3.5 column as gpt-3.5-turbo; OpenAI does not name the snapshot.

Evidence:

- **74.9** gpt-3.5-turbo, GSM8K, CoT Hub test scripts. Reported by Chain-of-Thought Hub (Fu et al., Table 1). arxiv_2305.17306, PDF p.3: "gpt-3.5-turbo ? RLHF 74.9* - 67.3* 70.1* 48.1 54.4*" (claim cothub.gpt-35-turbo.gsm8k)
- **57.1** GPT-3.5, GSM8K, 5-shot. Reported by OpenAI (GPT-4 TR, Table 2). arxiv_2303.08774, PDF p.7: "GSM-8K [60] 92.0%* 57.1% 58.8% 87.3%" (claim gpt4-tr.gpt-35.gsm8k)

C3

Claude Opus 4.5, SWE-bench Verified vs SWE-bench Pro. Class C: Different benchmarks. Difference: 25.5 points.
80.9 on SWE-bench Verified and 55.4 on SWE-bench Pro (Moonshot internal evaluation). Two benchmarks; not a discrepancy.
Evidence:

- **80.9** Claude Opus 4.5, SWE-bench Verified, as reported (unstarred), Claude Opus 4.5 with extended thinking per Kimi's baseline definition. Reported by Moonshot AI (Kimi K2.5 TR, Table 4). arxiv_2602.02276, PDF p.12: "SWE-Bench Verified 76.8 80.9 80.0 76.2 73.1" (claim kimi-k25.claude-opus-45.swebench-verified)
- **55.4** Claude Opus 4.5, SWE-bench Pro (public), Moonshot internal evaluation (*), *Claude Opus 4.5 with extended thinking. Reported by Moonshot AI (Kimi K2.5 TR, Table 4). arxiv_2602.02276, PDF p.12: "SWE-Bench Pro (public) 50.7 55.4* 55.6" (claim kimi-k25.claude-opus-45.swebench-pro)

C4

DeepSeek-R1, MMLU vs MMLU-Redux. Class C: Different test sets and prompts. Difference: 2.1 points.
92.9 on MMLU-Redux (ZeroEval prompt) and 90.8 on MMLU (simple-evals prompt), DeepSeek-R1 v1 Table 4.
Evidence:

- **92.9** DeepSeek-R1, MMLU-Redux, ZeroEval prompt, 0-shot. Reported by DeepSeek (DeepSeek-R1 TR v1, Table 4). arxiv_2501.12948v1, PDF p.13: "MMLU-Redux (EM) 88.9 88.0 89.1 86.7 - 92.9" (claim deepseek-r1-v1.deepseek-r1.mmlu-redux)
- **90.8** DeepSeek-R1, MMLU, pass@1. Reported by DeepSeek (DeepSeek-R1 TR v1, Table 4). arxiv_2501.12948v1, PDF p.13: "MMLU (Pass@1) 88.3 87.2 88.5 85.2 91.8 90.8" (claim deepseek-r1-v1.deepseek-r1.mmlu)

C5

GPT-4 / ChatGPT, HumanEval vs HumanEval+. Class C: Different test suites. Difference: 12.2 points.
EvalPlus: GPT-4 88.4 on the original tests, 76.2 with the extended tests; ChatGPT 73.2 and 63.4.
Evidence:

- **88.4** GPT-4, HumanEval (EvalPlus base tests), greedy pass@1. Reported by EvalPlus (Liu et al., Table 3). arxiv_2305.01210, PDF p.7: "GPT-4 [49] N/A base 88.4 +extra 76.2" (claim evalplus.gpt-4.humaneval-base)
- **76.2** GPT-4, HumanEval+ (EvalPlus extended tests), greedy pass@1. Reported by EvalPlus (Liu et al., Table 3). arxiv_2305.01210, PDF p.7: "GPT-4 [49] N/A base 88.4 +extra 76.2" (claim evalplus.gpt-4.humaneval-plus)
- **73.2** ChatGPT, HumanEval (EvalPlus base tests), greedy pass@1. Reported by EvalPlus (Liu et al., Table 3). arxiv_2305.01210, PDF p.7: "ChatGPT [48] N/A base 73.2 69.4 88.6 94.0 +extra 63.4 62.5 82.1 91.1" (claim evalplus.chatgpt.humaneval-base)
- **63.4** ChatGPT, HumanEval+ (EvalPlus extended tests), greedy pass@1. Reported by EvalPlus (Liu et al., Table 3). arxiv_2305.01210, PDF p.7: "ChatGPT [48] N/A base 73.2 69.4 88.6 94.0 +extra 63.4 62.5 82.1 91.1" (claim evalplus.chatgpt.humaneval-plus)

C6

GLM-130B, MMLU (57 subjects) vs HELM MMLU (5 subjects). Class C: Different test sets. Difference: 10.4 points.
44.8 in the GLM-130B paper (5-shot, 57 subjects); 34.4 in HELM, whose MMLU is a 5-subject subset scored by exact match.
Evidence:

- **44.8** GLM-130B, MMLU (57 subjects), 5-shot. Reported by Tsinghua/Zhipu (GLM-130B paper, Section 5.2). arxiv_2210.02414, PDF p.8: "achieving an accuracy of 44.8 when the training has to end" (claim glm130b-tr.glm-130b.mmlu)
- **34.4** GLM (130B), MMLU (HELM 5-subject subset), HELM exact match, 5 subjects x 3 trials. Reported by Stanford HELM classic v0.3.0. helm_classic_v0.3.0_mmlu, JSON: "{ "value": "GLM (130B)", "description": "", "markdown": false }, { "value": 0.34397660818713455," (claim helm-v030.glm-130b.mmlu)

C7

Mistral-7B-v0.1, GSM8K vs GSM1k. Class C: Different test sets; difference not significant. Difference: 2.7 points.

Scale AI measures 39.1 on GSM8K and 36.4 on GSM1k for Mistral-7B-v0.1, p = 0.078 in the paper's own test. The paper's maximum drop across all models is 8 points.

Evidence:

- **39.1** Mistral-7B-v0.1, GSM8K, GSM1k standard prompt. Reported by Scale AI (GSM1k, Appendix F). arxiv_2405.00332, PDF p.25: "Mistral-7B-v0.1 0.027 0.391 0.364 1.421 0.078" (claim `gsm1k.mistral-7b-v01.gsm8k`)
- **36.4** Mistral-7B-v0.1, GSM1k, GSM1k standard prompt. Reported by Scale AI (GSM1k, Appendix F). arxiv_2405.00332, PDF p.25: "Mistral-7B-v0.1 0.027 0.391 0.364 1.421 0.078" (claim `gsm1k.mistral-7b-v01.gsm1k`)
- **0.08** Mistral-7B-v0.1, GSM8K minus GSM1k, p-value as printed; the source states a two-tailed two-proportion Z-test, although 0.078 for Z = 1.421 equals the one-sided value (two-sided would be 0.155). Reported by Scale AI (GSM1k, Appendix F). arxiv_2405.00332, PDF p.25: "Mistral-7B-v0.1 0.027 0.391 0.364 1.421 0.078" (claim `gsm1k.mistral-7b-v01.pvalue`)
- **8.0** all models evaluated, GSM8K minus GSM1k, maximum drop. Reported by Scale AI (GSM1k abstract). arxiv_2405.00332, PDF p.1: "we observe accuracy drops of up to 8%" (claim `gsm1k.max-drop`)

C8

GPT-4 (March 2023) vs gpt-4-0125-preview, HumanEval. Class C: Different checkpoints; disclosed. Difference: 19.6 points.

The Claude 3 model card compares against GPT-4's March 2023 HumanEval figure (67.0) and states in footnote 6 that higher scores were reported for a newer GPT-4T; simple-evals lists 86.6 for gpt-4-0125-preview.

Evidence:

- **86.6** gpt-4-0125-preview, HumanEval, simple-evals. Reported by OpenAI (simple-evals README, commit 6ead091). openai_simple_evals_readme, MD: "gpt-4-0125-preview assistant 85.4 41.4 64.5 86.6 85.1 81.5 n/a" (claim `simple-evals-2025-07-09.gpt-4-0125.humaneval`)
- **67.0** GPT-4, HumanEval, 0-shot (as reported in the GPT-4 TR). Reported by Anthropic (Claude 3 model card, Table 1). anthropic_claude3_card, PDF p.6: "0-shot 84.9% 73.0% 75.9% 67.0%6 48.1% 74.4% 71.9% 67.7%" (claim `claude3-card.gpt4.humaneval.reported`)

C9

GPT-5.5, SWE-Bench Pro. Class C: Protocol not stated for one number. Difference: 0.8 points.

OpenAI's GPT-5.5 post gives 58.6 at xhigh effort; OpenAI's GPT-5.6 post lists GPT-5.5 at 59.4 without stating the setting.

Evidence:

- **59.4** GPT-5.5, SWE-Bench Pro, as listed in the GPT-5.6 post. Reported by OpenAI, GPT-5.6 launch post (Jul 9, 2026). openai_gpt56_blog, HTML: "SWE-Bench Pro 64.6% - 63.4% 62.7% 59.4%" (claim `m2026.gpt56.gpt-5-5.swe-bench-pro`)
- **58.6** GPT-5.5, SWE-Bench Pro (Public), reasoning effort xhigh. Reported by OpenAI, Introducing GPT-5.5 (Apr 23, 2026). openai_gpt55_blog, HTML: "On SWE-Bench Pro , which evaluates real-world GitHub issue resolution, it reaches 58.6%" (claim `m2026.gpt55.gpt-5-5.swe-bench-pro`)

C10

QwQ-32B-Preview, GPQA. Class C: Protocol and subset not stated by the vendor. Difference: 10.7 points.

Qwen's post states 65.2 on GPQA without naming the subset or protocol; DeepSeek measures 54.5 on GPQA Diamond, pass@1.

Evidence:

- **65.2** QwQ-32B-Preview, GPQA, not stated. Reported by Alibaba Qwen (QwQ-32B-Preview post). qwen_qwq32b_preview_blog, HTML: "achieving impressive scores of 65.2% on GPQA, showcasing its graduate-level scientific reasoning capabilities; 50.0% on AIME, highlighting its strong mathematical problem-solving skills; 90.6% on MATH-500" (claim `m2024.qwq.qwq-32b-preview.gpqa.unspecified`)
- **54.5** QwQ-32B-Preview, GPQA Diamond, pass@1. Reported by DeepSeek (DeepSeek-R1 report v1, Table 5). arxiv_2501.12948v1, PDF p.14: "QwQ-32B-Preview 50.0 60.0 90.6 54.5 41.9 1316" (claim `m2024.r1v1.qwq-32b-preview.gpqa-diamond.pass1`)

C11

Codex (size not named in the abstract) vs Codex-12B, HumanEval, 100 samples per problem. Class C: Model identity inferred, not stated. Difference: 2.11 points.

The Codex abstract says 'our model' solves 70.2% of problems with 100 samples per problem without naming its size; Table 1 gives Codex-12B pass@100 = 72.31. The abstract's single-sample figure matches

Codex-12B's pass@1 (28.81 in Table 1), which suggests but does not establish that both numbers refer to the same model.

Evidence:

- **72.31** Codex-12B, HumanEval, pass@100. Reported by OpenAI (Codex TR, Table 1). arxiv_2107.03374, PDF p.7: "CODEX-12B 28.81% 46.81% 72.31%" (claim `codex-tr.table1.codex-12b.humaneval.pass100`)
- **70.2** Codex ("our model" in abstract; size not stated), HumanEval, 100 samples per problem (abstract; not labelled pass@100). Reported by OpenAI (Codex TR, abstract). arxiv_2107.03374, PDF p.1: "we solve 70.2% of our problems with 100 samples per problem" (claim `codex-tr.abstract.codex-12b.humaneval.pass100`)
- **28.81** Codex-12B, HumanEval, pass@1 (0-shot docstring prompt, unbiased estimator over sampled completions, temperature chosen per k). Reported by Chen et al. (Codex paper), Table 1. arxiv_2107.03374, PDF p.7: "CODEX-12B 28.81% 46.81% 72.31%" (claim `m2021.chen-codex.codex-12b.humaneval.pass-at-1`)

C12

Grok 3 (Think), AIME 2025. Class C: Protocol of one value not stated. Difference: 16.0 points.

xAI's launch chart shows two values for Grok 3 (Think): 93.3, which the text identifies as cons@64, and 77.3, whose setting the page does not define. A competitor bar coded O3MH shows a single value, 86.5; the pinned page does not resolve the code to a model name. The headline number is the cons@64 value.

Evidence:

- **93.3** Grok 3 (Think), AIME 2025, cons@64. Reported by xAI (Grok 3 launch post, chart data). xai_grok3_blog, HTML: ""group":"AIME'25","data":[{"model":"GROK_3_R","score":77.3,"scoreMax":93.3}" (claim `grok3-blog.grok3-think.aime25.cons64`)
- **77.3** Grok 3 (Think), AIME 2025, chart 'score' value (lower test-time compute than cons@64; exact setting not stated). Reported by xAI (Grok 3 launch post, chart data). xai_grok3_blog, HTML: ""group":"AIME'25","data":[{"model":"GROK_3_R","score":77.3,"scoreMax":93.3}" (claim `grok3-blog.grok3-think.aime25.base`)
- **86.5** chart code 'O3MH' (label mapping not in pinned source; presumably o3-mini (high)), AIME 2025, single value, no consensus extension. Reported by xAI (Grok 3 launch post, chart data). xai_grok3_blog, HTML: "{"model":"O3MH","score":86.5,"scoreMax":86.5}" (claim `grok3-blog.o3-mini-high.aime25`)

FIG 3: PRIMARY-SOURCE CROPS FOR CASES A1, A2-A4 AND B7

Rendered from the SHA-256-pinned PDFs at build time (scripts/build_forensic_analysis.py); physical PDF page numbers

A1 Meta LLaMA-1 TR, Table 7 (PDF p.6): the Minerva 540B row prints 68.5 in the GSM8k column

		MATH	+maj1@k	GSM8k	+maj1@k
PaLM	8B	1.5	-	4.1	-
	62B	4.4	-	33.0	-
	540B	8.8	-	56.5	-
Minerva	8B	14.1	25.4	16.2	28.4
	62B	27.6	43.4	52.4	68.5
	540B	**33.6**	**50.3**	**68.5**	**78.5**
LLaMA	7B	2.9	6.9	11.0	18.1
	13B	3.9	8.8	17.8	29.3
	33B	7.1	15.2	35.6	53.1
	65B	10.6	20.5	50.9	69.7

Table 7: **Model performance on quantitative reason-**

A1 Google Minerva TR, Table 3 (PDF p.8): Minerva 540B GSM8k 58.8; Minerva 62B maj1@k GSM8k 68.5

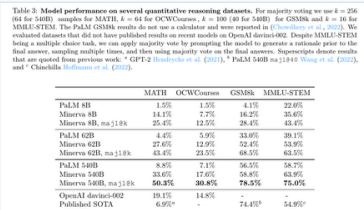


Table 3: **Model performance on several quantitative reasoning datasets.** For majority voting we use $k = 256$ (64 for 540B) samples for MATH, $k = 64$ for OCWCourses, $k = 100$ (40 for 540B) for GSM8k and $k = 16$ for MMLU-STEM. The PaLM GSM8k results do not use a calculator and were reported in (Chowdhery et al., 2022). We evaluated datasets that did not have published results on recent models on OpenAI davinci-002. Despite MMLU-STEM being a multiple choice task, we can apply majority vote by prompting the model to generate a rationale prior to the final answer, sampling multiple times, and then using majority vote on the final answers. Superscripts denote results that are quoted from previous work: [a] GPT-2 Hendrycks et al. (2021), [b] PaLM 540B maj1@40 Wang et al. (2022), and [c] Chinchilla Hoffmann et al. (2022).

	MATH	OCWCourses	GSM8k	MMLU-STEM
PaLM 8B	1.5%	1.5%	4.1%	22.0%
Minerva 8B	14.1%	7.7%	16.2%	35.6%
Minerva 8B, maj1@k	25.4%	12.5%	28.4%	43.4%
PaLM 62B	4.4%	5.9%	33.0%	39.1%
Minerva 62B	27.6%	12.9%	52.4%	53.9%
Minerva 62B, maj1@k	43.4%	23.5%	68.5%	63.5%
PaLM 540B	8.8%	7.1%	56.5%	58.7%
Minerva 540B	33.6%	17.6%	58.8%	63.9%
Minerva 540B, maj1@k	**50.3%**	**30.8%**	**78.5%**	**75.0%**
OpenAI davinci-002	19.1%	14.8%	-	-
Published SOTA	6.9%[a]	-	74.4%[b]	54.9%[c]

A2-A4 Mistral AI, Mistral 7B TR, Table 2 (PDF p.4)

Model	Modality	MMLU	HellaSwag	WinoG	PIQA	Arc-e	Arc-c	NQ	TriviaQA	HumanEval	MBPP	MATH	GSM8K
LLaMA 2 7B	Pretrained	44.4%	77.1%	69.5%	77.9%	68.7%	43.2%	24.7%	63.8%	11.6%	26.1%	3.9%	16.0%
LLaMA 2 13B	Pretrained	55.6%	**80.7%**	72.9%	80.8%	75.2%	48.8%	**29.0%**	**69.6%**	18.9%	35.4%	6.0%	34.3%
Code-Llama 7B	Finetuned	36.9%	62.9%	62.3%	72.8%	59.4%	34.5%	11.0%	34.9%	**31.1%**	**52.5%**	5.2%	20.8%
Mistral 7B	Pretrained	**60.1%**	**81.3%**	**75.3%**	**83.0%**	**80.0%**	**55.5%**	**28.8%**	**69.9%**	**30.5%**	47.5%	**13.1%**	**52.2%**

Table 2: Comparison of Mistral 7B with Llama. Mistral 7B outperforms Llama 2 13B on all metrics, and

A2-A4 Mistral AI, Mixtral TR, Table 2 (PDF p.4): same models, same declared protocols

Model	Active Params	MMLU	HellaS	WinoG	PIQA	Arc-e	Arc-c	NQ	TriQA	HumanE	MBPP	Math	GSM8K
LLaMA 2 7B	7B	44.4%	77.1%	69.5%	77.9%	68.7%	43.2%	17.5%	56.6%	11.6%	26.1%	3.9%	16.0%
LLaMA 2 13B	13B	55.6%	80.7%	72.9%	80.8%	75.2%	48.8%	16.7%	64.0%	18.9%	35.4%	6.0%	34.3%
LLaMA 1 33B	33B	56.8%	83.7%	76.2%	82.2%	79.6%	54.4%	24.1%	68.5%	25.0%	40.9%	8.4%	44.1%
LLaMA 2 70B	70B	69.9%	**85.4%**	**80.4%**	82.6%	79.9%	56.5%	25.4%	**73.0%**	29.3%	49.8%	13.8%	69.6%
Mistral 7B	7B	62.5%	81.0%	74.2%	82.2%	80.5%	54.9%	23.2%	62.5%	26.2%	50.2%	12.7%	50.0%
Mixtral 8x7B	13B	**70.6%**	84.4%	77.2%	**83.6%**	**83.1%**	**59.7%**	**30.6%**	71.5%	**40.2%**	**60.7%**	**28.4%**	**74.4%**

Table 2: Comparison of Mixtral with Llama. Mixtral outperforms or matches Llama 2 70B performance on

B7 Google Gemini 1.0 TR, Table 2 (PDF p.8): MMLU and GSM8K rows

	Gemini Ultra	Gemini Pro	GPT-4	GPT-3.5	PaLM 2-L	Claude 2	Inflect-ion-2	Grok 1	LLAMA-2
MMLU Multiple-choice questions in 57 subjects (professional & academic) (Hendrycks et al., 2021a)	**90.04%** CoT@32*	79.13% CoT@8*	87.29% CoT@32 (via API**)	70% 5-shot	78.4% 5-shot	78.5% 5-shot CoT	79.6% 5-shot	73.0% 5-shot	68.0%***
	83.7% 5-shot	71.8% 5-shot	86.4% 5-shot (reported)						
GSM8K	**94.4%**	86.5%	92.0%	57.1%	80.0%	88.0%	81.4%	62.9%	56.8%

4. Concordant cross-reports

Pairs where two documents report the same number for the same model and benchmark. They are listed to show that cross-reporting is often exact, which makes the class A cases more notable.

Model	Benchmark	Value	Reported by

o1-preview	GPQA Diamond	73.3	OpenAI ('Learning to Reason with LLMs', Appendix A); OpenAI (simple-evals README, commit 6ead091)
Claude 3.5 Sonnet	MMLU	88.3	Anthropic (Claude 3.5 Sonnet model card addendum, Table 1); Meta (Llama 3 Herd TR, Table 2)
Claude 3.5 Sonnet (New)	MMLU-Pro	78.0	Anthropic (October 2024 addendum, Table 7); DeepSeek (DeepSeek-R1 TR v1, Table 4)
Claude 3.5 Sonnet (New)	GPQA Diamond	65.0	Anthropic (October 2024 addendum, Table 7); DeepSeek (DeepSeek-R1 TR v1, Table 4)
gpt-4o	GPQA (subset not stated)	49.9	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13); OpenAI (simple-evals README, commit 6ead091, 2025-07-09)

5. How much can these benchmarks resolve?

A benchmark with N independent items cannot distinguish arbitrarily small differences. The table gives, for each benchmark, the score change from one item and the standard error of a score at an illustrative accuracy p under a binomial model. The last column is the smallest difference between two independently scored models that a two-sided unpaired z-test at the 5% level resolves at that p. A paired test on the same items (McNemar) needs per-item results and is usually tighter, so this column is a conservative bound, not a property of any model. N is a verified claim.

Benchmark	N	1 item (pts)	p	SE (pts)	95% CI half-width (pts)	Unpaired min. difference (pts)
MMLU	14,042	0.01	0.9	0.25	0.5	0.7
MMLU-Pro	12,032	0.01	0.8	0.36	0.71	1.01
GSM8K	1,319	0.08	0.95	0.6	1.18	1.66
MATH (test)	5,000	0.02	0.9	0.42	0.83	1.18
HumanEval	164	0.61	0.9	2.34	4.59	6.49
GPQA Diamond	198	0.51	0.8	2.84	5.57	7.88
SWE-bench Verified	500	0.2	0.7	2.05	4.02	5.68
AIME (one year)	30	3.33	0.8	7.3	14.31	20.24

Two related facts verified in the sources: Anthropic's Claude 3.7 Sonnet score is computed on a 489-task subset of SWE-bench Verified and OpenAI's o3-mini system card on a 477-task subset, each excluding tasks that do not run on the vendor's infrastructure (claims `claude37-blog.claude-37-sonnet.swebench-verified.parallel`, `o3mini-card.o3-mini-tools.swebench-verified`). Headline SWE-bench Verified scores from different vendors are therefore not always computed on the same tasks.

6. Source registry

Source id	Title	Pinned address	Bytes	SHA-256
`aa_claude_opus55`	Artificial Analysis: Claude Opus 5.5 model page (capture 2026-10-18)	wayback	1,495,287	`fc04145c4f5db47d`
`aa_claude_sonnet55`	Artificial Analysis: Claude Sonnet 5.5 model page (capture 2026-10-16)	wayback	1,491,756	`50b45d698db279ac`
`anthropic_claude2_card`	Anthropic Claude 2 Model Card (July 2023)	url	847,078	`48cda59721bd7dbd`
`anthropic_claude35_addendum`	Anthropic Claude 3.5 Sonnet Model Card Addendum (June 2024)	url	406,339	`ac396c7f155dcbfb`
`anthropic_claude35_chart`	Anthropic Claude 3.5 Sonnet Launch Benchmark Chart	url	51,173	`173342703f6260a4`
`anthropic_claude35_oct_addendum`	Anthropic Claude 3.5 Sonnet (new) & Haiku Model Card Addendum (Oct 2024)	url	2,598,697	`5425dde49b649916`
`anthropic_claude37_blog`	Anthropic 'Claude 3.7 Sonnet and Claude Code' (Feb 24, 2025)	wayback	141,174	`2f37a20dc0c9a4eb`
`anthropic_claude37_system_card`	Anthropic Claude 3.7 Sonnet System Card (Feb 2025)	wayback	5,974,368	`82947a46fc2d1b9a`
`anthropic_claude3_card`	Anthropic Claude 3 Model Card (March 2024)	url	28,241,567	`a96214e5d495dec7`
`anthropic_claude4_blog`	Anthropic 'Introducing Claude 4' (May 22, 2025)	wayback	265,050	`5a42f1f06cab87e7`

anthropic_claude_haiku45_blog	Anthropic 'Introducing Claude Haiku 4.5' (Oct 15, 2025)	wayback	136,230	18c34f34931ee37d
anthropic_claude_haiku55_page	Anthropic Claude Haiku 5.5 page (capture 2026-10-07)	wayback	221,135	95fc91fee05e2bf5
anthropic_claude_opus41_blog	Anthropic 'Claude Opus 4.1' (Aug 5, 2025)	wayback	169,472	4714e78ce7acf6f7
anthropic_claude_opus45_blog	Anthropic 'Introducing Claude Opus 4.5' (Nov 24, 2025)	wayback	261,656	a425e50a8c0ed619
anthropic_claude_opus46_blog	Anthropic Introducing Claude Opus 4.6 (capture 2026-02-05)	wayback	311,977	1f3971e73f708289
anthropic_claude_opus46_system_card	Anthropic Claude Opus 4.6 System Card (Feb 2026)	wayback	13,208,951	4db67c6b6aea0a87
anthropic_claude_opus47_blog	Anthropic Introducing Claude Opus 4.7 (capture 2026-04-16)	wayback	337,925	681c3276e37aabec
anthropic_claude_opus47_system_card	Anthropic Claude Opus 4.7 System Card (Apr 2026)	wayback	14,231,870	a7729a0e5eb61dc6
anthropic_claude_opus55_page	Anthropic Claude Opus 5.5 page (capture 2026-10-19)	wayback	464,353	2b1955c50b158f99
anthropic_claude_sonnet45_blog	Anthropic 'Introducing Claude Sonnet 4.5' (Sep 29, 2025)	wayback	198,365	27c811e7a5c57dcc
anthropic_claude_sonnet46_blog	Anthropic Introducing Claude Sonnet 4.6 (capture 2026-02-17)	wayback	234,802	6d3ffddd8edfb8d7
anthropic_claude_sonnet46_system_card	Anthropic Claude Sonnet 4.6 System Card (Feb 2026)	wayback	8,503,340	672339ab327e8798
anthropic_claude_sonnet55_page	Anthropic Claude Sonnet 5.5 page (capture 2026-10-19)	wayback	285,547	94e03fb416ab71fd
anthropic_claude_sonnet5_blog	Anthropic Introducing Claude Sonnet 5 (capture 2026-06-30)	wayback	212,428	65f16cedeec319bf
anthropic_claude_sonnet5_system_card	Anthropic Claude Sonnet 5 System Card (Jun 2026)	wayback	10,786,341	33573adb9f1871b9
arxiv_2009.03300	MMLU (Hendrycks et al., ICLR 2021)	arxiv	2,074,939	d3a7b0f8589f5365
arxiv_2009.03300v1	MMLU (Hendrycks et al., ICLR 2021), first arXiv version	arxiv	1,855,079	5a4759cf69b9b4c3
arxiv_2103.03874	MATH Dataset (Hendrycks et al., 2021)	arxiv	2,030,715	a6f448173e57d808
arxiv_2103.03874v1	MATH Dataset (Hendrycks et al., 2021), first arXiv version	arxiv	2,366,153	639e4e54f93910b9
arxiv_2107.03374	Codex & HumanEval (Chen et al., 2021)	arxiv	1,885,177	ebae72ea0e8a5eb2
arxiv_2107.03374v1	Codex & HumanEval (Chen et al., 2021), first arXiv version	arxiv	1,877,355	7085950bcce34fee
arxiv_2110.14168	GSM8K & Verifiers (Cobbe et al., 2021)	arxiv	2,935,719	a52417d8fcd006de
arxiv_2112.11446	DeepMind Gopher 280B (Rae et al., 2021)	arxiv	3,412,401	b0b535d60cf933e9
arxiv_2112.11446v1	DeepMind Gopher 280B (Rae et al., 2021), first arXiv version	arxiv	3,403,850	a7aa3b0c47a8cfda
arxiv_2201.11903	Chain-of-Thought Prompting (Wei et al., 2022)	arxiv	891,773	7d9f878c23b460e4
arxiv_2201.11903v1	Chain-of-Thought Prompting (Wei et al., 2022), first arXiv version	arxiv	1,454,242	3d40493ec4bcba92
arxiv_2203.11171	Self-Consistency CoT (Wang et al., 2022)	arxiv	10,505,941	1a49ce0373afc89d
arxiv_2203.11171v1	Self-Consistency CoT (Wang et al., 2022), first arXiv version	arxiv	6,139,595	ed1ef6f3ff521a53
arxiv_2203.13474	CodeGen (Nijkamp et al., 2022; ICLR 2023)	arxiv	747,283	5901c27415d74500
arxiv_2203.15556	DeepMind Chinchilla 70B (Hoffmann et al., 2022)	arxiv	6,004,349	3fd3632a8ef48171
arxiv_2204.02311	Google PaLM 540B (Chowdhery et al., 2022)	arxiv	2,711,295	1e88f1f3027cd1ab
arxiv_2204.02311v1	Google PaLM 540B (Chowdhery et al., 2022), first arXiv version	arxiv	2,636,382	692b101e3d0abf37
arxiv_2204.06745	GPT-NeoX-20B (Black et al., 2022)	arxiv	3,528,614	6456e4b5113424fd
arxiv_2206.14858	Google Minerva 540B (Lewkowycz et al., 2022)	arxiv	1,255,498	b01ec8c8502ca3de
arxiv_2206.14858v1	Google Minerva 540B (Lewkowycz et al., 2022), first arXiv version	arxiv	1,255,814	ae50532a477d28c5

arxiv_2207.10397	CodeT (Chen et al., 2022)	arxiv	1,394,376	ada1c4b233dda1b2
arxiv_2207.10397v1	CodeT (Chen et al., 2022), first arXiv version	arxiv	714,120	fdb68a1bb43206a4
arxiv_2210.02414	GLM-130B (Zeng et al., 2022)	arxiv	4,743,570	82d5b9634b3d2601
arxiv_2210.02414v1	GLM-130B (Zeng et al., 2022), first arXiv version	arxiv	4,441,730	f6a943d7febff5cf
arxiv_2210.11416	Google Flan-PaLM 540B (Chung et al., 2022)	arxiv	1,557,309	771f758c1b711c2a
arxiv_2210.11416v1	Google Flan-PaLM 540B (Chung et al., 2022), first arXiv version	arxiv	1,170,532	4a14798b6b85f112
arxiv_2211.05100	BigScience BLOOM 176B (Le Scao et al., 2022)	arxiv	3,029,243	8033c3e611aaa120
arxiv_2211.05100v1	BigScience BLOOM 176B (Le Scao et al., 2022), first arXiv version	arxiv	3,017,040	3ef17485d9f1db85
arxiv_2211.09085	Meta Galactica 120B (Taylor et al., 2022)	arxiv	3,664,672	612939ed35e76201
arxiv_2211.09110	Stanford HELM (Liang et al., 2022; TMLR 2023)	arxiv	6,045,688	30970bda8958a1a8
arxiv_2211.09110v1	Stanford HELM (Liang et al., 2022; TMLR 2023), first arXiv version	arxiv	6,099,449	eb365cc3cbb46cfb
arxiv_2212.12017	OPT-IML (Iyer et al., 2022)	arxiv	1,077,867	3e41ff1c8f04280f
arxiv_2212.12017v1	OPT-IML (Iyer et al., 2022), first arXiv version	arxiv	950,744	1579f240be68c2f0
arxiv_2301.03988	SantaCoder (Allal et al., 2023)	arxiv	1,814,883	d292c0e85bdd8194
arxiv_2301.13688	The Flan Collection (Longpre et al., 2023)	arxiv	667,714	1dedd2a2a9602490
arxiv_2302.13971	Meta LLaMA-1 65B (Touvron et al., 2023)	arxiv	726,566	2e663675ae36ad12
arxiv_2303.08774	OpenAI GPT-4 Technical Report (2023)	arxiv	5,245,564	c33a66dadca2388d
arxiv_2303.08774v1	OpenAI GPT-4 Technical Report (2023), first arXiv version	arxiv	5,229,731	053056a10114d22e
arxiv_2304.12244v1	WizardLM (Xu et al., 2023), v3, first arXiv version	arxiv	1,455,236	dadcdc3859056383
arxiv_2304.12244v3	WizardLM (Xu et al., 2023), v3	arxiv	1,199,147	e5fc7a244d34703b
arxiv_2305.01210	EvalPlus / HumanEval+ (Liu et al., 2023)	arxiv	970,804	52d35d25a10c38da
arxiv_2305.01210v1	EvalPlus / HumanEval+ (Liu et al., 2023), first arXiv version	arxiv	876,335	11a90271658c962b
arxiv_2305.10403	Google PaLM 2 Technical Report (2023)	arxiv	5,412,215	485141e0b2073dc1
arxiv_2305.10403v1	Google PaLM 2 Technical Report (2023), first arXiv version	arxiv	5,335,273	c1f529392e6952f9
arxiv_2305.17306	Chain-of-Thought Hub (Fu et al., 2023)	arxiv	425,870	884d44bc625cabee
arxiv_2306.04757	INSTRUCTEVAL (Chia et al., 2023)	arxiv	773,091	eb4273734c2d491c
arxiv_2306.08568	WizardCoder (Luo et al., 2023)	arxiv	474,693	8e4e2486c0f6a7ab
arxiv_2307.09288	Meta LLaMA-2 70B (Touvron et al., 2023)	arxiv	13,661,300	1df284ce95f78300
arxiv_2307.09288v1	Meta LLaMA-2 70B (Touvron et al., 2023), first arXiv version	arxiv	13,658,713	03c022db7d7ff846
arxiv_2308.12950	Meta Code Llama (Roziere et al., 2023)	arxiv	1,623,794	05ed05d2d76c420b
arxiv_2308.12950v1	Meta Code Llama (Roziere et al., 2023), first arXiv version	arxiv	1,544,856	fcb2cb6f093ff26b
arxiv_2309.16609	Qwen Technical Report (Bai et al., 2023)	arxiv	1,994,623	5d4a9c3f6da9e9fa
arxiv_2310.06770	SWE-bench (Jimenez et al., 2023)	arxiv	4,533,722	f7e8e1df64129742
arxiv_2310.06825	Mistral 7B (Jiang et al., 2023)	arxiv	3,749,788	dfbac4e7035344b3
arxiv_2310.10631	Llemma (Azerbayev et al., 2023)	arxiv	617,447	718ce22f51039ee9

arxiv_2311.12022	GPQA: Graduate-Level Google-Proof Q&A (Rein et al., 2023)	arxiv	617,128	58a85a21b653a01d
arxiv_2311.16867	The Falcon Series of Open Language Models (Almazrouei et al., 2023)	arxiv	1,248,651	fcdc2b0830717ff3
arxiv_2312.11805	Google Gemini 1.0 Report (2023)	arxiv	27,484,659	8ca95b27851b09c3
arxiv_2312.11805v1	Google Gemini 1.0 Report (2023), first arXiv version	arxiv	26,940,257	f3c5b175d26159b9
arxiv_2401.02954	DeepSeek LLM: Scaling Open-Source Language Models with Longtermism (2024)	arxiv	7,474,751	9be2b427133468d4
arxiv_2401.04088	Mixtral of Experts (Jiang et al., 2024)	arxiv	2,475,990	f8bbf0e9d979b7a8
arxiv_2401.14196	DeepSeek-Coder (Guo et al., 2024)	arxiv	1,909,074	366ce88bc7d8cc77
arxiv_2402.03300	DeepSeekMath (Shao et al., 2024)	arxiv	1,843,099	500ce2fae51dd4e0
arxiv_2403.05530	Google Gemini 1.5 Report (2024)	arxiv	7,184,542	abcd3b8e4b54721f
arxiv_2403.05530v1	Google Gemini 1.5 Report, v1 (Mar 2024)	arxiv	7,228,817	526d13d0b188c0d0
arxiv_2404.14219	Phi-3 Technical Report, v1 (Apr 2024)	arxiv	2,049,157	54fe7b2ddc32ff3d
arxiv_2404.14219v1	Phi-3 Technical Report, v1 (Apr 2024), first arXiv version	arxiv	2,049,157	54fe7b2ddc32ff3d
arxiv_2404.14219v4	Phi-3 Technical Report, v4 (Aug 2024, adds Phi-3.5)	arxiv	3,908,493	16c0bec322ce5c3b
arxiv_2405.00332	GSM1k: Careful Examination of LLM Performance on Grade School Arithmetic (Zhang et al., 2024)	arxiv	7,180,592	c8d45dd92db7fd76
arxiv_2405.04434	DeepSeek-V2 Technical Report (2024)	arxiv	1,586,880	f149f9c09aa8b46c
arxiv_2405.04434v1	DeepSeek-V2 Technical Report (2024), first arXiv version	arxiv	1,576,608	e930bb96dd1f0ecb
arxiv_2406.01574	MMLU-Pro (Wang et al., 2024)	arxiv	959,520	e8f695db151fc882
arxiv_2406.04127	MMLU-Redux: Are We Done with MMLU? (Gema et al., 2024)	arxiv	747,909	a45720c5df8c7e7f
arxiv_2406.11931	DeepSeek-Coder-V2 (2024)	arxiv	396,316	de4e77c86e6320ff
arxiv_2407.10671	Alibaba Qwen2 Technical Report (2024)	arxiv	493,144	32d7276673eec396
arxiv_2407.10671v1	Alibaba Qwen2 Technical Report (2024), first arXiv version	arxiv	463,171	adbdabac242285ba
arxiv_2407.21783	Meta Llama 3 Herd of Models (2024)	arxiv	9,833,173	481f1599468f95a0
arxiv_2407.21783v1	Meta Llama 3 Herd of Models (2024), first arXiv version	arxiv	9,832,414	d7df3d6f537f0250
arxiv_2411.14405	Marco-o1 (2024)	arxiv	5,900,146	85fa713f8d0efa52
arxiv_2412.08905	Phi-4 Technical Report (2024)	arxiv	1,016,588	be5577cfb815a861
arxiv_2412.15115	Qwen2.5 Technical Report (Dec 2024)	arxiv	1,935,303	367bbeef22d21b6a
arxiv_2412.19437	DeepSeek-V3 Technical Report (2024)	arxiv	1,887,366	812a3fd645c80725
arxiv_2412.19437v1	DeepSeek-V3 Technical Report (2024), first arXiv version	arxiv	1,887,607	a67e18a4ca6e2c81
arxiv_2501.12599	Kimi k1.5: Scaling Reinforcement Learning with LLMs (v1)	arxiv	949,553	cec23018118ba8e9
arxiv_2501.12948	DeepSeek-R1 Technical Report, v2 (revised)	arxiv	5,034,998	b191b0a365a64b4a
arxiv_2501.12948v1	DeepSeek-R1 Technical Report, v1 (Jan 2025)	arxiv	1,312,189	52d8ca3ac93e88ce
arxiv_2505.09388	Alibaba Qwen3 Technical Report (2025)	arxiv	779,424	84a5e2b1fa04bb77
arxiv_2506.10910	Mistral Magistral Report (2025)	arxiv	6,095,316	e4c6c5b21745cb16
arxiv_2506.13585	MiniMax-M1 (v1)	arxiv	1,584,812	d355a6a41a26b85c
arxiv_2507.06261	Gemini 2.5 Technical Report (v1)	arxiv	10,486,108	3c7f2ea70210804f

arxiv_2507.20534	Moonshot Kimi K2 Report (2025)	arxiv	5,872,335	035062b5e1ab865f
arxiv_2507.20534v1	Moonshot Kimi K2 Report (2025), first arXiv version	arxiv	6,335,018	b838917cf0a475d2
arxiv_2508.06471	Zhipu GLM-4.5 Technical Report (2025)	arxiv	4,067,812	b5eec92371521f58
arxiv_2508.10925	OpenAI gpt-oss-120b & gpt-oss-20b Model Card (2025)	arxiv	3,049,591	8839e1efdf835be0
arxiv_2512.02556	DeepSeek-V3.2 Technical Report (2025)	arxiv	980,616	2bec0671778769c1
arxiv_2602.02276	Moonshot Kimi K2.5 Report (2026)	arxiv	8,484,545	889025c0fcbacb8b
arxiv_2602.02276v1	Moonshot Kimi K2.5 Report (2026), first arXiv version	arxiv	9,084,004	06811e2d0e7a68cf
arxiv_2602.15763	Zhipu GLM-5 Technical Report (2026)	arxiv	6,177,650	e20742ff36e08dc3
arxiv_2602.15763v1	Zhipu GLM-5 Technical Report (2026), first arXiv version	arxiv	6,177,891	4a97eab759d4576b
deepseek_r1_lite_news	DeepSeek-R1-Lite Release (Nov 20, 2024)	wayback	19,625	701c5616bb24a0b1
google_gemini1_blog	Google 'Introducing Gemini' (Dec 6, 2023)	wayback	216,484	dac9440b220bb039
google_gemini25_blog	Google 'Gemini 2.5: Our most intelligent AI model' (Mar 25, 2025)	wayback	378,420	6171d361d5c37f34
google_gemini2_blog	Google Introducing Gemini 2.0: our new AI model for the agentic era (Dec 11, 2024)	wayback	223,730	6eec2ceb5ecbf7bc
google_gemini2_devblog	Google for Developers: The next chapter of the Gemini era for developers (Dec 11, 2024)	wayback	82,313	79ba34e1c0cbfaa6
google_gemini31pro_model_card	Google DeepMind Gemini 3.1 Pro Model Card (capture 2026-02-19)	wayback	241,414	34c1e158682960cc
google_gemini35flash_model_card	Google DeepMind Gemini 3.5 Flash Model Card (capture 2026-05-19)	wayback	148,789	68c8162745c777df
google_gemini3_blog	Google 'A new era of intelligence with Gemini 3' (Nov 18, 2025)	wayback	371,526	941931e4115f21de
google_gemini3_flash_blog	Google 'Gemini 3 Flash' (Dec 17, 2025)	wayback	372,037	fccc7dd673a4d0c7
google_gemini4argon_model_card	Google DeepMind Gemini 4 Argon Model Card (capture 2026-10-13)	wayback	166,304	f843dd130b5fbc35
helm_classic_v0.3.0_code_humaneval	Stanford HELM classic v0.3.0 results: HumanEval	helm	8,182	3d049121b54479f0
helm_classic_v0.3.0_gsm	Stanford HELM classic v0.3.0 results: GSM8K	helm	111,409	b5ba779c6e3c99ba
helm_classic_v0.3.0_mmlu	Stanford HELM classic v0.3.0 results: MMLU (5-subject subset)	helm	894,405	5d6ddb24c7f6d59c
hf_cais_mmlu_size	Hugging Face datasets-server: cais/mmlu split sizes	url	44,120	405af384545acb4b
hf_deepseek_r1_0528_jun	Hugging Face model card deepseek-ai/DeepSeek-R1-0528 (capture 2025-06-08)	wayback	196,397	0d763c29e6e61c60
hf_deepseek_v31	Hugging Face model card deepseek-ai/DeepSeek-V3.1 (capture 2025-08-25)	wayback	180,479	d61cf90c914af6a7
hf_deepseek_v32_exp	Hugging Face model card deepseek-ai/DeepSeek-V3.2-Exp	wayback	132,462	f88ee7cc253df3dd
hf_deepseek_v3_0324_apr	Hugging Face model card deepseek-ai/DeepSeek-V3-0324 (capture 2025-04-11)	wayback	194,859	cbcb55a1b98aca73
hf_deepseek_v4_pro_model_card	Hugging Face deepseek-ai/DeepSeek-V4-Pro model card (capture 2026-04-24)	wayback	145,815	fc4b241836c8aefe
hf_glm52_model_card	Hugging Face zai-org/GLM-5.2 model card (capture 2026-06-16)	wayback	225,396	71aea2a7d47f243d
hf_kimi_k2_thinking	Hugging Face model card moonshotai/Kimi-K2-Thinking (capture 2025-11-06)	wayback	140,146	8450d035829bd0c1
hf_openai_gsm8k_size	Hugging Face datasets-server: openai/gsm8k split sizes	url	1,363	a9bf445fa38f4477
hf_qwen35_397b_model_card	Hugging Face Qwen/Qwen3.5-397B-A17B model card (capture 2026-02-16)	wayback	326,269	c91ab4a6aa8f5842
hf_qwen38_2p4t_model_card	Hugging Face Qwen/Qwen3.8-2.4T-A95B model card (capture 2026-08-12)	wayback	294,102	e9cd5a9b65a7663a
hf_qwen3_235b_thinking_2507	Hugging Face model card Qwen/Qwen3-235B-A22B-Thinking-2507 (capture 2025-07-25)	wayback	144,443	cfeb831c72143bdb
hf_qwen_72b_card	Hugging Face model card Qwen/Qwen-72B (Nov 30, 2023)	wayback	121,494	245715eb24632e94

meta_llama33_model_card_c908f5b	Meta Llama 3.3 MODEL_CARD.md (meta-llama/llama-models commit c908f5b, 2024-12-06; Markdown with HTML tables, parsed as html)	github	17,360	4c864d7d77c14a03
meta_llama3_model_card_92a325e	Meta Llama 3 MODEL_CARD.md (meta-llama/llama3 commit 92a325e, 2024-04-18; Markdown with HTML tables, parsed as html)	github	22,373	d09edb5826e3ca75
meta_llama4_blog	Meta 'The Llama 4 herd' (Apr 5, 2025)	wayback	191,239	16e3cef2acb82d7c
meta_llama4_model_card_038acb9	Meta Llama 4 MODEL_CARD.md (llama-models commit 038acb9, 2025-04-05; Markdown with HTML tables, staged as html for tag stripping)	github	23,764	a1739083c8a496ca
mistral_large2_blog	Mistral AI Large Enough (Mistral Large 2, Jul 24, 2024)	wayback	21,111	ebbbeb926375540e
mistral_large_blog	Mistral AI Au Large (Mistral Large, Feb 26, 2024)	wayback	19,655	d4ce448c84ac0679
mistral_mixtral8x22b_blog	Mistral AI Cheaper, Better, Faster, Stronger (Mixtral 8x22B, Apr 17, 2024)	wayback	16,757	0524a8a327caf9ee
moonshot_kimi_k25_blog	Moonshot AI Kimi K2.5 launch blog (capture 2026-01-27)	wayback	128,787	547d444afdf40ef2
openai_gpt41_blog	OpenAI 'Introducing GPT-4.1 in the API' (Apr 14, 2025)	wayback	549,656	498949e05173b8c1
openai_gpt45_blog	OpenAI 'Introducing GPT-4.5' (Feb 27, 2025)	wayback	414,524	5fab0848d5cb8ce8
openai_gpt45_system_card	OpenAI GPT-4.5 System Card (Feb 27, 2025)	wayback	4,094,103	5bb6cbb4812b5c0c
openai_gpt4o_blog	OpenAI 'Hello GPT-4o' (May 13, 2024)	wayback	215,907	6df4ea9e597589e1
openai_gpt51_blog	OpenAI 'GPT-5.1' (Nov 12, 2025)	wayback	441,027	3effefb4d19c4035
openai_gpt52_blog	OpenAI 'Introducing GPT-5.2' (Dec 11, 2025)	wayback	632,670	b863850d9038fc39
openai_gpt53codex_blog	OpenAI Introducing GPT-5.3-Codex (capture 2026-02-05)	wayback	408,052	d1d61c6ee66503ad
openai_gpt54_blog	OpenAI Introducing GPT-5.4 (capture 2026-03-05)	wayback	671,852	bffd5b5e196fb2ea
openai_gpt55_blog	OpenAI Introducing GPT-5.5 (capture 2026-04-23)	wayback	669,652	982c290e29e418a4
openai_gpt55_instant_blog	OpenAI GPT-5.5 Instant (capture 2026-05-05)	wayback	493,411	112f2aef711c5e9d
openai_gpt56_blog	OpenAI GPT-5.6 (capture 2026-07-09)	wayback	1,823,896	9e3e0e3b8a466682
openai_gpt5_blog	OpenAI 'Introducing GPT-5' (Aug 7, 2025)	wayback	781,432	204b5f8706ba6356
openai_gpt6_astra_blog	OpenAI GPT-6 Astra (capture 2026-09-03)	wayback	2,828,533	b166d23f2e6ea50a
openai_o1_blog	OpenAI 'Learning to Reason with LLMs' (Sep 12, 2024)	wayback	1,242,106	7899910fe8d601f4
openai_o1mini_blog	OpenAI o1-mini: advancing cost-efficient reasoning (Sep 12, 2024)	wayback	700,506	6dab2b0043ce88f4
openai_o3_o4mini_blog	OpenAI 'Introducing OpenAI o3 and o4-mini' (Apr 16, 2025)	wayback	1,112,240	f5ebf0a27492009f
openai_o3mini_blog	OpenAI 'OpenAI o3-mini' (Jan 31, 2025)	wayback	573,406	0b09043d1846f962
openai_o3mini_swe_chart_png	OpenAI o3-mini launch post: SWE-bench Verified chart image	wayback	87,205	1d9b8bcb20217911
openai_o3mini_system_card	OpenAI o3-mini System Card (Feb 10, 2025)	wayback	4,128,205	6e0fa5c2590378f3
openai_simple_evals_readme	OpenAI simple-evals README (commit 6ead091, 2025-07-09)	github	13,160	d9647c4a0e34cd9b
openai_simple_evals_readme_2024-05-13	OpenAI simple-evals README (commit 294cb1f, 2024-05-13, GPT-4o launch day)	github	8,303	d900dda848d0da27
openai_swebench_verified_blog	OpenAI Introducing SWE-bench Verified (Aug 13, 2024)	wayback	319,140	904c000d230a05f6
qwen_qwen25max_blog	Qwen 'Qwen2.5-Max: Exploring the Intelligence of Large-scale MoE Model' (Jan 28, 2025)	wayback	22,544	a96efa4b73a0055a
qwen_qwq32b_blog	Qwen 'QwQ-32B: Embracing the Power of Reinforcement Learning' (Mar 5, 2025)	wayback	35,843	de73ca1fd910bdca
qwen_qwq32b_preview_blog	Qwen QwQ: Reflect Deeply on the Boundaries of the Unknown (QwQ-32B-Preview, Nov 28, 2024)	wayback	60,975	37d5c7d0949ad72e
xai_grok15_blog	xAI Announcing Grok-1.5 (Mar 28, 2024)	wayback	22,502	13ade540c65b17a5
xai_grok1_blog	xAI Announcing Grok (Grok-1 benchmark table)	wayback	28,729	5a005c7d3509cb9b

xai_grok2_blog	xAI Grok-2 Beta Release (Aug 13, 2024)	wayback	50,429	cbedaf529de8a3bd
xai_grok3_blog	xAI 'Grok 3 Beta: The Age of Reasoning Agents' (Feb 19, 2025)	wayback	95,121	5da9330b526eacc2
xai_grok45_blog	xAI Grok 4.5 announcement (capture 2026-07-08)	wayback	216,469	dd273cc1df700c85
xai_grok4_blog	xAI 'Grok 4' (Jul 9, 2025)	wayback	99,003	fbacb2b295dad50d
zai_glm47_blog	Z.ai 'GLM-4.7' blog (Dec 22, 2025)	wayback	237,871	7c6a2e4968a9d45c

Month-by-Month Dossier (2021-01 to 2026-10)

Generated from `data/monthly_benchmarks_2021_2026.json`. Each score links to its pinned source and cites the ledger claim that verifies it. A score's month is the model's release month as stated by a source or, if no source states it, the month of the reporting document.

2021-01

January 2021. 4 verified scores for 3 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
GPT-3 X-Large (few-shot) [davinci, 175B]	MMLU	few-shot, up to 5 demonstration examples (5 fixed dev examples per subject)	43.9	Hendrycks et al. (MMLU paper), Table 1	arxiv_2009.03300 p.5	m2021.hendrycks-mmlu.gpt-3-x-large.mmlu.few-shot-5
UnifiedQA (11B, T5 backbone)	MMLU	no further tuning on MMLU (transfer from QA fine-tuning); not few-shot	48.9	Hendrycks et al. (MMLU paper), Table 1	arxiv_2009.03300 p.5	m2021.hendrycks-mmlu.unifiedqa-11b.mmlu.transfer-no-further-tuning
davinci (175B)	GSM8K	EM, 5 in-context examples (# train = 5), mean over 3 trials	9.0	Stanford CRFM HELM classic v0.3.0	helm_classic_v0.3.0_gsm json	m2021.helm-v0.3.0.davinci-175b.gsm8k.em-5-in-context
davinci (175B)	MMLU (HELM 5-subject subset)	EM, 5 in-context examples (# train = 5), mean over trials; MMLU = 5 subjects only	42.2	Stanford CRFM HELM classic v0.3.0	helm_classic_v0.3.0_mmlu json	m2021.helm-v0.3.0.davinci-175b.mmlu.em-5-in-context

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	48.9	UnifiedQA (11B, T5 backbone)	no further tuning on MMLU (transfer from QA fine-tuning); not few-shot	2021-01	m2021.hendrycks-mmlu.unifiedqa-11b.mmlu.transfer-no-further-tuning

2021-02

February 2021. No verified score is dated to this month.

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	48.9	UnifiedQA (11B, T5 backbone)	no further tuning on MMLU (transfer from QA fine-tuning); not few-shot	2021-01	m2021.hendrycks-mmlu.unifiedqa-11b.mmlu.transfer-no-further-tuning

2021-03

March 2021. 5 verified scores for 4 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
GPT-2 1.5B (AMPS-pretrained, MATH fine-tuned)	MATH	pretrained on AMPS, then fine-tuned on MATH	6.9	Hendrycks et al. (MATH paper), Table 2	arxiv_2103.03874 p.5	m2021.hendrycks-math.gpt-2-1.5b.math.amps-pretrain-math-ft

GPT-3 175B* (few-shot)	MATH	few-shot, 8 problems with final answers prepended, temperature 0; no AMPS pretraining	5.2	Hendrycks et al. (MATH paper), Table 2	arxiv_2103.03874 p.5	m2021.hendrycks-math.gpt-3-175b.math.few-shot-8
GPT-Neo 1.3B	HumanEval	pass@1 (0-shot docstring prompt, unbiased estimator over sampled completions, temperature chosen per k)	4.79	Chen et al. (Codex paper), Table 1	arxiv_2107.03374 p.7	m2021.chen-codex.gpt-neo-1.3b.humaneval.pass-at-1
GPT-Neo 2.7B	HumanEval	pass@1 (0-shot docstring prompt, unbiased estimator over sampled completions, temperature chosen per k)	6.41	Chen et al. (Codex paper), Table 1	arxiv_2107.03374 p.7	m2021.chen-codex.gpt-neo-2.7b.humaneval.pass-at-1
GPT-Neo 2.7B	HumanEval	pass@100 (0-shot docstring prompt, unbiased estimator over sampled completions, temperature chosen per k)	21.37	Chen et al. (Codex paper), Table 1	arxiv_2107.03374 p.7	m2021.chen-codex.gpt-neo-2.7b.humaneval.pass-at-100

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	48.9	UnifiedQA (11B, T5 backbone)	no further tuning on MMLU (transfer from QA fine-tuning); not few-shot	2021-01	m2021.hendrycks-mmlu.unifiedqa-11b.mmlu.transfer-no-further-tuning

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Measuring Mathematical Problem Solving With the MATH Dataset, v2, PDF page 5: Table 2: MATH accuracies across subjects.

2021-04

April 2021. No verified score is dated to this month.

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	48.9	UnifiedQA (11B, T5 backbone)	no further tuning on MMLU (transfer from QA fine-tuning); not few-shot	2021-01	m2021.hendrycks-mmlu.unifiedqa-11b.mmlu.transfer-no-further-tuning

2021-05

May 2021. No verified score is dated to this month.

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	48.9	UnifiedQA (11B, T5 backbone)	no further tuning on MMLU (transfer from QA fine-tuning); not few-shot	2021-01	m2021.hendrycks-mmlu.unifiedqa-11b.mmlu.transfer-no-further-tuning

2021-06

June 2021. 3 verified scores for 2 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
GPT-J (6B)	MMLU (HELM 5-subject subset)	EM, 5 in-context examples (# train = 5), mean over trials; MMLU = 5 subjects only	24.9	Stanford CRFM HELM classic v0.3.0	helm_classic_v0.3.0_mmlu json	m2021.helm-v0.3.0.gpt-j-6b.mmlu.em-5-in-context
GPT-J 6B	HumanEval	pass@1 (0-shot docstring prompt, unbiased estimator over sampled completions, temperature chosen per k)	11.62	Chen et al. (Codex paper), Table 1	arxiv_2107.03374 p.7	m2021.chen-codex.gpt-j-6b.humaneval.pass-at-1
GPT-J 6B	HumanEval	pass@100 (0-shot docstring prompt, unbiased estimator over sampled completions, temperature chosen per k)	27.74	Chen et al. (Codex paper), Table 1	arxiv_2107.03374 p.7	m2021.chen-codex.gpt-j-6b.humaneval.pass-at-100

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	48.9	UnifiedQA (11B, T5 backbone)	no further tuning on MMLU (transfer from QA fine-tuning); not few-shot	2021-01	`m2021.hendrycks-mmlu.unifiedqa-11b.mmlu.transfer-no-further-tuning`

2021-07

July 2021. 5 verified scores for 3 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Codex-12B	HumanEval	pass@1 (0-shot docstring prompt, unbiased estimator over sampled completions, temperature chosen per k)	28.81	Chen et al. (Codex paper), Table 1	arxiv_2107.03374 p.7	`m2021.chen-codex.codex-12b.humaneval.pass-at-1`
Codex-12B	HumanEval	pass@100 (0-shot docstring prompt, unbiased estimator over sampled completions, temperature chosen per k)	72.31	Chen et al. (Codex paper), Table 1	arxiv_2107.03374 p.7	`m2021.chen-codex.codex-12b.humaneval.pass-at-100`
Codex-S	HumanEval	single sample (pass@1)	37.7	Chen et al. (Codex paper), Section 1 / Figure 1	arxiv_2107.03374 p.2	`m2021.chen-codex.codex-s-12b.humaneval.pass-at-1`
Codex-S	HumanEval	at least one correct in 100 samples (pass@100-style oracle)	77.5	Chen et al. (Codex paper), Section 1	arxiv_2107.03374 p.2	`m2021.chen-codex.codex-s.humaneval.oracle-100-samples`
GPT-3 (size not stated)	HumanEval	single sample, abstract	0.0	Chen et al. (Codex paper), abstract	arxiv_2107.03374 p.1	`m2021.chen-codex.gpt-3.humaneval.abstract`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	48.9	UnifiedQA (11B, T5 backbone)	no further tuning on MMLU (transfer from QA fine-tuning); not few-shot	2021-01	`m2021.hendrycks-mmlu.unifiedqa-11b.mmlu.transfer-no-further-tuning`
HumanEval	37.7	Codex-S	single sample (pass@1)	2021-07	`m2021.chen-codex.codex-s-12b.humaneval.pass-at-1`

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Evaluating Large Language Models Trained on Code, v2, PDF page 7: Table 1: Codex, GPT-Neo, & TabNine evaluations for HumanEval.

2021-08

August 2021. 2 verified scores for 1 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
J1-Jumbo v1 (178B)	GSM8K	EM, 5 in-context examples (# train = 5), mean over 3 trials	5.4	Stanford CRFM HELM classic v0.3.0	helm_classic_v0.3.0_gsm json	`m2021.helm-v0.3.0.j1-jumbo-v1-178b.gsm8k.em-5-in-context`
J1-Jumbo v1 (178B)	MMLU (HELM 5-subject subset)	EM, 5 in-context examples (# train = 5), mean over trials; MMLU = 5 subjects only	25.9	Stanford CRFM HELM classic v0.3.0	helm_classic_v0.3.0_mmlu json	`m2021.helm-v0.3.0.j1-jumbo-v1-178b.mmlu.em-5-in-context`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	48.9	UnifiedQA (11B, T5 backbone)	no further tuning on MMLU (transfer from QA fine-tuning); not few-shot	2021-01	`m2021.hendrycks-mmlu.unifiedqa-11b.mmlu.transfer-no-further-tuning`
HumanEval	37.7	Codex-S	single sample (pass@1)	2021-07	`m2021.chen-codex.codex-s-12b.humaneval.pass-at-1`

2021-09

September 2021. No verified score is dated to this month.

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	48.9	UnifiedQA (11B, T5 backbone)	no further tuning on MMLU (transfer from QA fine-tuning); not few-shot	2021-01	`m2021.hendrycks-mmlu.unifiedqa-11b.mmlu.transfer-no-further-tuning`
HumanEval	37.7	Codex-S	single sample (pass@1)	2021-07	`m2021.chen-codex.codex-s-12b.humaneval.pass-at-1`

2021-10

October 2021. 4 verified scores for 3 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
GPT-3 175B finetuned plus an additional 175B verifier	GSM8K	fine-tuned generator + 175B verifier reranking	55.0	Wang et al. (Self-Consistency, arXiv 2203.11171v4), Table 2	arxiv_2203.11171 p.5	`m2021.wang-selfconsistency.gpt-3-175b-ft-verifier.gsm8k.verifier`
GPT-3 175B finetuned with 7.5k examples	GSM8K	fine-tuned on the 7.5k GSM8K training set (single sample)	35.0	Wang et al. (Self-Consistency, arXiv 2203.11171v4), Table 2	arxiv_2203.11171 p.5	`m2021.wang-selfconsistency.gpt-3-175b-ft.gsm8k.finetuned-7.5k`
TNLG v2 (530B)	GSM8K	EM, 5 in-context examples (# train = 5), mean over 3 trials	14.6	Stanford CRFM HELM classic v0.3.0	helm_classic_v0.3.0_gsm json	`m2021.helm-v0.3.0.tnlg-v2-530b.gsm8k.em-5-in-context`
TNLG v2 (530B)	MMLU (HELM 5-subject subset)	EM, 5 in-context examples (# train = 5), mean over trials; MMLU = 5 subjects only	46.9	Stanford CRFM HELM classic v0.3.0	helm_classic_v0.3.0_mmlu json	`m2021.helm-v0.3.0.tnlg-v2-530b.mmlu.em-5-in-context`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	48.9	UnifiedQA (11B, T5 backbone)	no further tuning on MMLU (transfer from QA fine-tuning); not few-shot	2021-01	`m2021.hendrycks-mmlu.unifiedqa-11b.mmlu.transfer-no-further-tuning`
HumanEval	37.7	Codex-S	single sample (pass@1)	2021-07	`m2021.chen-codex.codex-s-12b.humaneval.pass-at-1`

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Training Verifiers to Solve Math Word Problems, v2, PDF page 8: Figure 5: A comparison between finetuning and verification using 6B and 175B.

2021-11

November 2021. No verified score is dated to this month.

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	48.9	UnifiedQA (11B, T5 backbone)	no further tuning on MMLU (transfer from QA fine-tuning); not few-shot	2021-01	`m2021.hendrycks-mmlu.unifiedqa-11b.mmlu.transfer-no-further-tuning`
HumanEval	37.7	Codex-S	single sample (pass@1)	2021-07	`m2021.chen-codex.codex-s-12b.humaneval.pass-at-1`
MATH	6.9	GPT-2 1.5B (AMPS-pretrained, MATH fine-tuned)	pretrained on AMPS, then fine-tuned on MATH	2021-11	`m2021.hendrycks-math.gpt-2-1.5b.math.amps-pretrain-math-ft`

2021-12

December 2021. 2 verified scores for 2 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim

Gopher (280B)	MMLU	5-shot	60.0	Rae et al. (Gopher), Table 5	arxiv_2112.11446 p.11	`m2021.gopher.gopher-280b.mmlu.5-shot`
Gopher (5-shot) 280B	MMLU mathematics subset (5 subjects)	5-shot	30.6	Taylor et al. (Galactica), Table 8	arxiv_2211.09085 p.15	`m2021.galactica.gopher-280b.mmlu-math-subset.5-shot`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	60.0	Gopher (280B)	5-shot	2021-12	`m2021.gopher.gopher-280b.mmlu.5-shot`
HumanEval	37.7	Codex-S	single sample (pass@1)	2021-07	`m2021.chen-codex.codex-s-12b.humaneval.pass-at-1`
MATH	6.9	GPT-2 1.5B (AMPS-pretrained, MATH fine-tuned)	pretrained on AMPS, then fine-tuned on MATH	2021-11	`m2021.hendrycks-math.gpt-2-1.5b.math.amps-pretrain-math-ft`

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Scaling Language Models: Methods, Analysis & Insights from Training Gopher, v2, PDF page 11: Table 5: Massive Multitask Language Understanding (MMLU).

2022-01

January 2022. 15 verified scores for 2 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
code-davinci-002	GSM8K	HELM v0.3.0 EM, 5 in-context examples	56.8	Stanford CRFM, HELM classic v0.3.0 results JSON	helm_classic_v0.3.0_gsm json	`m2022.helm.code-davinci-002.gsm8k`
code-davinci-002	GSM8K	HELM v0.1.0 (paper text)	52.1	Liang et al. (Stanford CRFM), HELM arXiv:2211.09110v2 Sec 1.1	arxiv_2211.09110 p.8	`m2022.helmpaper.code-davinci-002.gsm8k`
code-davinci-002	GSM8K	8-shot chain-of-thought, greedy	63.1	Wei et al. (Google), arXiv:2201.11903v6 Table 1	arxiv_2201.11903 p.20	`m2022.wei.code-davinci-002.gsm8k.8shot-cot`
code-davinci-002	GSM8K	8-shot standard prompting, greedy	19.7	Wei et al. (Google), arXiv:2201.11903v6 Table 1	arxiv_2201.11903 p.20	`m2022.wei.code-davinci-002.gsm8k.8shot-standard`
code-davinci-002	HumanEval	0-shot pass@1 (unbiased estimator over n samples) on HumanEval with the example input-output cases removed from the prompts	47.0	Chen et al. (Microsoft), CodeT, arXiv:2207.10397v2 Table 2	arxiv_2207.10397 p.5	`m2022.codet.code-davinci-002.humaneval.pass1-baseline`
code-davinci-002	HumanEval	HELM v0.3.0 pass@1	46.3	Stanford CRFM, HELM classic v0.3.0 results JSON	helm_classic_v0.3.0_code_humaneval json	`m2022.helm.code-davinci-002.humaneval`
code-davinci-002	MMLU	5-shot CoT, validation set (Chung et al. Table 16)	64.5	Chung et al. (Google), arXiv:2210.11416v5 Table 16	arxiv_2210.11416 p.40	`m2022.flan.code-davinci-002.mmlu.5shot-cot-val`
code-davinci-002	MMLU	5-shot direct, validation set (Chung et al. Table 16)	68.2	Chung et al. (Google), arXiv:2210.11416v5 Table 16	arxiv_2210.11416 p.40	`m2022.flan.code-davinci-002.mmlu.5shot-direct-val`
text-davinci-002	GSM8K	HELM v0.3.0 EM, 5 in-context examples	41.5	Stanford CRFM, HELM classic v0.3.0 results JSON	helm_classic_v0.3.0_gsm json	`m2022.helm.text-davinci-002.gsm8k`
text-davinci-002	GSM8K	HELM v0.1.0 (paper text)	35.0	Liang et al. (Stanford CRFM), HELM arXiv:2211.09110v2 Sec 1.1	arxiv_2211.09110 p.8	`m2022.helmpaper.text-davinci-002.gsm8k`
text-davinci-002	GSM8K	8-shot chain-of-thought, greedy	46.9	Wei et al. (Google), arXiv:2201.11903v6 Table 1	arxiv_2201.11903 p.20	`m2022.wei.text-davinci-002.gsm8k.8shot-cot`
text-davinci-002	GSM8K	8-shot standard prompting, greedy	15.6	Wei et al. (Google), arXiv:2201.11903v6 Table 1	arxiv_2201.11903 p.20	`m2022.wei.text-davinci-002.gsm8k.8shot-standard`

text-davinci-002	MMLU	5-shot direct, validation set (Chung et al. Table 16)	63.1	Chung et al. (Google), arXiv:2210.11416v5 Table 16	arxiv_2210.11416 p.40	m2022.flan.text-davinci-002.mmlu.5shot-direct-val
text-davinci-002	MMLU (HELM 5-subject subset)	HELM v0.3.0 EM, 5-shot	56.8	Stanford CRFM, HELM classic v0.3.0 results JSON	helm_classic_v0.3.0_mmlu json	m2022.helm.text-davinci-002.mmlu
text-davinci-002	MMLU (HELM 5-subject subset)	HELM v0.1.0 (paper text)	57.0	Liang et al. (Stanford CRFM), HELM arXiv:2211.09110v2 Sec 8	arxiv_2211.09110 p.68	m2022.helmpaper.text-davinci-002.mmlu

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	60.0	Gopher (280B)	5-shot	2021-12	m2021.gopher.gopher-280b.mmlu.5-shot
HumanEval	37.7	Codex-S	single sample (pass@1)	2021-07	m2021.chen-codex.codex-s-12b.humaneval.pass-at-1
MATH	6.9	GPT-2 1.5B (AMPS-pretrained, MATH fine-tuned)	pretrained on AMPS, then fine-tuned on MATH	2021-11	m2021.hendrycks-math.gpt-2-1.5b.math.amps-pretrain-math-ft

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, v6, PDF page 20: Table 1: Chain of thought prompting outperforms standard prompting for various large language.
- Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, v6, PDF page 5: Figure 4: Chain-of-thought prompting enables.

2022-02

February 2022. 3 verified scores for 1 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
GPT-NeoX-20B	GSM8K	HELM v0.3.0 EM, 5 in-context examples	5.3	Stanford CRFM, HELM classic v0.3.0 results JSON	helm_classic_v0.3.0_gsm json	m2022.helm.gpt-neox-20b.gsm8k
GPT-NeoX-20B	HumanEval	pass@1 (BLOOM Table 9, quoted from prior work)	15.4	BigScience, arXiv:2211.05100v4 Table 9	arxiv_2211.05100 p.34	m2022.bloom.gpt-neox-20b.humaneval.pass1
GPT-NeoX-20B	MMLU (HELM 5-subject subset)	HELM v0.3.0 EM, 5-shot	27.6	Stanford CRFM, HELM classic v0.3.0 results JSON	helm_classic_v0.3.0_mmlu json	m2022.helm.gpt-neox-20b.mmlu

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	60.0	Gopher (280B)	5-shot	2021-12	m2021.gopher.gopher-280b.mmlu.5-shot
HumanEval	37.7	Codex-S	single sample (pass@1)	2021-07	m2021.chen-codex.codex-s-12b.humaneval.pass-at-1
MATH	6.9	GPT-2 1.5B (AMPS-pretrained, MATH fine-tuned)	pretrained on AMPS, then fine-tuned on MATH	2021-11	m2021.hendrycks-math.gpt-2-1.5b.math.amps-pretrain-math-ft

2022-03

March 2022. 5 verified scores for 2 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Chinchilla 70B	MMLU	5-shot, 57-task average	67.6	Hoffmann et al. (DeepMind), arXiv:2203.15556v1 Table 6	arxiv_2203.15556 p.11	m2022.chinchilla.chinchilla-70b.mmlu.5shot

Chinchilla 70B	MMLU	abstract figure, protocol not stated	67.5	Hoffmann et al. (DeepMind), arXiv:2203.15556v1 abstract	arxiv_2203.15556 p.1	`m2022.chinchilla.chinchilla-70b.mmlu.abstract`
Chinchilla 70B	MMLU Mathematics subset (5 subjects)	5-shot (Galactica Table 8)	35.7	Taylor et al. (Meta), arXiv:2211.09085v1 Table 8	arxiv_2211.09085 p.15	`m2022.galactica.chinchilla.mmlu-math.5shot`
code-davinci-002	GSM8K	few-shot CoT, greedy (Wang et al.)	60.1	Wang et al. (Google), arXiv:2203.11171v4 Table 2	arxiv_2203.11171 p.5	`m2022.wang.code-davinci-002.gsm8k.cot-greedy`
code-davinci-002	GSM8K	few-shot CoT, self-consistency 40 samples (Wang et al.)	78.0	Wang et al. (Google), arXiv:2203.11171v4 Table 2	arxiv_2203.11171 p.5	`m2022.wang.code-davinci-002.gsm8k.sc-40`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	67.6	Chinchilla 70B	5-shot, 57-task average	2022-03	`m2022.chinchilla.chinchilla-70b.mmlu.5shot`
HumanEval	37.7	Codex-S	single sample (pass@1)	2021-07	`m2021.chen-codex.codex-s-12b.humaneval.pass-at-1`
MATH	6.9	GPT-2 1.5B (AMPS-pretrained, MATH fine-tuned)	pretrained on AMPS, then fine-tuned on MATH	2021-11	`m2021.hendrycks-math.gpt-2-1.5b.math.amps-pretrain-math-ft`

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Self-Consistency Improves Chain of Thought Reasoning in Language Models, v4, PDF page 5: Table 2: Arithmetic reasoning accuracy by self-consistency compared to chain-of-thought prompting.; Table 3: Commonsense and symbolic reasoning accuracy by self-consistency compared to chain-.
- Training Compute-Optimal Large Language Models, v1, PDF page 11: Table 6: Massive Multitask Language Understanding (MMLU).
- Training Compute-Optimal Large Language Models, v1, PDF page 12: Figure 6: MMLU results compared to Gopher We find that Chinchilla outperforms Gopher by 7.6%.; Table 7: Reading comprehension.

2022-04

April 2022. 15 verified scores for 4 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Anthropic-LM v4-s3 (52B)	GSM8K	HELM v0.3.0 EM, 5 in-context examples	17.1	Stanford CRFM, HELM classic v0.3.0 results JSON	helm_classic_v0.3.0_gsm json	`m2022.helm.anthropic-lm-v4-s3-52b.gsm8k`
Anthropic-LM v4-s3 (52B)	MMLU (HELM 5-subject subset)	HELM v0.3.0 EM, 5-shot	48.1	Stanford CRFM, HELM classic v0.3.0 results JSON	helm_classic_v0.3.0_mmlu json	`m2022.helm.anthropic-lm-v4-s3-52b.mmlu`
Anthropic-LM v4-s3 (52B)	MMLU (HELM 5-subject subset)	HELM v0.1.0 (paper text)	49.8	Liang et al. (Stanford CRFM), HELM arXiv:2211.09110v2 Sec 8	arxiv_2211.09110 p.68	`m2022.helmpaper.anthropic-lm-52b.mmlu`
PaLM 540B	GSM8K	8-shot CoT, no calculator (PaLM paper)	54.0	Chowdhery et al. (Google), arXiv:2204.02311v5 Table 10	arxiv_2204.02311 p.21	`m2022.palm.palm-540b.gsm8k.8shot-cot`
PaLM 540B	GSM8K	8-shot CoT + external calculator (PaLM paper)	58.0	Chowdhery et al. (Google), arXiv:2204.02311v5 Table 10	arxiv_2204.02311 p.21	`m2022.palm.palm-540b.gsm8k.8shot-cot-calc`
PaLM 540B	GSM8K	few-shot CoT, greedy (Wang et al.)	56.5	Wang et al. (Google), arXiv:2203.11171v4 Table 2	arxiv_2203.11171 p.5	`m2022.wang.palm-540b.gsm8k.cot-greedy`
PaLM 540B	GSM8K	few-shot CoT, self-consistency 40 samples (Wang et al.)	74.4	Wang et al. (Google), arXiv:2203.11171v4 Table 2	arxiv_2203.11171 p.5	`m2022.wang.palm-540b.gsm8k.sc-40`
PaLM 540B	GSM8K	8-shot CoT, greedy (Wei et al.)	56.9	Wei et al. (Google), arXiv:2201.11903v6 Table 1	arxiv_2201.11903 p.20	`m2022.wei.palm-540b.gsm8k.8shot-cot`
PaLM 540B	HumanEval	0-shot pass@1	26.2	Chowdhery et al. (Google), arXiv:2204.02311v5 Table 12	arxiv_2204.02311 p.25	`m2022.palm.palm-540b.humaneval.0shot-pass1`

PaLM 540B	MATH	fixed 4-shot prompt (Minerva Table 3)	8.8	Lewkowycz et al. (Google), arXiv:2206.14858v2 Table 3	arxiv_2206.14858 p.8	m2022.minerva.palm-540b.math.4shot
PaLM 540B	MMLU	5-shot CoT, test set (Chung et al. Table 4)	64.5	Chung et al. (Google), arXiv:2210.11416v5 Table 4	arxiv_2210.11416 p.8	m2022.flan.palm-540b.mmlu.5shot-cot-test
PaLM 540B	MMLU	5-shot, test set	69.3	Chowdhery et al. (Google), arXiv:2204.02311v5 Table 6	arxiv_2204.02311 p.13	m2022.palm.palm-540b.mmlu.5shot
PaLM 540B	MMLU-STEM	standard 5-shot (Minerva Table 3)	58.7	Lewkowycz et al. (Google), arXiv:2206.14858v2 Table 3	arxiv_2206.14858 p.8	m2022.minerva.palm-540b.mmlu-stem.5shot
PaLM-Coder 540B	HumanEval	0-shot pass@1	36.0	Chowdhery et al. (Google), arXiv:2204.02311v5 Table 12	arxiv_2204.02311 p.25	m2022.palm.palm-coder-540b.humaneval.0shot-pass1
code-davinci-001 (Davinci Codex)	HumanEval	0-shot pass@1, Google's API measurement	36.0	Chowdhery et al. (Google), arXiv:2204.02311v5 Table 12	arxiv_2204.02311 p.25	m2022.palm.code-davinci-001.humaneval.0shot-pass1

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	67.6	Chinchilla 70B	5-shot, 57-task average	2022-03	m2022.chinchilla.chinchilla-70b.mmlu.5shot
HumanEval	37.7	Codex-S	single sample (pass@1)	2021-07	m2021.chen-codex.codex-s-12b.humaneval.pass-at-1
GSM8K	58.0	PaLM 540B	8-shot CoT + external calculator (PaLM paper)	2022-04	m2022.palm.palm-540b.gsm8k.8shot-cot-calc
MATH	6.9	GPT-2 1.5B (AMPS-pretrained, MATH fine-tuned)	pretrained on AMPS, then fine-tuned on MATH	2021-11	m2021.hendrycks-math.gpt-2-1.5b.math.amps-pretrain-math-ft

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- PaLM: Scaling Language Modeling with Pathways, v5, PDF page 22: Figure 10: Via chain of thought prompting, PaLM achieves strong performance on a range of arithmetic and.
- PaLM: Scaling Language Modeling with Pathways, v5, PDF page 23: Figure 11: Examples from the PaLM-Coder 540B model.
- PaLM: Scaling Language Modeling with Pathways, v5, PDF page 25: Table 12: Results obtained by the PaLM 540B and PaLM-Coder 540B models across code synthesis and.; Figure 12: Scaling results for HumanEval, MBPP, GSM8K-Python, TransCoder, and DeepFix datasets on.

2022-05

May 2022. 4 verified scores for 1 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
OPT-175B	GSM8K	HELM v0.3.0 EM, 5 in-context examples	4.0	Stanford CRFM, HELM classic v0.3.0 results JSON	helm_classic_v0.3.0_gsm json	m2022.helm.opt-175b.gsm8k
OPT-175B	MMLU	0-shot, test set (OPT-IML Table 14)	27.3	Iyer et al. (Meta), OPT-IML, arXiv:2212.12017v3 Table 14	arxiv_2212.12017 p.21	m2022.optiml.opt-175b.mmlu.0shot
OPT-175B	MMLU	5-shot, test set (OPT-IML Table 14)	34.2	Iyer et al. (Meta), OPT-IML, arXiv:2212.12017v3 Table 14	arxiv_2212.12017 p.21	m2022.optiml.opt-175b.mmlu.5shot
OPT-175B	MMLU (HELM 5-subject subset)	HELM v0.3.0 EM, 5-shot	31.8	Stanford CRFM, HELM classic v0.3.0 results JSON	helm_classic_v0.3.0_mmlu json	m2022.helm.opt-175b.mmlu

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	67.6	Chinchilla 70B	5-shot, 57-task average	2022-03	m2022.chinchilla.chinchilla-70b.mmlu.5shot
HumanEval	37.7	Codex-S	single sample (pass@1)	2021-07	m2021.chen-codex.codex-s-12b.humaneval.pass-at-1
GSM8K	58.0	PaLM 540B	8-shot CoT + external calculator (PaLM paper)	2022-04	m2022.palm.palm-540b.gsm8k.8shot-cot-calc

MATH	6.9	GPT-2 1.5B (AMPS-pretrained, MATH fine-tuned)	pretrained on AMPS, then fine-tuned on MATH	2021-11	`m2021.hendrycks-math.gpt-2-1.5b.math.amps-pretrain-math-ft`

2022-06

June 2022. 8 verified scores for 3 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Minerva 540B	GSM8K	Wei et al. CoT prompt, no majority vote	58.8	Lewkowycz et al. (Google), arXiv:2206.14858v2 Table 3	arxiv_2206.14858 p.8	`m2022.minerva.minerva-540b.gsm8k.cot`
Minerva 540B	GSM8K	Wei et al. CoT prompt, maj1@40	78.5	Lewkowycz et al. (Google), arXiv:2206.14858v2 Table 3	arxiv_2206.14858 p.8	`m2022.minerva.minerva-540b.gsm8k.maj1at40`
Minerva 540B	MATH	fixed 4-shot, no majority vote	33.6	Lewkowycz et al. (Google), arXiv:2206.14858v2 Table 3	arxiv_2206.14858 p.8	`m2022.minerva.minerva-540b.math.4shot`
Minerva 540B	MATH	fixed 4-shot, maj1@64	50.3	Lewkowycz et al. (Google), arXiv:2206.14858v2 Table 3	arxiv_2206.14858 p.8	`m2022.minerva.minerva-540b.math.maj1at64`
Minerva 540B	MMLU-STEM	standard 5-shot, highest-scoring answer	63.9	Lewkowycz et al. (Google), arXiv:2206.14858v2 Table 3	arxiv_2206.14858 p.8	`m2022.minerva.minerva-540b.mmlu-stem.5shot`
Minerva 540B	MMLU-STEM	CoT prompt, maj1@16	75.0	Lewkowycz et al. (Google), arXiv:2206.14858v2 Table 3	arxiv_2206.14858 p.8	`m2022.minerva.minerva-540b.mmlu-stem.maj1at16`
Minerva 62B	GSM8K	Wei et al. CoT prompt, maj1@100	68.5	Lewkowycz et al. (Google), arXiv:2206.14858v2 Table 3	arxiv_2206.14858 p.8	`m2022.minerva.minerva-62b.gsm8k.maj1at100`
OpenAI davinci-002 (variant unstated)	MATH	fixed 4-shot, T = 0.2 (Minerva evaluation)	19.1	Lewkowycz et al. (Google), arXiv:2206.14858v2 Table 3	arxiv_2206.14858 p.8	`m2022.minerva.openai-davinci-002.math`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	67.6	Chinchilla 70B	5-shot, 57-task average	2022-03	`m2022.chinchilla.chinchilla-70b.mmlu.5shot`
HumanEval	37.7	Codex-S	single sample (pass@1)	2021-07	`m2021.chen-codex.codex-s-12b.humaneval.pass-at-1`
GSM8K	78.5	Minerva 540B	Wei et al. CoT prompt, maj1@40	2022-06	`m2022.minerva.minerva-540b.gsm8k.maj1at40`
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	`m2022.minerva.minerva-540b.math.maj1at64`

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Solving Quantitative Reasoning Problems with Language Models, v2, PDF page 7: Figure 4: Performance on MATH and MMLU-STEM by subtopic.
- Solving Quantitative Reasoning Problems with Language Models, v2, PDF page 8: Table 3: Model performance on several quantitative reasoning datasets.

2022-07

July 2022. 4 verified scores for 1 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
BLOOM 176B	GSM8K	HELM v0.3.0 EM, 5 in-context examples	9.5	Stanford CRFM, HELM classic v0.3.0 results JSON	helm_classic_v0.3.0_gsm json	`m2022.helm.bloom-176b.gsm8k`
BLOOM 176B	HumanEval	pass@1	15.52	BigScience, arXiv:2211.05100v4 Table 9	arxiv_2211.05100 p.34	`m2022.bloom.bloom-176b.humaneval.pass1`
BLOOM 176B	HumanEval	pass@100	55.45	BigScience, arXiv:2211.05100v4 Table 9	arxiv_2211.05100 p.34	`m2022.bloom.bloom-176b.humaneval.pass100`
BLOOM 176B	MMLU (HELM 5-subject subset)	HELM v0.3.0 EM, 5-shot	29.9	Stanford CRFM, HELM classic v0.3.0 results JSON	helm_classic_v0.3.0_mmlu json	`m2022.helm.bloom-176b.mmlu`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	67.6	Chinchilla 70B	5-shot, 57-task average	2022-03	`m2022.chinchilla.chinchilla-70b.mmlu.5shot`

HumanEval	37.7	Codex-S	single sample (pass@1)	2021-07	`m2021.chen-codex.codex-s-12b.humaneval.pass-at-1`
GSM8K	78.5	Minerva 540B	Wei et al. CoT prompt, maj1@40	2022-06	`m2022.minerva.minerva-540b.gsm8k.maj1at40`
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	`m2022.minerva.minerva-540b.math.maj1at64`

2022-08

August 2022. 3 verified scores for 1 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
GLM-130B	GSM8K	HELM v0.3.0 EM, 5 in-context examples	6.1	Stanford CRFM, HELM classic v0.3.0 results JSON	helm_classic_v0.3.0_gsm json	`m2022.helm.glm-130b.gsm8k`
GLM-130B	MMLU	5-shot, after 400B tokens	44.8	Zeng et al. (Tsinghua/Zhipu), arXiv:2210.02414v2 Sec 5.2	arxiv_2210.02414 p.8	`m2022.glm.glm-130b.mmlu.5shot`
GLM-130B	MMLU (HELM 5-subject subset)	HELM v0.3.0 EM, 5-shot	34.4	Stanford CRFM, HELM classic v0.3.0 results JSON	helm_classic_v0.3.0_mmlu json	`m2022.helm.glm-130b.mmlu`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	67.6	Chinchilla 70B	5-shot, 57-task average	2022-03	`m2022.chinchilla.chinchilla-70b.mmlu.5shot`
HumanEval	37.7	Codex-S	single sample (pass@1)	2021-07	`m2021.chen-codex.codex-s-12b.humaneval.pass-at-1`
GSM8K	78.5	Minerva 540B	Wei et al. CoT prompt, maj1@40	2022-06	`m2022.minerva.minerva-540b.gsm8k.maj1at40`
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	`m2022.minerva.minerva-540b.math.maj1at64`

2022-09

September 2022. 1 verified scores for 1 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
code-davinci-002 + CodeT	HumanEval	0-shot, CodeT reranking of 100 samples (T=0.8), pass@1, on HumanEval with example input-output cases removed from the prompts	65.8	Chen et al. (Microsoft), CodeT, arXiv:2207.10397v2 Table 2	arxiv_2207.10397 p.5	`m2022.codet.code-davinci-002+codet.humaneval.pass1`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	67.6	Chinchilla 70B	5-shot, 57-task average	2022-03	`m2022.chinchilla.chinchilla-70b.mmlu.5shot`
HumanEval	37.7	Codex-S	single sample (pass@1)	2021-07	`m2021.chen-codex.codex-s-12b.humaneval.pass-at-1`
GSM8K	78.5	Minerva 540B	Wei et al. CoT prompt, maj1@40	2022-06	`m2022.minerva.minerva-540b.gsm8k.maj1at40`
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	`m2022.minerva.minerva-540b.math.maj1at64`

2022-10

October 2022. 4 verified scores for 1 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Flan-PaLM 540B	GSM8K	CoT + self-consistency (train set held-in)	83.9	Chung et al. (Google), arXiv:2210.11416v5 Sec 4.1	arxiv_2210.11416 p.7	`m2022.flan.flan-palm-540b.gsm8k.cot-sc`
Flan-PaLM 540B	MMLU	5-shot CoT + self-consistency, test set	75.2	Chung et al. (Google), arXiv:2210.11416v5 Table 4	arxiv_2210.11416 p.8	`m2022.flan.flan-palm-540b.mmlu.5shot-cot-sc-test`

Flan-PaLM 540B	MMLU	5-shot direct, test set	72.2	Chung et al. (Google), arXiv:2210.11416v5 Table 4	arxiv_2210.11416 p.8	m2022.flan.flan-palm-540b.mmlu.5shot-direct-test
Flan-PaLM 540B	MMLU	5-shot direct, validation set	73.5	Chung et al. (Google), arXiv:2210.11416v5 Table 5	arxiv_2210.11416 p.11	m2022.flan.flan-palm-540b.mmlu.5shot-direct-val

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	75.2	Flan-PaLM 540B	5-shot CoT + self-consistency, test set	2022-10	m2022.flan.flan-palm-540b.mmlu.5shot-cot-sc-test
HumanEval	37.7	Codex-S	single sample (pass@1)	2021-07	m2021.chen-codex.codex-s-12b.humaneval.pass-at-1
GSM8K	83.9	Flan-PaLM 540B	CoT + self-consistency (train set held-in)	2022-10	m2022.flan.flan-palm-540b.gsm8k.cot-sc
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	m2022.minerva.minerva-540b.math.maj1at64

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- GLM-130B: An Open Bilingual Pre-trained Model, v2, PDF page 8: Figure 6: GLM-130B on MMLU.; Figure 7: BIG-bench-lite evalua-.; Table 4: Details on BIG-.
- Scaling Instruction-Finetuned Language Models, v5, PDF page 2: Table 1: Average 5-shot MMLU scores (%) for 57 tasks with model and human accuracy comparisons.
- Scaling Instruction-Finetuned Language Models, v5, PDF page 8: Table 4: Flan-PaLM outperforms PaLM on all evaluation benchmarks.; Figure 5: Jointly finetuning on non-CoT and CoT data improves performance on both evaluations, compared.

2022-11

November 2022. 11 verified scores for 5 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
GPT-3.5	GSM8K	5-shot (CoT is stated only for the GPT-4 cell; not stated for GPT-3.5)	57.1	OpenAI, GPT-4 Technical Report arXiv:2303.08774v6 Table 2	arxiv_2303.08774 p.7	m2022.gpt4tr.gpt-3.5.gsm8k.5shot
GPT-3.5 (snapshot not named)	HumanEval	0-shot	48.1	OpenAI, GPT-4 Technical Report arXiv:2303.08774v6 Table 2	arxiv_2303.08774 p.7	m2022.gpt4tr.gpt-3.5.humaneval.0shot
GPT-3.5 (snapshot not named)	MMLU	5-shot	70.0	OpenAI, GPT-4 Technical Report arXiv:2303.08774v6 Table 2	arxiv_2303.08774 p.7	m2022.gpt4tr.gpt-3.5.mmlu.5shot
Galactica 120B	MATH	5-shot mCoT	20.4	Taylor et al. (Meta), arXiv:2211.09085v1 Table 9	arxiv_2211.09085 p.15	m2022.galactica.gal-120b.math.5shot-mcot
Galactica 120B	MATH	token prompt	16.6	Taylor et al. (Meta), arXiv:2211.09085v1 Table 9	arxiv_2211.09085 p.15	m2022.galactica.gal-120b.math.work
Galactica 120B	MMLU Mathematics subset (5 subjects)	token, no few-shot examples	41.3	Taylor et al. (Meta), arXiv:2211.09085v1 Table 8	arxiv_2211.09085 p.15	m2022.galactica.gal-120b.mmlu-math.work
gpt-3.5-turbo	GSM8K	few-shot CoT (CoT Hub own run)	74.9	Fu et al., Chain-of-Thought Hub, arXiv:2305.17306v1 Table 1	arxiv_2305.17306 p.3	m2022.cothub.gpt-3.5-turbo.gsm8k.fewshot-cot
gpt-3.5-turbo	MMLU	few-shot CoT (CoT Hub own run)	67.3	Fu et al., Chain-of-Thought Hub, arXiv:2305.17306v1 Table 1	arxiv_2305.17306 p.3	m2022.cothub.gpt-3.5-turbo.mmlu.fewshot-cot
text-davinci-003	GSM8K	HELM v0.3.0 EM, 5 in-context examples	50.6	Stanford CRFM, HELM classic v0.3.0 results JSON	helm_classic_v0.3.0_gsm json	m2022.helm.text-davinci-003.gsm8k
text-davinci-003	MMLU	5-shot direct, validation set (Chung et al. Table 16)	64.8	Chung et al. (Google), arXiv:2210.11416v5 Table 16	arxiv_2210.11416 p.40	m2022.flan.text-davinci-003.mmlu.5shot-direct-val
text-davinci-003	MMLU (HELM 5-subject subset)	HELM v0.3.0 EM, 5-shot	56.9	Stanford CRFM, HELM classic v0.3.0 results JSON	helm_classic_v0.3.0_mmlu json	m2022.helm.text-davinci-003.mmlu

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	75.2	Flan-PaLM 540B	5-shot CoT + self-consistency, test set	2022-10	`m2022.flan.flan-palm-540b.mmlu.5shot-cot-sc-test`
HumanEval	65.8	code-davinci-002 + CodeT	0-shot, CodeT reranking of 100 samples (T=0.8), pass@1, on HumanEval with example input-output cases removed from the prompts	2022-11	`m2022.codet.code-davinci-002+codet.humaneval.pass1`
GSM8K	83.9	Flan-PaLM 540B	CoT + self-consistency (train set held-in)	2022-10	`m2022.flan.flan-palm-540b.gsm8k.cot-sc`
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	`m2022.minerva.minerva-540b.math.maj1at64`

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Galactica: A Large Language Model for Science, v1, PDF page 15: Table 8: Results on Mathematics MMLU. Galactica is evaluated without few-shot examples.; Table 9: Results on MATH. With both the chain-of-thought and token prompts, Galactica exceeds.

2022-12

December 2022. 2 verified scores for 1 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
OPT-IML-Max 175B	MMLU	0-shot, test set	49.1	Iyer et al. (Meta), OPT-IML, arXiv:2212.12017v3 Table 14	arxiv_2212.12017 p.21	`m2022.optiml.opt-iml-max-175b.mmlu.0shot`
OPT-IML-Max 175B	MMLU	5-shot, test set	47.1	Iyer et al. (Meta), OPT-IML, arXiv:2212.12017v3 Table 14	arxiv_2212.12017 p.21	`m2022.optiml.opt-iml-max-175b.mmlu.5shot`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	75.2	Flan-PaLM 540B	5-shot CoT + self-consistency, test set	2022-10	`m2022.flan.flan-palm-540b.mmlu.5shot-cot-sc-test`
HumanEval	65.8	code-davinci-002 + CodeT	0-shot, CodeT reranking of 100 samples (T=0.8), pass@1, on HumanEval with example input-output cases removed from the prompts	2022-11	`m2022.codet.code-davinci-002+codet.humaneval.pass1`
GSM8K	83.9	Flan-PaLM 540B	CoT + self-consistency (train set held-in)	2022-10	`m2022.flan.flan-palm-540b.gsm8k.cot-sc`
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	`m2022.minerva.minerva-540b.math.maj1at64`

2023-01

January 2023. 1 verified scores for 1 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
SantaCoder 1.1B (final)	HumanEval (MultiPL-E, Python)	pass@100	49.0	BigCode (Allal et al. 2023)	arxiv_2301.03988 p.10	`m2023.allal2023.santacoder-1.1b.humaneval-python.pass100`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	75.2	Flan-PaLM 540B	5-shot CoT + self-consistency, test set	2022-10	`m2022.flan.flan-palm-540b.mmlu.5shot-cot-sc-test`
HumanEval	65.8	code-davinci-002 + CodeT	0-shot, CodeT reranking of 100 samples (T=0.8), pass@1, on HumanEval with example input-output cases removed from the prompts	2022-11	`m2022.codet.code-davinci-002+codet.humaneval.pass1`
GSM8K	83.9	Flan-PaLM 540B	CoT + self-consistency (train set held-in)	2022-10	`m2022.flan.flan-palm-540b.gsm8k.cot-sc`
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	`m2022.minerva.minerva-540b.math.maj1at64`

2023-02

February 2023. 8 verified scores for 2 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
LLaMA 65B	GSM8K	maj1@k, k=100	69.7	Meta (Touvron et al. 2023a)	arxiv_2302.13971 p.6	`m2023.touvron2023a.llama-65b.gsm8k.maj1at100`
LLaMA 65B	GSM8K	without maj1@k (shot count not stated)	50.9	Meta (Touvron et al. 2023a)	arxiv_2302.13971 p.6	`m2023.touvron2023a.llama-65b.gsm8k.single`
LLaMA 65B	HumanEval	0-shot pass@1 (T=0.1)	23.7	Meta (Touvron et al. 2023a)	arxiv_2302.13971 p.6	`m2023.touvron2023a.llama-65b.humaneval.0shot-pass1`
LLaMA 65B	HumanEval	0-shot pass@100 (T=0.8)	79.3	Meta (Touvron et al. 2023a)	arxiv_2302.13971 p.6	`m2023.touvron2023a.llama-65b.humaneval.0shot-pass100`
LLaMA 65B	MATH	maj1@k, k=256	20.5	Meta (Touvron et al. 2023a)	arxiv_2302.13971 p.6	`m2023.touvron2023a.llama-65b.math.maj1at256`
LLaMA 65B	MATH	without maj1@k (shot count not stated)	10.6	Meta (Touvron et al. 2023a)	arxiv_2302.13971 p.6	`m2023.touvron2023a.llama-65b.math.single`
LLaMA 65B	MMLU	5-shot	63.4	Meta (Touvron et al. 2023a)	arxiv_2302.13971 p.7	`m2023.touvron2023a.llama-65b.mmlu.5shot`
LLaMA-I 65B	MMLU	5-shot	68.9	Meta (Touvron et al. 2023a)	arxiv_2302.13971 p.7	`m2023.touvron2023a.llama-i-65b.mmlu.5shot`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	75.2	Flan-PaLM 540B	5-shot CoT + self-consistency, test set	2022-10	`m2022.flan.flan-palm-540b.mmlu.5shot-cot-sc-test`
HumanEval	65.8	code-davinci-002 + CodeT	0-shot, CodeT reranking of 100 samples (T=0.8), pass@1, on HumanEval with example input-output cases removed from the prompts	2022-11	`m2022.codet.code-davinci-002+codet.humaneval.pass1`
GSM8K	83.9	Flan-PaLM 540B	CoT + self-consistency (train set held-in)	2022-10	`m2022.flan.flan-palm-540b.gsm8k.cot-sc`
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	`m2022.minerva.minerva-540b.math.maj1at64`

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- LLaMA: Open and Efficient Foundation Language Models, v1, PDF page 6: Table 7: Model performance on quantitative reason-.; Table 8: Model performance for code generation.
- LLaMA: Open and Efficient Foundation Language Models, v1, PDF page 7: Table 9: Massive Multitask Language Understanding (MMLU).; Table 10: Instruction finetuning - MMLU (5-shot).

2023-03

March 2023. 3 verified scores for 1 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
GPT-4	GSM8K	5-shot chain-of-thought (part of GSM8K train set in pre-training mix)	92.0	OpenAI (GPT-4 Technical Report)	arxiv_2303.08774 p.7	`m2023.openai2023.gpt-4.gsm8k.5shot-cot`
GPT-4	HumanEval	0-shot	67.0	OpenAI (GPT-4 Technical Report)	arxiv_2303.08774 p.7	`m2023.openai2023.gpt-4.humaneval.0shot`
GPT-4	MMLU	5-shot	86.4	OpenAI (GPT-4 Technical Report)	arxiv_2303.08774 p.7	`m2023.openai2023.gpt-4.mmlu.5shot`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	86.4	GPT-4	5-shot	2023-03	`m2023.openai2023.gpt-4.mmlu.5shot`
HumanEval	67.0	GPT-4	0-shot	2023-03	`m2023.openai2023.gpt-4.humaneval.0shot`
GSM8K	92.0	GPT-4	5-shot chain-of-thought (part of GSM8K train set in pre-training mix)	2023-03	`m2023.openai2023.gpt-4.gsm8k.5shot-cot`

MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	m2022.minerva.minerva-540b.math.maj1at64

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- GPT-4 Technical Report, v6, PDF page 7: Table 2: Performance of GPT-4 on academic benchmarks.

2023-04

April 2023. No verified score is dated to this month.
Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	86.4	GPT-4	5-shot	2023-03	m2023.openai2023.gpt-4.mmlu.5shot
HumanEval	67.0	GPT-4	0-shot	2023-03	m2023.openai2023.gpt-4.humaneval.0shot
GSM8K	92.0	GPT-4	5-shot chain-of-thought (part of GSM8K train set in pre-training mix)	2023-03	m2023.openai2023.gpt-4.gsm8k.5shot-cot
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	m2022.minerva.minerva-540b.math.maj1at64

2023-05

May 2023. 12 verified scores for 7 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Flan-PaLM 2	GSM8K	8-shot CoT + self-consistency, 40 paths	92.2	Google (PaLM 2 TR, Table 7)	arxiv 2305.10403 p.15	m2023.anil2023.flan-palm-2.gsm8k.8shot-cot-sc40
Flan-PaLM 2 (L)	MMLU	5-shot	81.2	Google (PaLM 2 TR, Table 16)	arxiv 2305.10403 p.44	m2023.anil2023.flan-palm-2-l.mmlu.5shot
GPT-4	GSM8K	as reported by OpenAI (SOTA column footnote b)	92.0	Google (PaLM 2 TR, Table 7)	arxiv 2305.10403 p.15	m2023.anil2023.gpt-4.gsm8k
GPT-4	HumanEval	pass@1 greedy	88.4	EvalPlus (Liu et al. 2023, v3)	arxiv 2305.01210 p.7	m2023.liu2023evalplus.gpt-4.humaneval.greedy
GPT-4	MATH	not stated for the GPT-4 column	42.5	Google (PaLM 2 TR, Table 7)	arxiv 2305.10403 p.15	m2023.anil2023.gpt-4.math.unstated
PaLM 2	GSM8K	8-shot CoT without self-consistency (decoding not stated)	80.7	Google (PaLM 2 TR, Table 7)	arxiv 2305.10403 p.15	m2023.anil2023.palm-2.gsm8k.8shot-cot
PaLM 2	GSM8K	8-shot CoT + self-consistency, 40 paths	91.0	Google (PaLM 2 TR, Table 7)	arxiv 2305.10403 p.15	m2023.anil2023.palm-2.gsm8k.8shot-cot-sc40
PaLM 2	MATH	4-shot CoT without self-consistency (decoding not stated)	34.3	Google (PaLM 2 TR, Table 7)	arxiv 2305.10403 p.15	m2023.anil2023.palm-2.math.4shot-cot
PaLM 2	MATH	4-shot CoT + self-consistency, 64 paths	48.8	Google (PaLM 2 TR, Table 7)	arxiv 2305.10403 p.15	m2023.anil2023.palm-2.math.4shot-cot-sc64
PaLM 2 (L)	MMLU	5-shot	78.3	Google (PaLM 2 TR, Table 16)	arxiv 2305.10403 p.44	m2023.anil2023.palm-2-l.mmlu.5shot
PaLM 2-S*	HumanEval	pass@1, greedy (described as few-shot coding task; shot count not given)	37.6	Google (PaLM 2 TR, Table 8)	arxiv 2305.10403 p.16	m2023.anil2023.palm-2-s-star.humaneval.pass1-greedy
SantaCoder 1.1B	HumanEval	pass@1 greedy	14.6	EvalPlus (Liu et al. 2023, v3)	arxiv 2305.01210 p.7	m2023.liu2023evalplus.santacoder-1.1b.humaneval.greedy

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	86.4	GPT-4	5-shot	2023-03	m2023.openai2023.gpt-4.mmlu.5shot
HumanEval	67.0	GPT-4	0-shot	2023-03	m2023.openai2023.gpt-4.humaneval.0shot
GSM8K	92.2	Flan-PaLM 2	8-shot CoT + self-consistency, 40 paths	2023-05	m2023.anil2023.flan-palm-2.gsm8k.8shot-cot-sc40
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	m2022.minerva.minerva-540b.math.maj1at64

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- PaLM 2 Technical Report, v3, PDF page 15: Table 7: Evaluation results on MATH, GSM8K, and MGSM with chain-of-thought prompting (Wei et al., 2022) /.

2023-06

June 2023. 4 verified scores for 3 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
StarCoder 15B	HumanEval	pass@1	33.6	Microsoft (Luo et al. 2023), quoting StarCoder paper	arxiv_2306.08568 p.6	`m2023.luo2023.starcoder-15b.humaneval.pass1`
Vicuna 13B	HumanEval	not stated in Table 5	15.2	Chia et al. 2023 (INSTRUCTEVAL v1)	arxiv_2306.04757 p.6	`m2023.chia2023.vicuna-13b.humaneval`
Vicuna 13B	MMLU	not stated in Table 5	49.7	Chia et al. 2023 (INSTRUCTEVAL v1)	arxiv_2306.04757 p.6	`m2023.chia2023.vicuna-13b.mmlu`
WizardCoder 15B	HumanEval	pass@1 (n samples, T=0.2, top_p=0.95)	57.3	Microsoft (Luo et al. 2023)	arxiv_2306.08568 p.6	`m2023.luo2023.wizardcoder-15b.humaneval.pass1`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	86.4	GPT-4	5-shot	2023-03	`m2023.openai2023.gpt-4.mmlu.5shot`
HumanEval	67.0	GPT-4	0-shot	2023-03	`m2023.openai2023.gpt-4.humaneval.0shot`
GSM8K	92.2	Flan-PaLM 2	8-shot CoT + self-consistency, 40 paths	2023-05	`m2023.anil2023.flan-palm-2.gsm8k.8shot-cot-sc40`
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	`m2022.minerva.minerva-540b.math.maj1at64`

2023-07

July 2023. 19 verified scores for 6 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Claude 1.3	GSM8K	0-shot CoT, T=1	85.2	Anthropic (Claude 2 model card)	anthropic_claude2_card p.10	`m2023.anthropic-claude2-card.claude-1.3.gsm8k.0shot-cot`
Claude 1.3	HumanEval (Codex P@1)	0-shot pass@1, T=1	56.0	Anthropic (Claude 2 model card)	anthropic_claude2_card p.10	`m2023.anthropic-claude2-card.claude-1.3.humaneval.0shot`
Claude 1.3	MMLU	5-shot CoT, T=1	77.0	Anthropic (Claude 2 model card)	anthropic_claude2_card p.10	`m2023.anthropic-claude2-card.claude-1.3.mmlu.5shot-cot`
Claude 2	GSM8K	0-shot CoT, T=1	88.0	Anthropic (Claude 2 model card)	anthropic_claude2_card p.10	`m2023.anthropic-claude2-card.claude-2.gsm8k.0shot-cot`
Claude 2	GSM8K	0-shot	88.0	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	`m2023.gemini2023.claude-2.gsm8k.0shot`
Claude 2	HumanEval	0-shot	70.0	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	`m2023.gemini2023.claude-2.humaneval.0shot`
Claude 2	HumanEval (Codex P@1)	0-shot pass@1, T=1	71.2	Anthropic (Claude 2 model card)	anthropic_claude2_card p.10	`m2023.anthropic-claude2-card.claude-2.humaneval.0shot`
Claude 2	MMLU	5-shot CoT, T=1	78.5	Anthropic (Claude 2 model card)	anthropic_claude2_card p.10	`m2023.anthropic-claude2-card.claude-2.mmlu.5shot-cot`
Claude 2	MMLU	5-shot CoT	78.5	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	`m2023.gemini2023.claude-2.mmlu.5shot-cot`
Claude Instant	GSM8K	0-shot CoT, T=1	80.9	Anthropic (Claude 2 model card)	anthropic_claude2_card p.10	`m2023.anthropic-claude2-card.claude-instant.gsm8k.0shot-cot`
Claude Instant	HumanEval (Codex P@1)	0-shot pass@1, T=1	52.8	Anthropic (Claude 2 model card)	anthropic_claude2_card p.10	`m2023.anthropic-claude2-card.claude-instant.humaneval.0shot`
Claude Instant	MMLU	5-shot CoT, T=1	73.4	Anthropic (Claude 2 model card)	anthropic_claude2_card p.10	`m2023.anthropic-claude2-card.claude-instant.mmlu.5shot-cot`
Llama 2 13B	MMLU	5-shot	54.8	Meta (Touvron et al. 2023b, Table 19)	arxiv_2307.09288 p.49	`m2023.touvron2023b.llama-2-13b.mmlu.5shot`
Llama 2 70B	GSM8K	8-shot	56.8	Meta (Touvron et al. 2023b, Table 4)	arxiv_2307.09288 p.8	`m2023.touvron2023b.llama-2-70b.gsm8k.8shot`

Llama 2 70B	HumanEval	0-shot pass@1	29.9	Meta (Touvron et al. 2023b, Table 4)	arxiv_2307.09288 p.8	m2023.touvron2023b.llama-2-70b.humaneval.0shot
Llama 2 70B	MATH	4-shot, maj1@1	13.5	Meta (Touvron et al. 2023b, Table 25)	arxiv_2307.09288 p.51	m2023.touvron2023b.llama-2-70b.math.4shot-maj1at1
Llama 2 70B	MMLU	5-shot	68.9	Meta (Touvron et al. 2023b, Table 4)	arxiv_2307.09288 p.8	m2023.touvron2023b.llama-2-70b.mmlu.5shot
Llama 2 70B	MMLU (clean subset)	contamination analysis, clean subset	68.0	Meta (Touvron et al. 2023b, Table 51)	arxiv_2307.09288 p.76	m2023.touvron2023b.llama-2-70b.mmlu.clean-subset
PaLM-2-L	MMLU	5-shot	78.3	Meta (Llama 2 paper), quoting Anil et al. 2023	arxiv_2307.09288 p.8	m2023.touvron2023b.palm-2-l.mmlu.5shot

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	86.4	GPT-4	5-shot	2023-03	m2023.openai2023.gpt-4.mmlu.5shot
HumanEval	71.2	Claude 2	0-shot pass@1, T=1	2023-07	m2023.anthropic-claude2-card.claude-2.humaneval.0shot
GSM8K	92.2	Flan-PaLM 2	8-shot CoT + self-consistency, 40 paths	2023-05	m2023.anil2023.flan-palm-2.gsm8k.8shot-cot-sc40
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	m2022.minerva.minerva-540b.math.maj1at64

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Anthropic Claude 2 Model Card (July 2023), None, PDF page 10: no table or figure caption on this page
- Llama 2: Open Foundation and Fine-Tuned Chat Models, v2, PDF page 8: Table 3: Overall performance on grouped academic benchmarks compared to open-source base models.; Table 4: Comparison to closed-source models on academic benchmarks.

2023-08

August 2023. 4 verified scores for 4 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Code Llama - Instruct 34B	HumanEval	0-shot pass@1 greedy	41.5	Meta (Roziere et al. 2023, v3)	arxiv_2308.12950 p.6	m2023.roziere2023.code-llama-instruct-34b.humaneval.pass1
Code Llama - Python 34B	HumanEval	0-shot pass@1 greedy	53.7	Meta (Roziere et al. 2023, v3)	arxiv_2308.12950 p.6	m2023.roziere2023.code-llama-python-34b.humaneval.pass1
Code Llama 34B	HumanEval	0-shot pass@1 greedy	48.8	Meta (Roziere et al. 2023, v3)	arxiv_2308.12950 p.6	m2023.roziere2023.code-llama-34b.humaneval.pass1
Llama 2 70B	HumanEval	0-shot pass@1	30.5	Meta (Code Llama paper, v3)	arxiv_2308.12950 p.6	m2023.roziere2023.llama-2-70b.humaneval.pass1

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	86.4	GPT-4	5-shot	2023-03	m2023.openai2023.gpt-4.mmlu.5shot
HumanEval	71.2	Claude 2	0-shot pass@1, T=1	2023-07	m2023.anthropic-claude2-card.claude-2.humaneval.0shot
GSM8K	92.2	Flan-PaLM 2	8-shot CoT + self-consistency, 40 paths	2023-05	m2023.anil2023.flan-palm-2.gsm8k.8shot-cot-sc40
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	m2022.minerva.minerva-540b.math.maj1at64

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Code Llama: Open Foundation Models for Code, v3, PDF page 6: Table 2: Code Llama pass@ scores on HumanEval and MBPP. The pass@1 scores of our models are.

2023-09

September 2023. 6 verified scores for 2 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
LLaMA 2 13B	MBPP (hand-verified subset, per 'Evaluation Differences' on p4)	3-shot	35.4	Mistral AI (Mistral 7B paper)	arxiv_2310.06825 p.4	`m2023.jiang2023mistral.llama-2-13b.mbpp.3shot`
LLaMA 2 13B	MMLU	5-shot (Mistral pipeline)	55.6	Mistral AI (Mistral 7B paper)	arxiv_2310.06825 p.4	`m2023.jiang2023mistral.llama-2-13b.mmlu.5shot`
Mistral 7B	GSM8K	8-shot maj@8	52.2	Mistral AI (Mistral 7B paper)	arxiv_2310.06825 p.4	`m2023.jiang2023mistral.mistral-7b.gsm8k.8shot-maj8`
Mistral 7B	HumanEval	0-shot	30.5	Mistral AI (Mistral 7B paper)	arxiv_2310.06825 p.4	`m2023.jiang2023mistral.mistral-7b.humaneval.0shot`
Mistral 7B	MATH	4-shot maj@4	13.1	Mistral AI (Mistral 7B paper)	arxiv_2310.06825 p.4	`m2023.jiang2023mistral.mistral-7b.math.4shot-maj4`
Mistral 7B	MMLU	5-shot	60.1	Mistral AI (Mistral 7B paper)	arxiv_2310.06825 p.4	`m2023.jiang2023mistral.mistral-7b.mmlu.5shot`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	86.4	GPT-4	5-shot	2023-03	`m2023.openai2023.gpt-4.mmlu.5shot`
HumanEval	71.2	Claude 2	0-shot pass@1, T=1	2023-07	`m2023.anthropic-claude2-card.claude-2.humaneval.0shot`
GSM8K	92.2	Flan-PaLM 2	8-shot CoT + self-consistency, 40 paths	2023-05	`m2023.anil2023.flan-palm-2.gsm8k.8shot-cot-sc40`
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	`m2022.minerva.minerva-540b.math.maj1at64`

2023-10

October 2023. 6 verified scores for 1 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
LLEMMA 34B	GSM8K	8-shot CoT, greedy	51.5	EleutherAI/Princeton (Azerbayev et al. 2023)	arxiv_2310.10631 p.5	`m2023.azerbayev2023.llemma-34b.gsm8k.8shot-greedy`
LLEMMA 34B	GSM8K	maj@100	69.3	EleutherAI/Princeton (Azerbayev et al. 2023)	arxiv_2310.10631 p.5	`m2023.azerbayev2023.llemma-34b.gsm8k.maj1at100`
LLEMMA 34B	GSM8k+Python (tool use)	pass@1	62.6	EleutherAI/Princeton (Azerbayev et al. 2023)	arxiv_2310.10631 p.6	`m2023.azerbayev2023.llemma-34b.gsm8k-python.pass1`
LLEMMA 34B	MATH	4-example prompt, greedy	25.0	EleutherAI/Princeton (Azerbayev et al. 2023)	arxiv_2310.10631 p.5	`m2023.azerbayev2023.llemma-34b.math.4shot-greedy`
LLEMMA 34B	MATH	maj@256	43.1	EleutherAI/Princeton (Azerbayev et al. 2023)	arxiv_2310.10631 p.5	`m2023.azerbayev2023.llemma-34b.math.maj1at256`
LLEMMA 34B	MMLU-STEM	4-example CoT prompt, greedy	49.0	EleutherAI/Princeton (Azerbayev et al. 2023)	arxiv_2310.10631 p.5	`m2023.azerbayev2023.llemma-34b.mmlu-stem.4shot-greedy`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	86.4	GPT-4	5-shot	2023-03	`m2023.openai2023.gpt-4.mmlu.5shot`
HumanEval	88.4	GPT-4	pass@1 greedy	2023-10	`m2023.liu2023evalplus.gpt-4.humaneval.greedy`
GSM8K	92.2	Flan-PaLM 2	8-shot CoT + self-consistency, 40 paths	2023-05	`m2023.anil2023.flan-palm-2.gsm8k.8shot-cot-sc40`
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	`m2022.minerva.minerva-540b.math.maj1at64`

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Mistral 7B, v1, PDF page 3: Figure 2: Rolling buffer cache.; Figure 3: Pre-fill and chunking.
- Mistral 7B, v1, PDF page 4: Figure 4: Performance of Mistral 7B and different Llama models on a wide range of benchmarks.; Table 2: Comparison of Mistral 7B with Llama.; Table 3: Comparison of Chat models.

2023-11

November 2023. 12 verified scores for 4 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Falcon-180B	MMLU	5-shot	70.6	TII (Falcon series paper, Table 19)	arxiv_2311.16867 p.34	m2023.almazrouei2023.falcon-180b.mmlu.5shot
GPT-4	GPQA Diamond	zero-shot CoT	35.7	Rein et al. 2023 (GPQA paper)	arxiv_2311.12022 p.19	m2023.rein2023.gpt-4.gpqa-diamond.0shot-cot
GPT-4	MATH	4-shot (as labelled by xAI)	42.5	xAI (Announcing Grok blog)	xai_grok1_blog html	m2023.xai-grok-blog.gpt-4.math.4shot
GPT-4	MMLU	5-shot	86.5	TII (Falcon series paper, Table 19), quoting GPT-4 paper	arxiv_2311.16867 p.34	m2023.almazrouei2023.gpt-4.mmlu.5shot-as-quoted
Grok-1	GSM8K	8-shot (chain-of-thought prompt)	62.9	xAI (Announcing Grok blog, Nov 3, 2023)	xai_grok1_blog html	m2023.xai-grok-blog.grok-1.gsm8k
Grok-1	HumanEval	0-shot pass@1	63.2	xAI (Announcing Grok blog, Nov 3, 2023)	xai_grok1_blog html	m2023.xai-grok-blog.grok-1.humaneval
Grok-1	MATH	4-shot	23.9	xAI (Announcing Grok blog, Nov 3, 2023)	xai_grok1_blog html	m2023.xai-grok-blog.grok-1.math
Grok-1	MMLU	5-shot	73.0	xAI (Announcing Grok blog, Nov 3, 2023)	xai_grok1_blog html	m2023.xai-grok-blog.grok-1.mmlu
Qwen-72B (base)	GSM8K	8-shot	78.9	Alibaba Qwen (Hugging Face model card, capture 2023-11-30)	hf_qwen_72b_card html	m2023.qwen-72b-card.qwen-72b.gsm8k.8shot
Qwen-72B (base)	HumanEval	0-shot	35.4	Alibaba Qwen (Hugging Face model card, capture 2023-11-30)	hf_qwen_72b_card html	m2023.qwen-72b-card.qwen-72b.humaneval.0shot
Qwen-72B (base)	MATH	4-shot	35.2	Alibaba Qwen (Hugging Face model card, capture 2023-11-30)	hf_qwen_72b_card html	m2023.qwen-72b-card.qwen-72b.math.4shot
Qwen-72B (base)	MMLU	5-shot	77.4	Alibaba Qwen (Hugging Face model card, capture 2023-11-30)	hf_qwen_72b_card html	m2023.qwen-72b-card.qwen-72b.mmlu.5shot

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	86.4	GPT-4	5-shot	2023-03	m2023.openai2023.gpt-4.mmlu.5shot
HumanEval	88.4	GPT-4	pass@1 greedy	2023-10	m2023.liu2023evalplus.gpt-4.humaneval.greedy
GSM8K	92.2	Flan-PaLM 2	8-shot CoT + self-consistency, 40 paths	2023-05	m2023.anil2023.flan-palm-2.gsm8k.8shot-cot-sc40
MATH	50.3	Minerva 540B	fixed 4-shot, maj1@64	2022-06	m2022.minerva.minerva-540b.math.maj1at64
GPQA Diamond	35.7	GPT-4	zero-shot CoT	2023-11	m2023.rein2023.gpt-4.gpqa-diamond.0shot-cot

2023-12

December 2023. 38 verified scores for 11 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
GPT-4	GSM8K	SFT & 5-shot CoT	92.0	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.gpt-4.gsm8k.sft-5shot-cot
GPT-4	HumanEval	0-shot (reported)	67.0	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.gpt-4.humaneval.0shot-reported
GPT-4	MATH	4-shot (via API, Nov 2023)	52.9	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.gpt-4.math.4shot-api
GPT-4	MATH	as reported by Zheng et al. (2023)	50.3	Gemini Team (Gemini 1.0 TR, Table 2), quoting Zheng et al. 2023	arxiv_2312.11805 p.8	m2023.gemini2023.gpt-4.math.zheng2023
GPT-4	MMLU	5-shot (reported)	86.4	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.gpt-4.mmlu.5shot-reported
GPT-4	MMLU	CoT@32 (via API, Nov 2023)	87.29	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.gpt-4.mmlu.cot32-api

GPT-4 (gpt-4-0613)	MMLU	CoT@32 (and uncertainty-routed CoT@32)	87.29	Gemini Team (Gemini 1.0 TR, Figure 9)	arxiv_2312.11805 p.74	m2023.gemini2023.gpt-4-0613.mmlu.cot32-fig9
Gemini Ultra	GSM8K	Maj1@32	94.4	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.gemini-ultra.gsm8k.maj1at32
Gemini Ultra	MATH	4-shot	53.2	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.gemini-ultra.math.4shot
Gemini Ultra	MMLU	5-shot	83.7	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.gemini-ultra.mmlu.5shot
Gemini Ultra	MMLU	CoT@32 uncertainty-routed	90.04	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.gemini-ultra.mmlu.cot32-uncertainty
Gemini Ultra	MMLU	uncertainty-routed CoT@32	90.04	Gemini Team (Gemini 1.0 TR, sec. 5.1.1)	arxiv_2312.11805 p.6	m2023.gemini2023.gemini-ultra.mmlu.text-claim
Gemini Ultra	MMLU	not stated in text (blog)	90.0	Google (Introducing Gemini blog, Dec 06, 2023)	google_gemini1_blog html	m2023.google-gemini-blog.gemini-ultra.mmlu
Gemini Ultra (post-trained API model)	HumanEval	0-shot (PT)	74.4	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.gemini-ultra.humaneval.0shot-pt
Grok 1	HumanEval	0-shot	63.2	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.grok-1.humaneval.0shot
Grok 1	MATH	4-shot	23.9	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.grok-1.math.4shot
Grok 1	MMLU	5-shot	73.0	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.grok-1.mmlu.5shot
LLAMA-2	GSM8K	5-shot (as labelled by Gemini TR)	56.8	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.llama-2.gsm8k.5shot
LLAMA-2	HumanEval	0-shot	29.9	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.llama-2.humaneval.0shot
LLAMA-2	MATH	4-shot	13.5	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.llama-2.math.4shot
LLAMA-2 (70B per Touvron et al. 2023b)	MMLU	decontaminated numbers from Touvron et al. (2023b)	68.0	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.llama-2.mmlu.decontaminated
LLaMA 2 70B	GSM8K	5-shot	53.6	Mistral AI (Mixtral paper, Table 3)	arxiv_2401.04088 p.5	m2023.jiang2024mixtral.llama-2-70b.gsm8k.5shot
LLaMA 2 70B	GSM8K	8-shot maj@8	69.6	Mistral AI (Mixtral paper, Table 2)	arxiv_2401.04088 p.4	m2023.jiang2024mixtral.llama-2-70b.gsm8k.8shot-maj8
LLaMA 2 70B	HumanEval	0-shot	29.3	Mistral AI (Mixtral paper, Table 2)	arxiv_2401.04088 p.4	m2023.jiang2024mixtral.llama-2-70b.humaneval.0shot
LLaMA 2 70B	MATH	4-shot maj@4	13.8	Mistral AI (Mixtral paper, Table 2)	arxiv_2401.04088 p.4	m2023.jiang2024mixtral.llama-2-70b.math.4shot-maj4
LLaMA 2 70B	MMLU	5-shot	69.9	Mistral AI (Mixtral paper, Table 2)	arxiv_2401.04088 p.4	m2023.jiang2024mixtral.llama-2-70b.mmlu.5shot
Mistral 7B	GSM8K	8-shot maj@8	50.0	Mistral AI (Mixtral paper, Table 2)	arxiv_2401.04088 p.4	m2023.jiang2024mixtral.mistral-7b.gsm8k.8shot-maj8
Mistral 7B	HumanEval	0-shot	26.2	Mistral AI (Mixtral paper, Table 2)	arxiv_2401.04088 p.4	m2023.jiang2024mixtral.mistral-7b.humaneval.0shot
Mistral 7B	MATH	4-shot maj@4	12.7	Mistral AI (Mixtral paper, Table 2)	arxiv_2401.04088 p.4	m2023.jiang2024mixtral.mistral-7b.math.4shot-maj4
Mistral 7B	MMLU	5-shot	62.5	Mistral AI (Mixtral paper, Table 2)	arxiv_2401.04088 p.4	m2023.jiang2024mixtral.mistral-7b.mmlu.5shot
Mixtral 8x7B	GSM8K	5-shot	58.4	Mistral AI (Mixtral paper, Table 3)	arxiv_2401.04088 p.5	m2023.jiang2024mixtral.mixtral-8x7b.gsm8k.5shot
Mixtral 8x7B	GSM8K	8-shot maj@8	74.4	Mistral AI (Mixtral paper, Table 2)	arxiv_2401.04088 p.4	m2023.jiang2024mixtral.mixtral-8x7b.gsm8k.8shot-maj8
Mixtral 8x7B	HumanEval	0-shot	40.2	Mistral AI (Mixtral paper, Table 2)	arxiv_2401.04088 p.4	m2023.jiang2024mixtral.mixtral-8x7b.humaneval.0shot
Mixtral 8x7B	MATH	4-shot maj@4	28.4	Mistral AI (Mixtral paper, Table 2)	arxiv_2401.04088 p.4	m2023.jiang2024mixtral.mixtral-8x7b.math.4shot-maj4
Mixtral 8x7B	MMLU	5-shot	70.6	Mistral AI (Mixtral paper, Table 2)	arxiv_2401.04088 p.4	m2023.jiang2024mixtral.mixtral-8x7b.mmlu.5shot
PaLM 2-L	GSM8K	5-shot	80.0	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.palm-2-l.gsm8k.5shot
PaLM 2-L	MATH	4-shot	34.4	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	m2023.gemini2023.palm-2-l.math.4shot

PaLM 2-L	MMLU	5-shot	78.4	Gemini Team (Gemini 1.0 TR, Table 2)	arxiv_2312.11805 p.8	`m2023.gemini2023.palm-2-l.mmlu.5shot`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	90.04	Gemini Ultra	CoT@32 uncertainty-routed	2023-12	`m2023.gemini2023.gemini-ultra.mmlu.cot32-uncertainty`
HumanEval	88.4	GPT-4	pass@1 greedy	2023-10	`m2023.liu2023evalplus.gpt-4.humaneval.greedy`
GSM8K	94.4	Gemini Ultra	Maj1@32	2023-12	`m2023.gemini2023.gemini-ultra.gsm8k.maj1at32`
MATH	53.2	Gemini Ultra	4-shot	2023-12	`m2023.gemini2023.gemini-ultra.math.4shot`
GPQA Diamond	35.7	GPT-4	zero-shot CoT	2023-11	`m2023.rein2023.gpt-4.gpqa-diamond.0shot-cot`

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Gemini: A Family of Highly Capable Multimodal Models, v5, PDF page 8: Table 2: Gemini performance on text benchmarks with external comparisons and PaLM 2-L.

2024-01

January 2024. 19 verified scores for 6 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Code Llama - Instruct 70B	HumanEval	0-shot pass@1, greedy	67.8	Meta (Code Llama report v3, Table 2)	arxiv_2308.12950 p.6	`m2024.codellama.code-llama-instruct-70b.humaneval.pass1`
DeepSeek 67B (Base)	MMLU	5-shot	71.3	DeepSeek (DeepSeek LLM report, base-model table)	arxiv_2401.02954 p.15	`m2024.dsllm.deepseek-67b-base.mmlu.5-shot`
DeepSeek 67B Chat	GSM8K	0-shot	84.1	DeepSeek (DeepSeek LLM report, Table 6)	arxiv_2401.02954 p.16	`m2024.dsllm.deepseek-67b-chat.gsm8k.0-shot`
DeepSeek 67B Chat	GSM8K	8-shot	84.1	DeepSeek (DeepSeek-V2 report, Table 3)	arxiv_2405.04434 p.19	`m2024.dsv2.deepseek-67b-chat.gsm8k.8-shot`
DeepSeek 67B Chat	HumanEval	not stated in Table 6 caption (HumanEval is not in the 0-shot list)	73.8	DeepSeek (DeepSeek LLM report, Table 6)	arxiv_2401.02954 p.16	`m2024.dsllm.deepseek-67b-chat.humaneval.0-shot`
DeepSeek 67B Chat	MATH	0-shot	32.6	DeepSeek (DeepSeek LLM report, Table 6)	arxiv_2401.02954 p.16	`m2024.dsllm.deepseek-67b-chat.math.0-shot`
DeepSeek 67B Chat	MMLU	0-shot	71.1	DeepSeek (DeepSeek LLM report, Table 6)	arxiv_2401.02954 p.16	`m2024.dsllm.deepseek-67b-chat.mmlu.0-shot`
DeepSeek-Coder-Instruct 33B	HumanEval (Python)	0-shot, greedy	79.3	DeepSeek (Guo et al. 2024)	arxiv_2401.14196 p.11	`m2023.guo2024.deepseek-coder-instruct-33b.humaneval.python`
GPT-4 (0125)	GPQA	0-shot CoT (subset not stated in Table 2)	41.4	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	`m2024.llama3herd.gpt-4-0125.gpqa.0-shot-cot`
GPT-4 (0125)	GSM8K	8-shot CoT	94.2	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	`m2024.llama3herd.gpt-4-0125.gsm8k.8-shot-cot`
GPT-4 (0125)	HumanEval	0-shot	86.6	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	`m2024.llama3herd.gpt-4-0125.humaneval.0-shot`
GPT-4 (0125)	MATH	0-shot CoT	64.5	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	`m2024.llama3herd.gpt-4-0125.math.0-shot-cot`

GPT-4 (0125)	MMLU	0-shot CoT	85.4	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.gpt-4-0125.mmlu.0-shot-cot
GPT-4 (0125)	MMLU	5-shot	85.1	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.gpt-4-0125.mmlu.5-shot
GPT-4 (0125)	MMLU-Pro	5-shot CoT	64.8	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.gpt-4-0125.mmlu-pro.5-shot-cot
gpt-4-0125-preview	GPQA	0-shot CoT, 'assistant' system prompt (simple-evals)	41.4	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	m2024.se0513.gpt-4-0125-preview.gpqa.0-shot-cot-assistant
gpt-4-0125-preview	HumanEval	0-shot CoT, 'assistant' system prompt (simple-evals)	86.6	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	m2024.se0513.gpt-4-0125-preview.humaneval.0-shot-cot-assistant
gpt-4-0125-preview	MATH	0-shot CoT, 'assistant' system prompt (simple-evals)	64.5	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	m2024.se0513.gpt-4-0125-preview.math.0-shot-cot-assistant
gpt-4-0125-preview	MMLU	0-shot CoT, 'assistant' system prompt (simple-evals)	85.4	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	m2024.se0513.gpt-4-0125-preview.mmlu.0-shot-cot-assistant

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	90.04	Gemini Ultra	CoT@32 uncertainty-routed	2023-12	m2023.gemini2023.gemini-ultra.mmlu.cot32-uncertainty
HumanEval	88.4	GPT-4	pass@1 greedy	2023-10	m2023.liu2023evalplus.gpt-4.humaneval.greedy
GSM8K	94.4	Gemini Ultra	Maj1@32	2023-12	m2023.gemini2023.gemini-ultra.gsm8k.maj1at32
MATH	53.2	Gemini Ultra	4-shot	2023-12	m2023.gemini2023.gemini-ultra.math.4shot
GPQA Diamond	35.7	GPT-4	zero-shot CoT	2023-11	m2023.rein2023.gpt-4.gpqa-diamond.0shot-cot

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Mixtral of Experts, v1, PDF page 4: Table 2: Comparison of Mixtral with Llama.; Figure 3: Results on MMLU, commonsense reasoning, world knowledge and reading comprehension,.
- Mixtral of Experts, v1, PDF page 5: Table 3: Comparison of Mixtral with Llama 2 70B and GPT-3.5.; Table 4: Comparison of Mixtral with Llama on Multilingual Benchmarks.; Figure 4: Long range performance of Mixtral.

2024-02

February 2024. 13 verified scores for 5 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
DeepSeekMath 7B (variant not named in the sentence)	MATH	self-consistency over 64 samples	60.9	DeepSeek (DeepSeekMath report, abstract)	arxiv_2402.03300 p.1	m2024.dsmath.deepseekmath-7b.math.sc64
DeepSeekMath-RL 7B	GSM8K	chain-of-thought, Top1	88.2	DeepSeek (DeepSeekMath report, Table 5)	arxiv_2402.03300 p.12	m2024.dsmath.deepseekmath-rl-7b.gsm8k.top1-cot
DeepSeekMath-RL 7B	MATH	chain-of-thought, Top1	51.7	DeepSeek (DeepSeekMath report, Table 5)	arxiv_2402.03300 p.12	m2024.dsmath.deepseekmath-rl-7b.math.top1-cot
Gemini 1.5 Pro	MATH	4-shot	67.7	Anthropic, quoting Google [9]	anthropic_claude35_addendum p.2	m2024.c35jun.gemini-1.5-pro.math.4-shot
Gemini 1.5 Pro	MMLU	5-shot	81.9	Anthropic, quoting the Gemini 1.5 report	anthropic_claude3_card p.6	m2024.c3card.gemini-1.5-pro.mmlu.5-shot
Gemini 1.5 Pro (Feb 2024)	GPQA	0-shot (subset not stated)	41.5	Google DeepMind (Gemini 1.5 report v1, Table 8)	arxiv_2403.05530v1 p.21	m2024.gem15v1.gemini-1.5-pro.gpqa.0-shot

Gemini 1.5 Pro (Feb 2024)	GSM8K	11-shot	91.7	Google DeepMind (Gemini 1.5 report v1, Table 8)	arxiv_2403.05530v1 p.21	m2024.gem15v1.gemini-1.5-pro.gsm8k.11-shot
Gemini 1.5 Pro (Feb 2024)	HumanEval	0-shot	71.9	Google DeepMind (Gemini 1.5 report v1, Table 8)	arxiv_2403.05530v1 p.21	m2024.gem15v1.gemini-1.5-pro.humaneval.0-shot
Gemini 1.5 Pro (Feb 2024)	MATH	4-shot Minerva prompt	58.5	Google DeepMind (Gemini 1.5 report v1, Table 8)	arxiv_2403.05530v1 p.21	m2024.gem15v1.gemini-1.5-pro.math.4-shot-minerva
Gemini 1.5 Pro (Feb 2024)	MMLU	5-shot	81.9	Google DeepMind (Gemini 1.5 report v1, Table 8)	arxiv_2403.05530v1 p.21	m2024.gem15v1.gemini-1.5-pro.mmlu.5-shot
Mistral Large	GSM8K	5-shot	81.0	xAI, reporting Mistral Large	xai_grok15_blog html	m2024.grok15.mistral-large.gsm8k.5-shot
Mistral Large	HumanEval	0-shot	45.1	xAI, reporting Mistral Large	xai_grok15_blog html	m2024.grok15.mistral-large.humaneval.0-shot
Mistral Large	MMLU	5-shot	81.2	xAI, reporting Mistral Large	xai_grok15_blog html	m2024.grok15.mistral-large.mmlu.5-shot

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	90.04	Gemini Ultra	CoT@32 uncertainty-routed	2023-12	m2023.gemini2023.gemini-ultra.mmlu.cot32-uncertainty
HumanEval	88.4	GPT-4	pass@1 greedy	2023-10	m2023.liu2023evalplus.gpt-4.humaneval.greedy
GSM8K	94.4	Gemini Ultra	Maj1@32	2023-12	m2023.gemini2023.gemini-ultra.gsm8k.maj1at32
MATH	60.9	DeepSeekMath 7B (variant not named in the sentence)	self-consistency over 64 samples	2024-02	m2024.dsmath.deepseekmath-7b.math.sc64
GPQA Diamond	35.7	GPT-4	zero-shot CoT	2023-11	m2023.rein2023.gpt-4.gpqa-diamond.0shot-cot

2024-03

March 2024. 13 verified scores for 3 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Claude 3 Opus	GPQA Diamond	0-shot CoT (mean of 10 rollouts)	50.4	Anthropic (Claude 3 model card, Table 1)	anthropic_claude3_card p.6	m2024.c3card.claude-3-opus.gpqa-diamond.0-shot-cot
Claude 3 Opus	GSM8K	0-shot CoT	95.0	Anthropic (Claude 3 model card, Table 1)	anthropic_claude3_card p.6	m2024.c3card.claude-3-opus.gsm8k.0-shot-cot
Claude 3 Opus	HumanEval	0-shot	84.9	Anthropic (Claude 3 model card, Table 1)	anthropic_claude3_card p.6	m2024.c3card.claude-3-opus.humaneval.0-shot
Claude 3 Opus	MATH	0-shot (CoT per footnote 5)	60.1	Anthropic (Claude 3 model card, Table 1)	anthropic_claude3_card p.6	m2024.c3card.claude-3-opus.math.0-shot-cot
Claude 3 Opus	MMLU	5-shot	86.8	Anthropic (Claude 3 model card, Table 1)	anthropic_claude3_card p.6	m2024.c3card.claude-3-opus.mmlu.5-shot
Claude 3 Opus	MMLU	5-shot CoT	88.2	Anthropic (Claude 3 model card, Table 1)	anthropic_claude3_card p.6	m2024.c3card.claude-3-opus.mmlu.5-shot-cot
GPT-4	GPQA Diamond	0-shot CoT	35.7	Anthropic (Claude 3 model card), quoting Rein et al. 2023 [1]	anthropic_claude3_card p.6	m2023.anthropic-claude3-card.gpt-4.gpqa-diamond.0shot-cot
GPT-4	MATH	0-shot	42.5	Anthropic (Claude 3 model card), quoting Bubeck et al. 2023 [39]	anthropic_claude3_card p.6	m2023.anthropic-claude3-card.gpt-4.math.0shot
GPT-4	MATH	4-shot CoT	52.9	Anthropic (Claude 3 model card), quoting Gemini TR	anthropic_claude3_card p.6	m2023.anthropic-claude3-card.gpt-4.math.4shot
Grok-1.5	GSM8K	8-shot, maj@1	90.0	xAI (Grok-1.5 post)	xai_grok15_blog html	m2024.grok15.grok-1.5.gsm8k.8-shot
Grok-1.5	HumanEval	0-shot pass@1	74.1	xAI (Grok-1.5 post)	xai_grok15_blog html	m2024.grok15.grok-1.5.humaneval.0-shot
Grok-1.5	MATH	4-shot, maj@1	50.6	xAI (Grok-1.5 post)	xai_grok15_blog html	m2024.grok15.grok-1.5.math.4-shot
Grok-1.5	MMLU	5-shot	81.3	xAI (Grok-1.5 post)	xai_grok15_blog html	m2024.grok15.grok-1.5.mmlu.5-shot

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	90.04	Gemini Ultra	CoT@32 uncertainty-routed	2023-12	`m2023.gemini2023.gemini-ultra.mmlu.cot32-uncertainty`
HumanEval	88.4	GPT-4	pass@1 greedy	2023-10	`m2023.liu2023evalplus.gpt-4.humaneval.greedy`
GSM8K	95.0	Claude 3 Opus	0-shot CoT	2024-03	`m2024.c3card.claude-3-opus.gsm8k.0-shot-cot`
MATH	60.9	DeepSeekMath 7B (variant not named in the sentence)	self-consistency over 64 samples	2024-02	`m2024.dsmath.deepseekmath-7b.math.sc64`
GPQA Diamond	50.4	Claude 3 Opus	0-shot CoT (mean of 10 rollouts)	2024-03	`m2024.c3card.claude-3-opus.gpqa-diamond.0-shot-cot`

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Anthropic Claude 3 Model Card (March 2024), None, PDF page 6: Table 1: We show evaluation results for reasoning, math, coding, reading comprehension, and question.
- Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context, v5, PDF page 30: Table 11: Evaluation results of Gemini 1.5 Pro, 1.5 Flash and Gemini 1.0 models on standard coding,.

2024-04

April 2024. 19 verified scores for 5 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Llama 3 70B (instruction tuned)	GPQA	0-shot	39.5	Meta (Llama 3 MODEL_CARD.md, commit 92a325e)	meta_llama3_model_card_92a325e html	`m2024.llama3card.llama-3-70b-instruct.gpqa.0-shot`
Llama 3 70B (instruction tuned)	GSM8K	8-shot CoT	93.0	Meta (Llama 3 MODEL_CARD.md, commit 92a325e)	meta_llama3_model_card_92a325e html	`m2024.llama3card.llama-3-70b-instruct.gsm8k.8-shot-cot`
Llama 3 70B (instruction tuned)	HumanEval	0-shot	81.7	Meta (Llama 3 MODEL_CARD.md, commit 92a325e)	meta_llama3_model_card_92a325e html	`m2024.llama3card.llama-3-70b-instruct.humaneval.0-shot`
Llama 3 70B (instruction tuned)	MATH	4-shot CoT	50.4	Meta (Llama 3 MODEL_CARD.md, commit 92a325e)	meta_llama3_model_card_92a325e html	`m2024.llama3card.llama-3-70b-instruct.math.4-shot-cot`
Llama 3 70B (instruction tuned)	MMLU	5-shot	82.0	Meta (Llama 3 MODEL_CARD.md, commit 92a325e)	meta_llama3_model_card_92a325e html	`m2024.llama3card.llama-3-70b-instruct.mmlu.5-shot`
Mixtral 8x22B (instructed)	GSM8K	maj@8	90.8	Mistral AI (Mixtral 8x22B post)	mistral_mixtral8x22b_blog html	`m2024.mixtral8x22b.mixtral-8x22b-instruct.gsm8k.maj8`
Mixtral 8x22B (instructed)	MATH	maj@4	44.6	Mistral AI (Mixtral 8x22B post)	mistral_mixtral8x22b_blog html	`m2024.mixtral8x22b.mixtral-8x22b-instruct.math.maj4`
Mixtral 8x22B Inst.	HumanEval	0-shot	75.0	DeepSeek (DeepSeek-V2 report, Table 3)	arxiv_2405.04434 p.19	`m2024.dsv2.mixtral-8x22b-instruct.humaneval.0-shot`
Mixtral 8x22B Inst.	MMLU	5-shot	77.8	DeepSeek (DeepSeek-V2 report, Table 3)	arxiv_2405.04434 p.19	`m2024.dsv2.mixtral-8x22b-instruct.mmlu.5-shot`
Phi-3-small 7b	GSM8K	0-shot CoT	88.9	Microsoft (Phi-3 report v1, Apr 2024)	arxiv_2404.14219 p.5	`m2024.phi3v1.phi-3-small.gsm8k.0-shot-cot`
Phi-3-small 7b	GSM8K	8-shot CoT	89.6	Microsoft (Phi-3 report v4, Aug 2024)	arxiv_2404.14219v4 p.6	`m2024.phi3v4.phi-3-small.gsm8k.8-shot-cot`
gpt-4-turbo-2024-04-09	GPQA	0-shot CoT, 'assistant' system prompt (simple-evals)	49.3	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	`m2024.se0513.gpt-4-turbo-2024-04-09.gpqa.0-shot-cot-assistant`
gpt-4-turbo-2024-04-09	GPQA	0-shot CoT, 'chatgpt' system prompt (simple-evals)	49.1	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	`m2024.se0513.gpt-4-turbo-2024-04-09.gpqa.0-shot-cot-chatgpt`
gpt-4-turbo-2024-04-09	HumanEval	0-shot CoT, 'assistant' system prompt (simple-evals)	88.2	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	`m2024.se0513.gpt-4-turbo-2024-04-09.humaneval.0-shot-cot-assistant`
gpt-4-turbo-2024-04-09	HumanEval	0-shot CoT, 'chatgpt' system prompt (simple-evals)	87.6	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	`m2024.se0513.gpt-4-turbo-2024-04-09.humaneval.0-shot-cot-chatgpt`

gpt-4-turbo-2024-04-09	MATH	0-shot CoT, 'assistant' system prompt (simple-evals)	73.4	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	m2024.se0513.gpt-4-turbo-2024-04-09.math.0-shot-cot-assistant
gpt-4-turbo-2024-04-09	MATH	0-shot CoT, 'chatgpt' system prompt (simple-evals)	72.2	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	m2024.se0513.gpt-4-turbo-2024-04-09.math.0-shot-cot-chatgpt
gpt-4-turbo-2024-04-09	MMLU	0-shot CoT, 'assistant' system prompt (simple-evals)	86.7	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	m2024.se0513.gpt-4-turbo-2024-04-09.mmlu.0-shot-cot-assistant
gpt-4-turbo-2024-04-09	MMLU	0-shot CoT, 'chatgpt' system prompt (simple-evals)	86.5	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	m2024.se0513.gpt-4-turbo-2024-04-09.mmlu.0-shot-cot-chatgpt

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	90.04	Gemini Ultra	CoT@32 uncertainty-routed	2023-12	m2023.gemini2023.gemini-ultra.mmlu.cot32-uncertainty
HumanEval	88.4	GPT-4	pass@1 greedy	2023-10	m2023.liu2023evalplus.gpt-4.humaneval.greedy
GSM8K	95.0	Claude 3 Opus	0-shot CoT	2024-03	m2024.c3card.claude-3-opus.gsm8k.0-shot-cot
MATH	60.9	DeepSeekMath 7B (variant not named in the sentence)	self-consistency over 64 samples	2024-02	m2024.dsmath.deepseekmath-7b.math.sc64
GPQA Diamond	50.4	Claude 3 Opus	0-shot CoT (mean of 10 rollouts)	2024-03	m2024.c3card.claude-3-opus.gpqa-diamond.0-shot-cot

2024-05

May 2024. 36 verified scores for 6 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
DeepSeek-V2 Chat (RL)	GSM8K	8-shot	92.2	DeepSeek (DeepSeek-V2 report, Table 3)	arxiv_2405.04434 p.19	m2024.dsv2.deepseek-v2-chat-rl.gsm8k.8-shot
DeepSeek-V2 Chat (RL)	HumanEval	0-shot	81.1	DeepSeek (DeepSeek-V2 report, Table 3)	arxiv_2405.04434 p.19	m2024.dsv2.deepseek-v2-chat-rl.humaneval.0-shot
DeepSeek-V2 Chat (RL)	MATH	4-shot	53.9	DeepSeek (DeepSeek-V2 report, Table 3)	arxiv_2405.04434 p.19	m2024.dsv2.deepseek-v2-chat-rl.math.4-shot
GPT-4o	AIME 2024	0-shot CoT (as labelled by Anthropic)	9.3	Anthropic, quoting OpenAI [9]	anthropic_claude35_oct_addendum p.6	m2024.c35oct.gpt-4o.aime-2024.0-shot-cot
GPT-4o	GPQA	0-shot CoT (subset not stated in Table 2)	53.6	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.gpt-4o.gpqa.0-shot-cot
GPT-4o	GSM8K	8-shot CoT	96.1	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.gpt-4o.gsm8k.8-shot-cot
GPT-4o	HumanEval	0-shot	90.2	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.gpt-4o.humaneval.0-shot
GPT-4o	MATH	0-shot CoT	76.6	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.gpt-4o.math.0-shot-cot
GPT-4o	MMLU	0-shot CoT	88.7	OpenAI (Hello GPT-4o post)	openai_gpt4o_blog html	m2024.gpt4oblog.gpt-4o.mmlu.0-shot-cot
GPT-4o	MMLU	5-shot, no CoT	87.2	OpenAI (Hello GPT-4o post)	openai_gpt4o_blog html	m2024.gpt4oblog.gpt-4o.mmlu.5-shot-no-cot
GPT-4o	MMLU	0-shot CoT	88.7	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.gpt-4o.mmlu.0-shot-cot
GPT-4o	MMLU	5-shot	89.1	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.gpt-4o.mmlu.5-shot

GPT-4o	MMLU-Pro	as reported by xAI ('May 2024 release')	72.6	xAI (Grok-2 post); origin not stated, footnote only says the scores are for the May 2024 release of GPT-4o	xai_grok2_blog html	m2024.grok2.gpt-4o.mmlu-pro.xai
GPT-4o	MMLU-Pro	5-shot CoT	74.0	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.gpt-4o.mmlu-pro.5-shot-cot
GPT-4o 0513	AIME 2024	pass@1, temperature 0.7, mean of 16 runs	9.3	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.gpt-4o-0513.aime-2024.pass@1
GPT-4o 0513	GPQA Diamond	pass@1	49.9	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.gpt-4o-0513.gpqa-diamond.pass@1
GPT-4o 0513	MATH-500	EM, greedy decoding	74.6	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.gpt-4o-0513.math-500.em
GPT-4o 0513	MMLU	EM (shots not stated for chat models)	87.2	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.gpt-4o-0513.mmlu.em
GPT-4o 0513	MMLU-Pro	EM	72.6	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.gpt-4o-0513.mmlu-pro.em
GPT-4o 0513	SWE-bench Verified	resolved, Agentless framework	38.8	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.gpt-4o-0513.swe-bench-verified.resolved
Gemini 1.5 Pro (May 2024)	HumanEval	0-shot	84.1	Google DeepMind (Gemini 1.5 report v5, Table 11)	arxiv_2403.05530 p.30	m2024.gem15v5.gemini-1.5-pro-may.humaneval.0-shot
Gemini 1.5 Pro (May 2024)	MATH	4-shot Minerva prompt (Table 11)	67.7	Google DeepMind (Gemini 1.5 report v5)	arxiv_2403.05530 p.4	m2024.gem15v5.gemini-1.5-pro-may.math.4-shot
Gemini 1.5 Pro (May 2024)	MMLU	5-shot	85.9	Google DeepMind (Gemini 1.5 report v5, Table 11)	arxiv_2403.05530 p.30	m2024.gem15v5.gemini-1.5-pro-may.mmlu.5-shot
gpt-4-1106(-vision)-preview	GPQA (subset not stated in README)	0-shot CoT, 'assistant' system prompt	42.5	OpenAI (simple-evals README, commit 294cb1f)	openai_simple_evals_readme_2024-05-13 md	m2023.simple-evals-2024-05-13.gpt-4-1106-preview.gpqa.0shot-cot-assistant
gpt-4-1106(-vision)-preview	HumanEval	0-shot CoT, 'assistant' system prompt	83.7	OpenAI (simple-evals README, commit 294cb1f)	openai_simple_evals_readme_2024-05-13 md	m2023.simple-evals-2024-05-13.gpt-4-1106-preview.humaneval.0shot-cot-assistant
gpt-4-1106(-vision)-preview	MATH	0-shot CoT, 'assistant' system prompt	64.3	OpenAI (simple-evals README, commit 294cb1f)	openai_simple_evals_readme_2024-05-13 md	m2023.simple-evals-2024-05-13.gpt-4-1106-preview.math.0shot-cot-assistant
gpt-4-1106(-vision)-preview	MMLU	0-shot CoT, 'assistant' system prompt	84.7	OpenAI (simple-evals README, commit 294cb1f)	openai_simple_evals_readme_2024-05-13 md	m2023.simple-evals-2024-05-13.gpt-4-1106-preview.mmlu.0shot-cot-assistant
gpt-4o	AIME 2024	pass@1	9.3	OpenAI (o1 launch post, Appendix A)	openai_o1_blog html	m2024.o1blog.gpt-4o.aime-2024.pass1
gpt-4o	GPQA	0-shot CoT, 'assistant' system prompt (simple-evals)	49.9	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	m2024.se0513.gpt-4o.gpqa.0-shot-cot-assistant
gpt-4o	GPQA	0-shot CoT, 'chatgpt' system prompt (simple-evals)	53.6	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	m2024.se0513.gpt-4o.gpqa.0-shot-cot-chatgpt
gpt-4o	HumanEval	0-shot CoT, 'assistant' system prompt (simple-evals)	91.0	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	m2024.se0513.gpt-4o.humaneval.0-shot-cot-assistant
gpt-4o	HumanEval	0-shot CoT, 'chatgpt' system prompt (simple-evals)	90.2	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	m2024.se0513.gpt-4o.humaneval.0-shot-cot-chatgpt

gpt-4o	MATH	0-shot CoT, 'assistant' system prompt (simple-evals)	76.6	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	m2024.se0513.gpt-4o.math.0-shot-cot-assistant
gpt-4o	MATH	0-shot CoT, 'chatgpt' system prompt (simple-evals)	76.6	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	m2024.se0513.gpt-4o.math.0-shot-cot-chatgpt
gpt-4o	MMLU	0-shot CoT, 'assistant' system prompt (simple-evals)	87.2	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	m2024.se0513.gpt-4o.mmlu.0-shot-cot-assistant
gpt-4o	MMLU	0-shot CoT, 'chatgpt' system prompt (simple-evals)	88.7	OpenAI (simple-evals README, commit 294cb1f, 2024-05-13)	openai_simple_evals_readme_2024-05-13 md	m2024.se0513.gpt-4o.mmlu.0-shot-cot-chatgpt

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	90.04	Gemini Ultra	CoT@32 uncertainty-routed	2023-12	m2023.gemini2023.gemini-ultra.mmlu.cot32-uncertainty
HumanEval	91.0	gpt-4o	0-shot CoT, 'assistant' system prompt (simple-evals)	2024-05	m2024.se0513.gpt-4o.humaneval.0-shot-cot-assistant
GSM8K	95.0	Claude 3 Opus	0-shot CoT	2024-03	m2024.c3card.claude-3-opus.gsm8k.0-shot-cot
MATH	76.6	gpt-4o	0-shot CoT, 'assistant' system prompt (simple-evals)	2024-05	m2024.se0513.gpt-4o.math.0-shot-cot-assistant
GPQA Diamond	50.4	Claude 3 Opus	0-shot CoT (mean of 10 rollouts)	2024-03	m2024.c3card.claude-3-opus.gpqa-diamond.0-shot-cot

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- DeepSeek-V2: A Strong, Economical, and Efficient Mixture-of-Experts Language Model, v5, PDF page 19: Table 3: Comparison among DeepSeek-V2 Chat (SFT), DeepSeek-V2 Chat (RL), and other.; Table 4: English open-ended conversation evaluations.

2024-06

June 2024. 25 verified scores for 4 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Claude 3.5 Sonnet	GPQA	as reported by xAI ('June 2024 release')	59.6	xAI (Grok-2 post); origin not stated, footnote only says the scores are for the June 2024 release of Claude 3.5 Sonnet	xai_grok2_blog html	m2024.grok2.claude-3.5-sonnet.gpqa.xai
Claude 3.5 Sonnet	GPQA	0-shot CoT (subset not stated in Table 2)	59.4	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.claude-3.5-sonnet.gpqa.0-shot-cot
Claude 3.5 Sonnet	GPQA Diamond	0-shot CoT	59.4	Anthropic (Claude 3.5 Sonnet addendum, June 2024, Table 1)	anthropic_claude35_addendum p.2	m2024.c35jun.claude-3.5-sonnet.gpqa-diamond.0-shot-cot
Claude 3.5 Sonnet	GSM8K	0-shot CoT	96.4	Anthropic (Claude 3.5 Sonnet addendum, June 2024, Table 1)	anthropic_claude35_addendum p.2	m2024.c35jun.claude-3.5-sonnet.gsm8k.0-shot-cot
Claude 3.5 Sonnet	GSM8K	0-shot (table marks it with ◊, overriding the row's 8-shot CoT)	96.4	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.claude-3.5-sonnet.gsm8k.0-shot
Claude 3.5 Sonnet	HumanEval	0-shot	92.0	Anthropic (Claude 3.5 Sonnet addendum, June 2024, Table 1)	anthropic_claude35_addendum p.2	m2024.c35jun.claude-3.5-sonnet.humaneval.0-shot
Claude 3.5 Sonnet	HumanEval	0-shot	92.0	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.claude-3.5-sonnet.humaneval.0-shot

Claude 3.5 Sonnet	MATH	0-shot CoT	71.1	Anthropic (Claude 3.5 Sonnet addendum, June 2024, Table 1)	anthropic_claude35_addendum p.2	m2024.c35jun.claude-3.5-sonnet.math.0-shot-cot
Claude 3.5 Sonnet	MATH	0-shot CoT	71.1	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.claude-3.5-sonnet.math.0-shot-cot
Claude 3.5 Sonnet	MMLU	0-shot CoT	88.3	Anthropic (Claude 3.5 Sonnet addendum, June 2024, Table 1)	anthropic_claude35_addendum p.2	m2024.c35jun.claude-3.5-sonnet.mmlu.0-shot-cot
Claude 3.5 Sonnet	MMLU	5-shot	88.7	Anthropic (Claude 3.5 Sonnet addendum, June 2024, Table 1)	anthropic_claude35_addendum p.2	m2024.c35jun.claude-3.5-sonnet.mmlu.5-shot
Claude 3.5 Sonnet	MMLU	5-shot CoT	90.4	Anthropic (Claude 3.5 Sonnet addendum, June 2024, Table 1)	anthropic_claude35_addendum p.2	m2024.c35jun.claude-3.5-sonnet.mmlu.5-shot-cot
Claude 3.5 Sonnet	MMLU	0-shot CoT	88.3	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.claude-3.5-sonnet.mmlu.0-shot-cot
Claude 3.5 Sonnet	MMLU	5-shot	89.9	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.claude-3.5-sonnet.mmlu.5-shot
Claude 3.5 Sonnet	MMLU-Pro	0-shot CoT	75.1	Anthropic (Claude 3.5 Sonnet (New) & Haiku addendum, Oct 2024, Table 7)	anthropic_claude35_oct_addendum p.6	m2024.c35oct.claude-3.5-sonnet.mmlu-pro.0-shot-cot
Claude 3.5 Sonnet	MMLU-Pro	as reported by xAI ('June 2024 release')	76.1	xAI (Grok-2 post); origin not stated, footnote only says the scores are for the June 2024 release	xai_grok2_blog html	m2024.grok2.claude-3.5-sonnet.mmlu-pro.xai
Claude 3.5 Sonnet	MMLU-Pro	5-shot CoT	77.0	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	m2024.llama3herd.claude-3.5-sonnet.mmlu-pro.5-shot-cot
Claude 3.5 Sonnet	SWE-bench Verified	pass@1, Anthropic agent scaffold	33.4	Anthropic (Oct 2024 addendum, Table 2)	anthropic_claude35_oct_addendum p.3	m2024.c35oct.claude-3.5-sonnet.swe-bench-verified.pass1
DS-Coder-V2-Instruct	GSM8K	greedy decoding, no tools or voting	94.9	DeepSeek (DeepSeek-Coder-V2 report, Table 9)	arxiv_2406.11931 p.13	m2024.dscoderv2.deepseek-coder-v2-instruct.gsm8k.greedy
DS-Coder-V2-Instruct	HumanEval	Python, greedy decoding	90.2	DeepSeek (DeepSeek-Coder-V2 report, Table 3)	arxiv_2406.11931 p.8	m2024.dscoderv2.deepseek-coder-v2-instruct.humaneval.greedy
DS-Coder-V2-Instruct	MATH	greedy decoding, no tools or voting	75.7	DeepSeek (DeepSeek-Coder-V2 report, Table 9)	arxiv_2406.11931 p.13	m2024.dscoderv2.deepseek-coder-v2-instruct.math.greedy
Qwen2-72B (base)	MMLU	base model (Qwen2 Table 2)	84.2	Alibaba (Qwen2 report, Table 2)	arxiv_2407.10671 p.9	m2024.qwen2.qwen2-72b-base.mmlu.base
Qwen2-72B-Instruct	GSM8K	not stated in table	93.2	Alibaba (Qwen2 report, Table 6)	arxiv_2407.10671 p.13	m2024.qwen2.qwen2-72b-instruct.gsm8k.instruct
Qwen2-72B-Instruct	HumanEval	not stated in table	86.0	Alibaba (Qwen2 report, Table 6)	arxiv_2407.10671 p.13	m2024.qwen2.qwen2-72b-instruct.humaneval.instruct
Qwen2-72B-Instruct	MATH	not stated in table	69.0	Alibaba (Qwen2 report, Table 6)	arxiv_2407.10671 p.13	m2024.qwen2.qwen2-72b-instruct.math.instruct

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	90.4	Claude 3.5 Sonnet	5-shot CoT	2024-06	m2024.c35jun.claude-3.5-sonnet.mmlu.5-shot-cot
HumanEval	92.0	Claude 3.5 Sonnet	0-shot	2024-06	m2024.c35jun.claude-3.5-sonnet.humaneval.0-shot
GSM8K	96.4	Claude 3.5 Sonnet	0-shot CoT	2024-06	m2024.c35jun.claude-3.5-sonnet.gsm8k.0-shot-cot

MATH	76.6	gpt-4o	0-shot CoT, 'assistant' system prompt (simple-evals)	2024-05	`m2024.se0513.gpt-4o.math.0-shot-cot-assistant`
GPQA Diamond	59.4	Claude 3.5 Sonnet	0-shot CoT	2024-06	`m2024.c35jun.claude-3.5-sonnet.gpqa-diamond.0-shot-cot`

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Anthropic Claude 3.5 Sonnet Model Card Addendum (June 2024), None, PDF page 2: Table 1: This table shows evaluation results for reasoning, math, coding, reading comprehension, and ques-.; Table 2: This table shows evaluation results for multimodal tasks.

2024-07

July 2024. 18 verified scores for 7 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
GPT4o-mini	GSM8K	not stated in table	93.2	Alibaba (Qwen2.5 report, Table 7)	arxiv_2412.15115 p.11	`m2024.qwen25.gpt-4o-mini.gsm8k.qwen-harness`
LLaMA-3.1 405B-Inst.	AIME 2024	pass@1, temperature 0.7, mean of 16 runs	23.3	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	`m2024.dsv3.llama-3.1-405b-instruct.aime-2024.pass@1`
LLaMA-3.1 405B-Inst.	MATH-500	EM, greedy decoding	73.8	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	`m2024.dsv3.llama-3.1-405b-instruct.math-500.em`
LLaMA-3.1 405B-Inst.	SWE-bench Verified	resolved, Agentless framework	24.5	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	`m2024.dsv3.llama-3.1-405b-instruct.swe-bench-verified.resolved`
Llama 3 405B (Llama 3.1 405B Instruct)	GPQA	0-shot CoT (subset not stated in Table 2)	51.1	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	`m2024.llama3herd.llama-3.1-405b.gpqa.0-shot-cot`
Llama 3 405B (Llama 3.1 405B Instruct)	GSM8K	8-shot CoT	96.8	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	`m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot`
Llama 3 405B (Llama 3.1 405B Instruct)	HumanEval	0-shot	89.0	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	`m2024.llama3herd.llama-3.1-405b.humaneval.0-shot`
Llama 3 405B (Llama 3.1 405B Instruct)	MATH	0-shot CoT	73.8	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	`m2024.llama3herd.llama-3.1-405b.math.0-shot-cot`
Llama 3 405B (Llama 3.1 405B Instruct)	MMLU	0-shot CoT	88.6	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	`m2024.llama3herd.llama-3.1-405b.mmlu.0-shot-cot`
Llama 3 405B (Llama 3.1 405B Instruct)	MMLU	5-shot	87.3	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	`m2024.llama3herd.llama-3.1-405b.mmlu.5-shot`
Llama 3 405B (Llama 3.1 405B Instruct)	MMLU-Pro	5-shot CoT	73.3	Meta (Llama 3 Herd of Models, Table 2)	arxiv_2407.21783 p.3	`m2024.llama3herd.llama-3.1-405b.mmlu-pro.5-shot-cot`
Llama 3 405B (pre-trained)	MMLU	pre-trained model; shot count not stated in Table 13	85.2	Meta (Table 13)	arxiv_2407.21783 p.31	`m2024.llama3herd.llama-3.1-405b-base.mmlu.pretrained`
Llama 3.1 405B Instruct	GPQA Diamond	0-shot CoT	49.0	Meta (Llama 3.3 MODEL_CARD.md, commit c908f5b)	meta_llama33_model_card_c908f5b html	`m2024.llama33card.llama-3.1-405b-instruct.gpqa-diamond.0-shot-cot`
Mistral Large 2 (pretrained)	MMLU	not stated	84.0	Mistral AI (Mistral Large 2 post)	mistral_large2_blog html	`m2024.mistrallarge2.mistral-large-2-base.mmlu.unspecified`
gpt-4o-mini-2024-07-18	GPQA	0-shot CoT (simple-evals), assistant system prompt	40.2	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	`m2024.se0709.gpt-4o-mini-2024-07-18.gpqa.0-shot-cot`
gpt-4o-mini-2024-07-18	HumanEval	0-shot CoT (simple-evals), assistant system prompt	87.2	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	`m2024.se0709.gpt-4o-mini-2024-07-18.humaneval.0-shot-cot`

gpt-4o-mini-2024-07-18	MATH	0-shot CoT (simple-evals), assistant system prompt	70.2	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	m2024.se0709.gpt-4o-mini-2024-07-18.math.0-shot-cot
gpt-4o-mini-2024-07-18	MMLU	0-shot CoT (simple-evals), assistant system prompt	82.0	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	m2024.se0709.gpt-4o-mini-2024-07-18.mmlu.0-shot-cot

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	90.4	Claude 3.5 Sonnet	5-shot CoT	2024-06	m2024.c35jun.claude-3.5-sonnet.mmlu.5-shot-cot
MMLU-Pro	77.0	Claude 3.5 Sonnet	5-shot CoT	2024-07	m2024.llama3herd.claude-3.5-sonnet.mmlu-pro.5-shot-cot
HumanEval	92.0	Claude 3.5 Sonnet	0-shot	2024-06	m2024.c35jun.claude-3.5-sonnet.humaneval.0-shot
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot
MATH	76.6	gpt-4o	0-shot CoT, 'assistant' system prompt (simple-evals)	2024-05	m2024.se0513.gpt-4o.math.0-shot-cot-assistant
GPQA Diamond	59.4	Claude 3.5 Sonnet	0-shot CoT	2024-06	m2024.c35jun.claude-3.5-sonnet.gpqa-diamond.0-shot-cot

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Qwen2 Technical Report, v4, PDF page 9: Table 2: Performance of the 70B+ models.
- The Llama 3 Herd of Models, v3, PDF page 3: Table 2: Performance of finetuned Llama 3 models on key benchmark evaluations.

2024-08

August 2024. 11 verified scores for 4 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
GPT-4o (checkpoint not stated)	SWE-bench Verified	best open-source scaffold (Agentless, per the post's scaffold chart)	33.2	OpenAI (Introducing SWE-bench Verified)	openai_swebench_verified_blog html	m2024.swebv.gpt-4o.swe-bench-verified.agentless
Grok-2	GPQA	not stated (subset not stated)	56.0	xAI (Grok-2 post)	xai_grok2_blog html	m2024.grok2.grok-2.gpqa.unspecified
Grok-2	HumanEval	pass@1	88.4	xAI (Grok-2 post)	xai_grok2_blog html	m2024.grok2.grok-2.humaneval.pass1
Grok-2	MATH	maj@1	76.1	xAI (Grok-2 post)	xai_grok2_blog html	m2024.grok2.grok-2.math.maj1
Grok-2	MMLU	0-shot CoT	87.5	xAI (Grok-2 post)	xai_grok2_blog html	m2024.grok2.grok-2.mmlu.0-shot-cot
Grok-2	MMLU-Pro	0-shot CoT	75.5	xAI (Grok-2 post)	xai_grok2_blog html	m2024.grok2.grok-2.mmlu-pro.0-shot-cot
Phi-3.5-MoE 16x3.8B	GSM8K	8-shot CoT	88.7	Microsoft (Phi-3 report v4, Aug 2024, Table 3)	arxiv_2404.14219v4 p.9	m2024.phi3v4.phi-3.5-moe.gsm8k.8-shot-cot
gpt-4o-2024-08-06	GPQA	0-shot CoT (simple-evals), assistant system prompt	53.1	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	m2024.se0709.gpt-4o-2024-08-06.gpqa.0-shot-cot
gpt-4o-2024-08-06	HumanEval	0-shot CoT (simple-evals), assistant system prompt	90.2	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	m2024.se0709.gpt-4o-2024-08-06.humaneval.0-shot-cot
gpt-4o-2024-08-06	MATH	0-shot CoT (simple-evals), assistant system prompt	75.9	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	m2024.se0709.gpt-4o-2024-08-06.math.0-shot-cot
gpt-4o-2024-08-06	MMLU	0-shot CoT (simple-evals), assistant system prompt	88.7	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	m2024.se0709.gpt-4o-2024-08-06.mmlu.0-shot-cot

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	90.4	Claude 3.5 Sonnet	5-shot CoT	2024-06	`m2024.c35jun.claude-3.5-sonnet.mmlu.5-shot-cot`
MMLU-Pro	77.0	Claude 3.5 Sonnet	5-shot CoT	2024-07	`m2024.llama3herd.claude-3.5-sonnet.mmlu-pro.5-shot-cot`
HumanEval	92.0	Claude 3.5 Sonnet	0-shot	2024-06	`m2024.c35jun.claude-3.5-sonnet.humaneval.0-shot`
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	`m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot`
MATH	76.6	gpt-4o	0-shot CoT, 'assistant' system prompt (simple-evals)	2024-05	`m2024.se0513.gpt-4o.math.0-shot-cot-assistant`
GPQA Diamond	59.4	Claude 3.5 Sonnet	0-shot CoT	2024-06	`m2024.c35jun.claude-3.5-sonnet.gpqa-diamond.0-shot-cot`
SWE-bench Verified	33.2	GPT-4o (checkpoint not stated)	best open-source scaffold (Agentless, per the post's scaffold chart)	2024-08	`m2024.swebv.gpt-4o.swe-bench-verified.agentless`

2024-09

September 2024. 33 verified scores for 8 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
DeepSeek V2.5-0905	GPQA Diamond	pass@1	41.3	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	`m2024.dsv3.deepseek-v2.5-0905.gpqa-diamond.pass@1`
Gemini 1.5 Pro (September 2024 release)	GPQA Diamond	0-shot CoT	59.1	Anthropic, quoting Google's September 2024 release	anthropic_claude35_oct_addendum p.6	`m2024.c35oct.gemini-1.5-pro-002.gpqa-diamond.0-shot-cot`
Gemini 1.5 Pro (September 2024 release)	MATH	4-shot CoT	86.5	Anthropic, quoting Google's September 2024 release	anthropic_claude35_oct_addendum p.6	`m2024.c35oct.gemini-1.5-pro-002.math.4-shot-cot`
Gemini 1.5 Pro (September 2024 release)	MMLU-Pro	0-shot CoT (as labelled by Anthropic)	75.8	Anthropic, quoting Google's September 2024 release	anthropic_claude35_oct_addendum p.6	`m2024.c35oct.gemini-1.5-pro-002.mmlu-pro.0-shot-cot`
OpenAI-o1-mini	AIME 2024	pass@1	63.6	DeepSeek (DeepSeek-R1 report v1, Table 4)	arxiv_2501.12948v1 p.13	`m2024.r1v1.openai-o1-mini.aime-2024.pass@1`
OpenAI-o1-mini	GPQA Diamond	pass@1	60.0	DeepSeek (DeepSeek-R1 report v1, Table 4)	arxiv_2501.12948v1 p.13	`m2024.r1v1.openai-o1-mini.gpqa-diamond.pass@1`
OpenAI-o1-mini	MATH-500	pass@1	90.0	DeepSeek (DeepSeek-R1 report v1, Table 4)	arxiv_2501.12948v1 p.13	`m2024.r1v1.openai-o1-mini.math-500.pass@1`
OpenAI-o1-mini	MMLU	pass@1	85.2	DeepSeek (DeepSeek-R1 report v1, Table 4)	arxiv_2501.12948v1 p.13	`m2024.r1v1.openai-o1-mini.mmlu.pass@1`
OpenAI-o1-mini	SWE-bench Verified	resolved	41.6	DeepSeek (DeepSeek-R1 report v1, Table 4)	arxiv_2501.12948v1 p.13	`m2024.r1v1.openai-o1-mini.swe-bench-verified.resolved`
Qwen2.5-72B (base)	MMLU	5-shot, base model (Table 2)	86.1	Alibaba (Qwen2.5 report, Table 2)	arxiv_2412.15115 p.8	`m2024.qwen25.qwen2.5-72b-base.mmlu.base`
Qwen2.5-72B-Instruct	GPQA	not stated in table	49.0	Alibaba (Qwen2.5 report, Table 6)	arxiv_2412.15115 p.11	`m2024.qwen25.qwen2.5-72b-instruct.gpqa.instruct`
Qwen2.5-72B-Instruct	GSM8K	not stated in table	95.8	Alibaba (Qwen2.5 report, Table 6)	arxiv_2412.15115 p.11	`m2024.qwen25.qwen2.5-72b-instruct.gsm8k.instruct`
Qwen2.5-72B-Instruct	HumanEval	not stated in table	86.6	Alibaba (Qwen2.5 report, Table 6)	arxiv_2412.15115 p.11	`m2024.qwen25.qwen2.5-72b-instruct.humaneval.instruct`

Qwen2.5-72B-Instruct	MATH	not stated in table	83.1	Alibaba (Qwen2.5 report, Table 6)	arxiv_2412.15115 p.11	`m2024.qwen25.qwen2.5-72b-instruct.math.instruct`
Qwen2.5-72B-Instruct	MMLU-Pro	not stated in table	71.1	Alibaba (Qwen2.5 report, Table 6)	arxiv_2412.15115 p.11	`m2024.qwen25.qwen2.5-72b-instruct.mmlu-pro.instruct`
o1 (Sept 2024 pre-release)	GPQA Diamond	cons@64	78.0	OpenAI (o1 launch post, Appendix A)	openai_o1_blog html	`m2024.o1blog.o1.gpqa-diamond.cons64`
o1 (Sept 2024 pre-release)	GPQA Diamond	pass@1	77.3	OpenAI (o1 launch post, Appendix A)	openai_o1_blog html	`m2024.o1blog.o1.gpqa-diamond.pass1`
o1 (Sept 2024 pre-release)	MATH-500	pass@1	94.8	OpenAI (o1 launch post, Appendix A)	openai_o1_blog html	`m2024.o1blog.o1.math.pass1`
o1 (Sept 2024 pre-release)	MMLU	pass@1	92.3	OpenAI (o1 launch post, Appendix A)	openai_o1_blog html	`m2024.o1blog.o1.mmlu.pass1`
o1-mini	AIME 2024	not stated in the sentence (o1 74.4% matches the o1 blog's pass@1)	70.0	OpenAI (o1-mini launch post)	openai_o1mini_blog html	`m2024.o1mini.o1-mini.aime-2024.unspecified`
o1-mini	GPQA	0-shot CoT (simple-evals), no system prompt (o-series)	60.0	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	`m2024.se0709.o1-mini.gpqa.0-shot-cot`
o1-mini	HumanEval	0-shot CoT (simple-evals), no system prompt (o-series)	92.4	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	`m2024.se0709.o1-mini.humaneval.0-shot-cot`
o1-mini	MATH-500	0-shot CoT (simple-evals), no system prompt (o-series); README footnote: MATH-500 for models on or after o1 (o1-preview/o1-mini status ambiguous)	90.0	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	`m2024.se0709.o1-mini.math.0-shot-cot`
o1-mini	MMLU	0-shot CoT (simple-evals), no system prompt (o-series)	85.2	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	`m2024.se0709.o1-mini.mmlu.0-shot-cot`
o1-preview	AIME 2024	pass@1	44.6	OpenAI (o1 launch post, Appendix A)	openai_o1_blog html	`m2024.o1blog.o1-preview.aime-2024.pass1`
o1-preview	GPQA	0-shot CoT (simple-evals), no system prompt (o-series)	73.3	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	`m2024.se0709.o1-preview.gpqa.0-shot-cot`
o1-preview	GPQA Diamond	cons@64	78.3	OpenAI (o1 launch post, Appendix A)	openai_o1_blog html	`m2024.o1blog.o1-preview.gpqa-diamond.cons64`
o1-preview	GPQA Diamond	pass@1	73.3	OpenAI (o1 launch post, Appendix A)	openai_o1_blog html	`m2024.o1blog.o1-preview.gpqa-diamond.pass1`
o1-preview	HumanEval	0-shot CoT (simple-evals), no system prompt (o-series)	92.4	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	`m2024.se0709.o1-preview.humaneval.0-shot-cot`
o1-preview	MATH-500	pass@1	85.5	OpenAI (o1 launch post, Appendix A)	openai_o1_blog html	`m2024.o1blog.o1-preview.math.pass1`
o1-preview	MATH-500	0-shot CoT (simple-evals), no system prompt (o-series); README footnote: MATH-500 for models on or after o1 (o1-preview/o1-mini status ambiguous)	85.5	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	`m2024.se0709.o1-preview.math.0-shot-cot`
o1-preview	MMLU	pass@1	90.8	OpenAI (o1 launch post, Appendix A)	openai_o1_blog html	`m2024.o1blog.o1-preview.mmlu.pass1`
o1-preview	MMLU	0-shot CoT (simple-evals), no system prompt (o-series)	90.8	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	`m2024.se0709.o1-preview.mmlu.0-shot-cot`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.3	o1 (Sept 2024 pre-release)	pass@1	2024-09	`m2024.o1blog.o1.mmlu.pass1`
MMLU-Pro	77.0	Claude 3.5 Sonnet	5-shot CoT	2024-07	`m2024.llama3herd.claude-3.5-sonnet.mmlu-pro.5-shot-cot`
HumanEval	92.0	Claude 3.5 Sonnet	0-shot	2024-06	`m2024.c35jun.claude-3.5-sonnet.humaneval.0-shot`
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	`m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot`
MATH	76.6	gpt-4o	0-shot CoT, 'assistant' system prompt (simple-evals)	2024-05	`m2024.se0513.gpt-4o.math.0-shot-cot-assistant`
MATH-500	94.8	o1 (Sept 2024 pre-release)	pass@1	2024-09	`m2024.o1blog.o1.math.pass1`
GPQA Diamond	78.3	o1-preview	cons@64	2024-09	`m2024.o1blog.o1-preview.gpqa-diamond.cons64`
SWE-bench Verified	33.2	GPT-4o (checkpoint not stated)	best open-source scaffold (Agentless, per the post's scaffold chart)	2024-08	`m2024.swebv.gpt-4o.swe-bench-verified.agentless`
AIME 2024	70.0	o1-mini	not stated in the sentence (o1 74.4% matches the o1 blog's pass@1)	2024-09	`m2024.o1mini.o1-mini.aime-2024.unspecified`

2024-10

October 2024. 23 verified scores for 3 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Claude 3.5 Haiku	GPQA Diamond	0-shot CoT	41.6	Anthropic (Oct 2024 addendum, Table 8)	anthropic_claude35_oct_addendum p.7	`m2024.c35oct.claude-3.5-haiku.gpqa-diamond.0-shot-cot`
Claude 3.5 Haiku	HumanEval	0-shot	88.1	Anthropic (Oct 2024 addendum, Table 8)	anthropic_claude35_oct_addendum p.7	`m2024.c35oct.claude-3.5-haiku.humaneval.0-shot`
Claude 3.5 Haiku	MATH	0-shot CoT	69.2	Anthropic (Oct 2024 addendum, Table 8)	anthropic_claude35_oct_addendum p.7	`m2024.c35oct.claude-3.5-haiku.math.0-shot-cot`
Claude 3.5 Haiku	MMLU	0-shot CoT	80.3	Anthropic (Oct 2024 addendum, Table 8)	anthropic_claude35_oct_addendum p.7	`m2024.c35oct.claude-3.5-haiku.mmlu.0-shot-cot`
Claude 3.5 Haiku	MMLU	5-shot CoT	80.9	Anthropic (Oct 2024 addendum, Table 8)	anthropic_claude35_oct_addendum p.7	`m2024.c35oct.claude-3.5-haiku.mmlu.5-shot-cot`
Claude 3.5 Haiku	MMLU-Pro	0-shot CoT	65.0	Anthropic (Oct 2024 addendum, Table 8)	anthropic_claude35_oct_addendum p.7	`m2024.c35oct.claude-3.5-haiku.mmlu-pro.0-shot-cot`
Claude 3.5 Haiku	SWE-bench Verified	pass@1, Anthropic agent scaffold	40.6	Anthropic (Oct 2024 addendum, Table 2)	anthropic_claude35_oct_addendum p.3	`m2024.c35oct.claude-3.5-haiku.swe-bench-verified.pass1`
Claude 3.5 Sonnet (New)	AIME 2024	0-shot CoT	16.0	Anthropic (Claude 3.5 Sonnet (New) & Haiku addendum, Oct 2024, Table 7)	anthropic_claude35_oct_addendum p.6	`m2024.c35oct.claude-3.5-sonnet-new.aime-2024.0-shot-cot`
Claude 3.5 Sonnet (New)	GPQA Diamond	0-shot CoT	65.0	Anthropic (Claude 3.5 Sonnet (New) & Haiku addendum, Oct 2024, Table 7)	anthropic_claude35_oct_addendum p.6	`m2024.c35oct.claude-3.5-sonnet-new.gpqa-diamond.0-shot-cot`
Claude 3.5 Sonnet (New)	HumanEval	0-shot	93.7	Anthropic (Claude 3.5 Sonnet (New) & Haiku addendum, Oct 2024, Table 7)	anthropic_claude35_oct_addendum p.6	`m2024.c35oct.claude-3.5-sonnet-new.humaneval.0-shot`
Claude 3.5 Sonnet (New)	MATH	0-shot CoT	78.3	Anthropic (Claude 3.5 Sonnet (New) & Haiku addendum, Oct 2024, Table 7)	anthropic_claude35_oct_addendum p.6	`m2024.c35oct.claude-3.5-sonnet-new.math.0-shot-cot`
Claude 3.5 Sonnet (New)	MMLU	0-shot CoT	89.3	Anthropic (Claude 3.5 Sonnet (New) & Haiku addendum, Oct 2024, Table 7)	anthropic_claude35_oct_addendum p.6	`m2024.c35oct.claude-3.5-sonnet-new.mmlu.0-shot-cot`
Claude 3.5 Sonnet (New)	MMLU	5-shot	88.7	Anthropic (Claude 3.5 Sonnet (New) & Haiku addendum, Oct 2024, Table 7)	anthropic_claude35_oct_addendum p.6	`m2024.c35oct.claude-3.5-sonnet-new.mmlu.5-shot`

Claude 3.5 Sonnet (New)	MMLU	5-shot CoT	90.5	Anthropic (Claude 3.5 Sonnet (New) & Haiku addendum, Oct 2024, Table 7)	anthropic_claude35_oct_addendum p.6	m2024.c35oct.claude-3.5-sonnet-new.mmlu.5-shot-cot
Claude 3.5 Sonnet (New)	MMLU-Pro	0-shot CoT	78.0	Anthropic (Claude 3.5 Sonnet (New) & Haiku addendum, Oct 2024, Table 7)	anthropic_claude35_oct_addendum p.6	m2024.c35oct.claude-3.5-sonnet-new.mmlu-pro.0-shot-cot
Claude 3.5 Sonnet (New)	SWE-bench Verified	pass@1, Anthropic agent scaffold	49.0	Anthropic (Oct 2024 addendum, Table 2)	anthropic_claude35_oct_addendum p.3	m2024.c35oct.claude-3.5-sonnet-new.swe-bench-verified.pass1
Claude-3.5-Sonnet-1022	AIME 2024	pass@1, temperature 0.7, mean of 16 runs	16.0	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.claude-3.5-sonnet-1022.aime-2024.pass@1
Claude-3.5-Sonnet-1022	GPQA Diamond	pass@1	65.0	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.claude-3.5-sonnet-1022.gpqa-diamond.pass@1
Claude-3.5-Sonnet-1022	HumanEval-Mul	pass@1, 8-language average	81.7	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.claude-3.5-sonnet-1022.humaneval-mul.pass@1
Claude-3.5-Sonnet-1022	MATH-500	EM, greedy decoding	78.3	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.claude-3.5-sonnet-1022.math-500.em
Claude-3.5-Sonnet-1022	MMLU	EM (shots not stated for chat models)	88.3	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.claude-3.5-sonnet-1022.mmlu.em
Claude-3.5-Sonnet-1022	MMLU-Pro	EM	78.0	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.claude-3.5-sonnet-1022.mmlu-pro.em
Claude-3.5-Sonnet-1022	SWE-bench Verified	resolved, Agentless framework	50.8	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.claude-3.5-sonnet-1022.swe-bench-verified.resolved

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.3	o1 (Sept 2024 pre-release)	pass@1	2024-09	m2024.o1blog.o1.mmlu.pass1
MMLU-Pro	78.0	Claude 3.5 Sonnet (New)	0-shot CoT	2024-10	m2024.c35oct.claude-3.5-sonnet-new.mmlu-pro.0-shot-cot
HumanEval	93.7	Claude 3.5 Sonnet (New)	0-shot	2024-10	m2024.c35oct.claude-3.5-sonnet-new.humaneval.0-shot
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot
MATH	78.3	Claude 3.5 Sonnet (New)	0-shot CoT	2024-10	m2024.c35oct.claude-3.5-sonnet-new.math.0-shot-cot
MATH-500	94.8	o1 (Sept 2024 pre-release)	pass@1	2024-09	m2024.o1blog.o1.math.pass1
GPQA Diamond	78.3	o1-preview	cons@64	2024-09	m2024.o1blog.o1-preview.gpqa-diamond.cons64
SWE-bench Verified	49.0	Claude 3.5 Sonnet (New)	pass@1, Anthropic agent scaffold	2024-10	m2024.c35oct.claude-3.5-sonnet-new.swe-bench-verified.pass1
AIME 2024	70.0	o1-mini	not stated in the sentence (o1 74.4% matches the o1 blog's pass@1)	2024-09	m2024.o1mini.o1-mini.aime-2024.unspecified

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Anthropic Claude 3.5 Sonnet (new) & Haiku Model Card Addendum (Oct 2024), None, PDF page 6: Table 7: This table shows evaluation results for the upgraded Claude 3.5 Sonnet and peer models on reason-.

2024-11

November 2024. 9 verified scores for 2 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim

Marco-o1-MCTS (step)	MGSM-En	CoT prompt	90.4	Alibaba (Marco-o1 report, Table 2)	arxiv_2411.14405 p.5	m2024.marcoo1.marco-o1-mcts-step.mgsm-en.cot
QwQ-32B-Preview	AIME (year not stated)	not stated	50.0	Alibaba Qwen (QwQ-32B-Preview post)	qwen_qwq32b_preview_blog html	m2024.qwq.qwq-32b-preview.aime.unspecified
QwQ-32B-Preview	AIME 2024	pass@1	50.0	DeepSeek (DeepSeek-R1 report v1, Table 5)	arxiv_2501.12948v1 p.14	m2024.r1v1.qwq-32b-preview.aime-2024.pass1
QwQ-32B-Preview	GPQA	not stated	65.2	Alibaba Qwen (QwQ-32B-Preview post)	qwen_qwq32b_preview_blog html	m2024.qwq.qwq-32b-preview.gpqa.unspecified
QwQ-32B-Preview	GPQA Diamond	pass@1	54.5	DeepSeek (DeepSeek-R1 report v1, Table 5)	arxiv_2501.12948v1 p.14	m2024.r1v1.qwq-32b-preview.gpqa-diamond.pass1
QwQ-32B-Preview	LiveCodeBench	not stated	50.0	Alibaba Qwen (QwQ-32B-Preview post)	qwen_qwq32b_preview_blog html	m2024.qwq.qwq-32b-preview.livecodebench.unspecified
QwQ-32B-Preview	LiveCodeBench	pass@1	41.9	DeepSeek (DeepSeek-R1 report v1, Table 5)	arxiv_2501.12948v1 p.14	m2024.r1v1.qwq-32b-preview.livecodebench.pass1
QwQ-32B-Preview	MATH-500	not stated	90.6	Alibaba Qwen (QwQ-32B-Preview post)	qwen_qwq32b_preview_blog html	m2024.qwq.qwq-32b-preview.math-500.unspecified
QwQ-32B-Preview	MATH-500	pass@1	90.6	DeepSeek (DeepSeek-R1 report v1, Table 5)	arxiv_2501.12948v1 p.14	m2024.r1v1.qwq-32b-preview.math-500.pass1

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.3	o1 (Sept 2024 pre-release)	pass@1	2024-09	m2024.o1blog.o1.mmlu.pass1
MMLU-Pro	78.0	Claude 3.5 Sonnet (New)	0-shot CoT	2024-10	m2024.c35oct.claude-3.5-sonnet-new.mmlu-pro.0-shot-cot
HumanEval	93.7	Claude 3.5 Sonnet (New)	0-shot	2024-10	m2024.c35oct.claude-3.5-sonnet-new.humaneval.0-shot
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot
MATH	78.3	Claude 3.5 Sonnet (New)	0-shot CoT	2024-10	m2024.c35oct.claude-3.5-sonnet-new.math.0-shot-cot
MATH-500	94.8	o1 (Sept 2024 pre-release)	pass@1	2024-09	m2024.o1blog.o1.math.pass1
GPQA Diamond	78.3	o1-preview	cons@64	2024-09	m2024.o1blog.o1-preview.gpqa-diamond.cons64
SWE-bench Verified	49.0	Claude 3.5 Sonnet (New)	pass@1, Anthropic agent scaffold	2024-10	m2024.c35oct.claude-3.5-sonnet-new.swe-bench-verified.pass1
AIME 2024	70.0	o1-mini	not stated in the sentence (o1 74.4% matches the o1 blog's pass@1)	2024-09	m2024.o1mini.o1-mini.aime-2024.unspecified

2024-12

December 2024. 26 verified scores for 8 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
DeepSeek-V3	AIME 2024	pass@1, temperature 0.7, mean of 16 runs	39.2	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.deepseek-v3.aime-2024.pass@1
DeepSeek-V3	GPQA Diamond	pass@1	59.1	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.deepseek-v3.gpqa-diamond.pass@1
DeepSeek-V3	HumanEval-Mul	pass@1, 8-language average	82.6	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.deepseek-v3.humaneval-mul.pass@1
DeepSeek-V3	MATH-500	EM, greedy decoding	90.2	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.deepseek-v3.math-500.em
DeepSeek-V3	MMLU	EM (shots not stated for chat models)	88.5	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.deepseek-v3.mmlu.em
DeepSeek-V3	MMLU-Pro	EM	75.9	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.deepseek-v3.mmlu-pro.em

DeepSeek-V3	SWE-bench Verified	resolved, Agentless framework	42.0	DeepSeek (DeepSeek-V3 report, Table 6, own harness)	arxiv_2412.19437 p.31	m2024.dsv3.deepseek-v3.swe-bench-verified.resolved
DeepSeek-V3 Base	GSM8K	8-shot EM	89.3	DeepSeek (DeepSeek-V3 report, base-model table)	arxiv_2412.19437 p.25	m2024.dsv3.deepseek-v3-base.gsm8k.8-shot
DeepSeek-V3 Base	MMLU	5-shot EM	87.1	DeepSeek (DeepSeek-V3 report, base-model table)	arxiv_2412.19437 p.25	m2024.dsv3.deepseek-v3-base.mmlu.5-shot
Gemini 2.0 Flash (research coding agent)	SWE-bench Verified	agent with code execution tools, samples hundreds of candidate solutions	51.8	Google (developers blog, Dec 11 2024)	google_gemini2_devblog html	m2024.gemini2dev.gemini-2.0-flash-agent.swe-bench-verified.agent
Llama-3.3 70B Instruct	GPQA Diamond	0-shot CoT	50.5	Meta (Llama 3.3 MODEL_CARD.md, commit c908f5b)	meta_llama33_model_card_c908f5b html	m2024.llama33card.llama-3.3-70b-instruct.gpqa-diamond.0-shot-cot
Llama-3.3 70B Instruct	HumanEval	0-shot pass@1	88.4	Meta (Llama 3.3 MODEL_CARD.md, commit c908f5b)	meta_llama33_model_card_c908f5b html	m2024.llama33card.llama-3.3-70b-instruct.humaneval.0-shot-pass@1
Llama-3.3 70B Instruct	MATH	0-shot CoT	77.0	Meta (Llama 3.3 MODEL_CARD.md, commit c908f5b)	meta_llama33_model_card_c908f5b html	m2024.llama33card.llama-3.3-70b-instruct.math.0-shot-cot
Llama-3.3 70B Instruct	MMLU	0-shot CoT	86.0	Meta (Llama 3.3 MODEL_CARD.md, commit c908f5b)	meta_llama33_model_card_c908f5b html	m2024.llama33card.llama-3.3-70b-instruct.mmlu.0-shot-cot
Llama-3.3 70B Instruct	MMLU-Pro	5-shot CoT	68.9	Meta (Llama 3.3 MODEL_CARD.md, commit c908f5b)	meta_llama33_model_card_c908f5b html	m2024.llama33card.llama-3.3-70b-instruct.mmlu-pro.5-shot-cot
Llama-3.3 70b instruct	MMLU	simple-evals (0-shot CoT), temperature 0.5	86.3	Microsoft (Phi-4 report, Table 1)	arxiv_2412.08905 p.2	m2024.phi4.llama-3.3-70b-instruct.mmlu.simple-evals
OpenAI-o1-1217	AIME 2024	pass@1	79.2	DeepSeek (DeepSeek-R1 report v1, Table 4)	arxiv_2501.12948v1 p.13	m2024.r1v1.openai-o1-1217.aime-2024.pass@1
OpenAI-o1-1217	GPQA Diamond	pass@1	75.7	DeepSeek (DeepSeek-R1 report v1, Table 4)	arxiv_2501.12948v1 p.13	m2024.r1v1.openai-o1-1217.gpqa-diamond.pass@1
OpenAI-o1-1217	MATH-500	pass@1	96.4	DeepSeek (DeepSeek-R1 report v1, Table 4)	arxiv_2501.12948v1 p.13	m2024.r1v1.openai-o1-1217.math-500.pass@1
OpenAI-o1-1217	MMLU	pass@1	91.8	DeepSeek (DeepSeek-R1 report v1, Table 4)	arxiv_2501.12948v1 p.13	m2024.r1v1.openai-o1-1217.mmlu.pass@1
OpenAI-o1-1217	SWE-bench Verified	resolved	48.9	DeepSeek (DeepSeek-R1 report v1, Table 4)	arxiv_2501.12948v1 p.13	m2024.r1v1.openai-o1-1217.swe-bench-verified.resolved
o1	GPQA	0-shot CoT (simple-evals), no system prompt (o-series)	75.7	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	m2024.se0709.o1.gpqa.0-shot-cot
o1	MATH-500	0-shot CoT (simple-evals), no system prompt (o-series)	96.4	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	m2024.se0709.o1.math-500.0-shot-cot
o1	MMLU	0-shot CoT (simple-evals), no system prompt (o-series)	91.8	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	m2024.se0709.o1.mmlu.0-shot-cot
phi-4 14b	MATH	simple-evals, temperature 0.5	80.4	Microsoft (Phi-4 report, Table 1)	arxiv_2412.08905 p.2	m2024.phi4.phi-4.math.simple-evals
phi-4 14b	MMLU	simple-evals (0-shot CoT), temperature 0.5	84.8	Microsoft (Phi-4 report, Table 1)	arxiv_2412.08905 p.2	m2024.phi4.phi-4.mmlu.simple-evals

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.3	o1 (Sept 2024 pre-release)	pass@1	2024-09	m2024.o1blog.o1.mmlu.pass1
MMLU-Pro	78.0	Claude 3.5 Sonnet (New)	0-shot CoT	2024-10	m2024.c35oct.claude-3.5-sonnet-new.mmlu-pro.0-shot-cot

HumanEval	93.7	Claude 3.5 Sonnet (New)	0-shot	2024-10	`m2024.c35oct.claude-3.5-sonnet-new.humaneval.0-shot`
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	`m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot`
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	`m2024.qwen25.qwen2.5-72b-instruct.math.instruct`
MATH-500	94.8	o1 (Sept 2024 pre-release)	pass@1	2024-09	`m2024.o1blog.o1.math.pass1`
GPQA Diamond	78.3	o1-preview	cons@64	2024-09	`m2024.o1blog.o1-preview.gpqa-diamond.cons64`
SWE-bench Verified	51.8	Gemini 2.0 Flash (research coding agent)	agent with code execution tools, samples hundreds of candidate solutions	2024-12	`m2024.gemini2dev.gemini-2.0-flash-agent.swe-bench-verified.agent`
AIME 2024	70.0	o1-mini	not stated in the sentence (o1 74.4% matches the o1 blog's pass@1)	2024-09	`m2024.o1mini.o1-mini.aime-2024.unspecified`

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- DeepSeek-V3 Technical Report, v2, PDF page 1: Figure 1: Benchmark performance of DeepSeek-V3 and its counterparts.
- DeepSeek-V3 Technical Report, v2, PDF page 31: Table 6: Comparison between DeepSeek-V3 and other representative chat models.

2025-01

January 2025. 19 verified scores for 6 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
DeepSeek-R1	AIME 2024	Pass@1	79.8	DeepSeek (DeepSeek-R1 TR v1, Table 4)	arxiv_2501.12948v1 p.13	`m2025.r1v1.deepseek-r1.aime24.pass1`
DeepSeek-R1	GPQA Diamond	Pass@1	71.5	DeepSeek (DeepSeek-R1 TR v1, Table 4)	arxiv_2501.12948v1 p.13	`m2025.r1v1.deepseek-r1.gpqa.pass1`
DeepSeek-R1	LiveCodeBench	Pass@1-COT	65.9	DeepSeek (DeepSeek-R1 TR v1, Table 4)	arxiv_2501.12948v1 p.13	`m2025.r1v1.deepseek-r1.lcb.pass1-cot`
DeepSeek-R1	MATH-500	Pass@1	97.3	DeepSeek (DeepSeek-R1 TR v1, Table 4)	arxiv_2501.12948v1 p.13	`m2025.r1v1.deepseek-r1.math500.pass1`
DeepSeek-R1	MMLU	Pass@1, simple-evals prompt (0-shot)	90.8	DeepSeek (DeepSeek-R1 TR v1, Table 4)	arxiv_2501.12948v1 p.13	`m2025.r1v1.deepseek-r1.mmlu.pass1`
DeepSeek-R1	MMLU-Pro	EM	84.0	DeepSeek (DeepSeek-R1 TR v1, Table 4)	arxiv_2501.12948v1 p.13	`m2025.r1v1.deepseek-r1.mmlu-pro.em`
DeepSeek-R1	SWE-bench Verified	Resolved	49.2	DeepSeek (DeepSeek-R1 TR v1, Table 4)	arxiv_2501.12948v1 p.13	`m2025.r1v1.deepseek-r1.swebench.resolved`
Kimi k1.5 (long-CoT)	AIME 2024	Pass@1	77.5	Moonshot AI (Kimi k1.5 TR v1, abstract)	arxiv_2501.12599 p.1	`m2025.kimi-k15.kimi-k1.5-long-cot.aime24.pass1`
Kimi k1.5 (long-CoT)	MATH-500	EM (Table 2: MATH-500 (EM))	96.2	Moonshot AI (Kimi k1.5 TR v1, abstract)	arxiv_2501.12599 p.1	`m2025.kimi-k15.kimi-k1.5-long-cot.math500.em`
OpenAI o3-mini (high)	AIME 2024	launch table	87.3	OpenAI (GPT-4.5 launch post, Feb 27 2025)	openai_gpt45_blog html	`m2025.gpt45-blog.o3-mini-high.aime24`
OpenAI o3-mini (high)	GPQA (subset not stated)	launch table	79.7	OpenAI (GPT-4.5 launch post, Feb 27 2025)	openai_gpt45_blog html	`m2025.gpt45-blog.o3-mini-high.gpqa`
OpenAI o3-mini (high)	MMMLU	launch table	81.1	OpenAI (GPT-4.5 launch post, Feb 27 2025)	openai_gpt45_blog html	`m2025.gpt45-blog.o3-mini-high.mmmlu`
OpenAI o3-mini (high)	SWE-bench Verified (477 of 500 tasks)	23/500 tasks omitted (footnote 2)	49.3	OpenAI (GPT-4.1 launch post, Apr 14 2025, appendix)	openai_gpt41_blog html	`m2025.gpt41-blog.o3-mini-high.swebench`
o3-mini (tools), non-final checkpoint	SWE-bench Verified (n=477 subset)	OpenAI internal tool scaffold, pass@1 over 4 tries	61.0	OpenAI (o3-mini System Card, Section 5.7.2)	openai_o3mini_system_card p.28	`m2025.o3mini-card.o3-mini-tools.swebench.n477`
o3-mini launch candidate	SWE-bench Verified (n=477 subset)	Agentless 1.0 scaffold	39.0	OpenAI (o3-mini System Card, Section 5.7.2)	openai_o3mini_system_card p.28	`m2025.o3mini-card.o3-mini.swebench.agentless.n477`
o3-mini-high	GPQA (subset not stated)	0-shot CoT (simple-evals)	77.2	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	`m2025.simple-evals.o3-mini-high.gpqa`

o3-mini-high	HumanEval	0-shot CoT (simple-evals)	97.6	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme.md	m2025.simple-evals.o3-mini-high.humaneval
o3-mini-high	MATH-500	0-shot CoT (simple-evals)	97.9	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme.md	m2025.simple-evals.o3-mini-high.math500
o3-mini-high	MMLU	0-shot CoT (simple-evals)	86.9	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme.md	m2025.simple-evals.o3-mini-high.mmlu

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.3	o1 (Sept 2024 pre-release)	pass@1	2024-09	m2024.o1blog.o1.mmlu.pass1
MMLU-Pro	84.0	DeepSeek-R1	EM	2025-01	m2025.r1v1.deepseek-r1.mmlu-pro.em
HumanEval	93.7	Claude 3.5 Sonnet (New)	0-shot	2024-10	m2024.c35oct.claude-3.5-sonnet-new.humaneval.0-shot
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	m2024.qwen25.qwen2.5-72b-instruct.math.instruct
MATH-500	97.3	DeepSeek-R1	Pass@1	2025-01	m2025.r1v1.deepseek-r1.math500.pass1
GPQA Diamond	78.3	o1-preview	cons@64	2024-09	m2024.o1blog.o1-preview.gpqa-diamond.cons64
SWE-bench Verified	51.8	Gemini 2.0 Flash (research coding agent)	agent with code execution tools, samples hundreds of candidate solutions	2024-12	m2024.gemini2dev.gemini-2.0-flash-agent.swe-bench-verified.agent
AIME 2024	79.8	DeepSeek-R1	Pass@1	2025-01	m2025.r1v1.deepseek-r1.aime24.pass1

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning, v1, PDF page 13: Table 4: Comparison between DeepSeek-R1 and other representative models.
- OpenAI o3-mini System Card (Feb 10, 2025), 20250212235848, PDF page 28: no table or figure caption on this page

2025-02

February 2025. 18 verified scores for 6 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Claude 3.7 Sonnet	SWE-bench Verified (n=489 subset)	high compute: parallel attempts, regression-test rejection, scoring-model selection	70.3	Anthropic (Claude 3.7 Sonnet launch post, appendix)	anthropic_claude37_blog.html	m2025.claude37-blog.claude-3.7-sonnet.swebench.parallel
Claude 3.7 Sonnet	SWE-bench Verified (n=489 subset)	without the high-compute scaffold	63.7	Anthropic (Claude 3.7 Sonnet launch post, appendix)	anthropic_claude37_blog.html	m2025.claude37-blog.claude-3.7-sonnet.swebench.single
GPT-4.5	AIME 2024	launch table	36.7	OpenAI (GPT-4.5 launch post, Feb 27 2025)	openai_gpt45_blog.html	m2025.gpt45-blog.gpt-4.5.aime24
GPT-4.5	GPQA (subset not stated)	launch table	71.4	OpenAI (GPT-4.5 launch post, Feb 27 2025)	openai_gpt45_blog.html	m2025.gpt45-blog.gpt-4.5.gpqa
GPT-4.5	MMLU (English, not translated)	0-shot	89.6	OpenAI (GPT-4.5 System Card, Table 16)	openai_gpt45_system_card.p.26	m2025.gpt45-card.gpt-4.5.mmlu-english
GPT-4.5	MMMLU	launch table	85.1	OpenAI (GPT-4.5 launch post, Feb 27 2025)	openai_gpt45_blog.html	m2025.gpt45-blog.gpt-4.5.mmmlu
GPT-4.5	SWE-bench Verified	'best internal performance'	38.0	OpenAI (GPT-4.5 launch post, Feb 27 2025)	openai_gpt45_blog.html	m2025.gpt45-blog.gpt-4.5.swebench

Grok 3 (Think)	AIME 2025	cons@64	93.3	xAI (Grok 3 Beta launch post, prose)	xai_grok3_blog html	m2025.grok3-blog.grok-3-think.aime25.cons64.prose
Grok 3 (Think) [chart code GROK_3_R]	AIME 2024	chart scoreMax (inferred highest test-time compute; the prose states cons@64 only for AIME 2025)	93.3	xAI (Grok 3 Beta launch post, Feb 19 2025, chart data)	xai_grok3_blog html	m2025.grok3-blog.grok-3-think.aime24.max
Grok 3 (Think) [chart code GROK_3_R]	AIME 2024	chart base bar (scoreMax is the highest test-time compute)	83.9	xAI (Grok 3 Beta launch post, Feb 19 2025, chart data)	xai_grok3_blog html	m2025.grok3-blog.grok-3-think.aime24.score
Grok 3 (Think) [chart code GROK_3_R]	GPQA	chart scoreMax (highest test-time compute)	84.6	xAI (Grok 3 Beta launch post, Feb 19 2025, chart data)	xai_grok3_blog html	m2025.grok3-blog.grok-3-think.gpqa.max
Grok 3 (Think) [chart code GROK_3_R]	GPQA	chart base bar	80.2	xAI (Grok 3 Beta launch post, Feb 19 2025, chart data)	xai_grok3_blog html	m2025.grok3-blog.grok-3-think.gpqa.score
Grok 3 (Think) [chart code GROK_3_R]	LiveCodeBench	chart scoreMax (highest test-time compute)	79.4	xAI (Grok 3 Beta launch post, Feb 19 2025, chart data)	xai_grok3_blog html	m2025.grok3-blog.grok-3-think.lcb.max
Grok 3 (Think) [chart code GROK_3_R]	LiveCodeBench	chart base bar	70.6	xAI (Grok 3 Beta launch post, Feb 19 2025, chart data)	xai_grok3_blog html	m2025.grok3-blog.grok-3-think.lcb.score
Grok 3 Beta (reasoning off)	MMLU-Pro	reasoning turned off	79.9	xAI (Grok 3 Beta launch post, HTML table)	xai_grok3_blog html	m2025.grok3-blog.grok-3-beta.mmlu-pro
gpt-4.5-preview-2025-02-27	GPQA (subset not stated)	0-shot CoT, assistant system prompt	69.5	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	m2025.simple-evals.gpt-4.5-preview.gpqa
gpt-4.5-preview-2025-02-27	HumanEval	0-shot CoT, assistant system prompt	88.6	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	m2025.simple-evals.gpt-4.5-preview.humaneval
gpt-4.5-preview-2025-02-27	MMLU	0-shot CoT, assistant system prompt	90.8	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	m2025.simple-evals.gpt-4.5-preview.mmlu

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.3	o1 (Sept 2024 pre-release)	pass@1	2024-09	m2024.o1blog.o1.mmlu.pass1
MMLU-Pro	84.0	DeepSeek-R1	EM	2025-01	m2025.r1v1.deepseek-r1.mmlu-pro.em
HumanEval	93.7	Claude 3.5 Sonnet (New)	0-shot	2024-10	m2024.c35oct.claude-3.5-sonnet-new.humaneval.0-shot
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	m2024.qwen25.qwen2.5-72b-instruct.math.instruct
MATH-500	97.3	DeepSeek-R1	Pass@1	2025-01	m2025.r1v1.deepseek-r1.math500.pass1
GPQA Diamond	78.3	o1-preview	cons@64	2024-09	m2024.o1blog.o1-preview.gpqa-diamond.cons64
SWE-bench Verified	51.8	Gemini 2.0 Flash (research coding agent)	agent with code execution tools, samples hundreds of candidate solutions	2024-12	m2024.gemini2dev.gemini-2.0-flash-agent.swe-bench-verified.agent
AIME 2024	93.3	Grok 3 (Think) [chart code GROK_3_R]	chart scoreMax (inferred highest test-time compute; the prose states cons@64 only for AIME 2025)	2025-02	m2025.grok3-blog.grok-3-think.aime24.max
AIME 2025	93.3	Grok 3 (Think)	cons@64	2025-02	m2025.grok3-blog.grok-3-think.aime25.cons64.prose

2025-03

March 2025. 13 verified scores for 4 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
DeepSeek-V3-0324	AIME (year not stated)	model card list	59.4	DeepSeek (DeepSeek-V3-0324 Hugging Face model card)	hf_deepseek_v3_0324_apr html	m2025.hf-v3-0324.deepseek-v3-0324.aime
DeepSeek-V3-0324	GPQA	model card list	68.4	DeepSeek (DeepSeek-V3-0324 Hugging Face model card)	hf_deepseek_v3_0324_apr html	m2025.hf-v3-0324.deepseek-v3-0324.gpqa
DeepSeek-V3-0324	LiveCodeBench (window not stated)	model card list	49.2	DeepSeek (DeepSeek-V3-0324 Hugging Face model card)	hf_deepseek_v3_0324_apr html	m2025.hf-v3-0324.deepseek-v3-0324.lcb
DeepSeek-V3-0324	MATH-500	Acc.; starred = taken from the model's own report/blog	94.0	Moonshot AI (Kimi K2 TR, Table 3)	arxiv_2507.20534 p.16	m2025.kimi-k2.deepseek-v3-0324.math500
DeepSeek-V3-0324	MMLU-Pro	model card list (predecessor → new)	81.2	DeepSeek (DeepSeek-V3-0324 Hugging Face model card)	hf_deepseek_v3_0324_apr html	m2025.hf-v3-0324.deepseek-v3-0324.mmlu-pro
Gemini 2.5 Pro (Experimental 03-25)	SWE-bench Verified	custom agent setup	63.8	Google (Gemini 2.5 launch post, Mar 25 2025)	google_gemini25_blog html	m2025.gemini25-blog.gemini-2.5-pro.swebench.custom-agent
Gemini2.5-Pro	AIME 2024	as reported by Qwen	92.0	Alibaba Qwen (Qwen3 TR, Table 11)	arxiv_2505.09388 p.14	m2025.qwen3.gemini-2.5-pro.aime24
Gemini2.5-Pro	AIME 2025	as reported by Qwen	86.7	Alibaba Qwen (Qwen3 TR, Table 11)	arxiv_2505.09388 p.14	m2025.qwen3.gemini-2.5-pro.aime25
Gemini2.5-Pro	GPQA Diamond	as reported by Qwen	84.0	Alibaba Qwen (Qwen3 TR, Table 11)	arxiv_2505.09388 p.14	m2025.qwen3.gemini-2.5-pro.gpqa
Gemini2.5-Pro	LiveCodeBench v5	as reported by Qwen	70.4	Alibaba Qwen (Qwen3 TR, Table 11)	arxiv_2505.09388 p.14	m2025.qwen3.gemini-2.5-pro.lcb-v5
QwQ-32B	AIME 2024	as reported by Qwen (Qwen3 TR)	79.5	Alibaba Qwen (Qwen3 TR, Table 13)	arxiv_2505.09388 p.16	m2025.qwen3.qwq-32b.aime24
QwQ-32B	GPQA Diamond	as reported by Qwen (Qwen3 TR)	65.6	Alibaba Qwen (Qwen3 TR, Table 13)	arxiv_2505.09388 p.16	m2025.qwen3.qwq-32b.gpqa
QwQ-32B	LiveCodeBench v5	as reported by Qwen (Qwen3 TR)	62.7	Alibaba Qwen (Qwen3 TR, Table 13)	arxiv_2505.09388 p.16	m2025.qwen3.qwq-32b.lcb-v5

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.3	o1 (Sept 2024 pre-release)	pass@1	2024-09	m2024.o1blog.o1.mmlu.pass1
MMLU-Pro	84.0	DeepSeek-R1	EM	2025-01	m2025.r1v1.deepseek-r1.mmlu-pro.em
HumanEval	93.7	Claude 3.5 Sonnet (New)	0-shot	2024-10	m2024.c35oct.claude-3.5-sonnet-new.humaneval.0-shot
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	m2024.qwen25.qwen2.5-72b-instruct.math.instruct
MATH-500	97.3	DeepSeek-R1	Pass@1	2025-01	m2025.r1v1.deepseek-r1.math500.pass1
GPQA Diamond	78.3	o1-preview	cons@64	2024-09	m2024.o1blog.o1-preview.gpqa-diamond.cons64
SWE-bench Verified	63.8	Gemini 2.5 Pro (Experimental 03-25)	custom agent setup	2025-03	m2025.gemini25-blog.gemini-2.5-pro.swebench.custom-agent
AIME 2024	93.3	Grok 3 (Think) [chart code GROK_3_R]	chart scoreMax (inferred highest test-time compute; the prose states cons@64 only for AIME 2025)	2025-02	m2025.grok3-blog.grok-3-think.aime24.max
AIME 2025	93.3	Grok 3 (Think)	cons@64	2025-02	m2025.grok3-blog.grok-3-think.aime25.cons64.prose

2025-04

April 2025. 25 verified scores for 12 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
GPT-4.1	AIME 2024	appendix table	48.1	OpenAI (GPT-4.1 launch post, Apr 14 2025, appendix)	openai_gpt41_blog html	m2025.gpt41-blog.gpt-4.1.aime24
GPT-4.1	GPQA Diamond	model-based answer extraction (footnote 1)	66.3	OpenAI (GPT-4.1 launch post, Apr 14 2025, appendix)	openai_gpt41_blog html	m2025.gpt41-blog.gpt-4.1.gpqa
GPT-4.1	MMLU	appendix table	90.2	OpenAI (GPT-4.1 launch post, Apr 14 2025, appendix)	openai_gpt41_blog html	m2025.gpt41-blog.gpt-4.1.mmlu
GPT-4.1	SWE-bench Verified (477 of 500 tasks)	23/500 tasks omitted (footnote 2)	54.6	OpenAI (GPT-4.1 launch post, Apr 14 2025, appendix)	openai_gpt41_blog html	m2025.gpt41-blog.gpt-4.1.swebench
Llama 4 Maverick (instruction tuned)	GPQA Diamond	0-shot, accuracy	69.8	Meta (Llama 4 MODEL_CARD.md, llama-models commit 038acb9)	meta_llama4_model_card_038acb9 html	m2025.llama4-card.llama-4-maverick.gpqa
Llama 4 Maverick (instruction tuned)	MMLU-Pro	0-shot, macro_avg/acc	80.5	Meta (Llama 4 MODEL_CARD.md, llama-models commit 038acb9)	meta_llama4_model_card_038acb9 html	m2025.llama4-card.llama-4-maverick.mmlu-pro
OpenAI o3 (high)	MMMLU (14-language average)	reasoning effort high	88.8	OpenAI (gpt-oss Model Card, Table 2)	arxiv_2508.10925 p.10	m2025.gpt-oss.o3-high.mmmlu-avg
Qwen3-235B-A22B (Non-thinking)	MATH-500	non-thinking mode	91.2	Alibaba Qwen (Qwen3 TR, Table 12)	arxiv_2505.09388 p.14	m2025.qwen3.qwen3-235b-nothink.math500
Qwen3-235B-A22B (Thinking)	AIME 2024	thinking mode	85.7	Alibaba Qwen (Qwen3 TR, Table 11)	arxiv_2505.09388 p.14	m2025.qwen3.qwen3-235b-think.aime24
Qwen3-235B-A22B (Thinking)	AIME 2025	thinking mode	81.5	Alibaba Qwen (Qwen3 TR, Table 11)	arxiv_2505.09388 p.14	m2025.qwen3.qwen3-235b-think.aime25
Qwen3-235B-A22B (Thinking)	GPQA Diamond	thinking mode	71.1	Alibaba Qwen (Qwen3 TR, Table 11)	arxiv_2505.09388 p.14	m2025.qwen3.qwen3-235b-think.gpqa
Qwen3-235B-A22B (Thinking)	LiveCodeBench v5	thinking mode	70.7	Alibaba Qwen (Qwen3 TR, Table 11)	arxiv_2505.09388 p.14	m2025.qwen3.qwen3-235b-think.lcb-v5
Qwen3-235B-A22B (Thinking)	MATH-500	thinking mode	98.0	Alibaba Qwen (Qwen3 TR, Table 11)	arxiv_2505.09388 p.14	m2025.qwen3.qwen3-235b-think.math500
Qwen3-235B-A22B (Thinking)	MMLU-Redux	thinking mode	92.7	Alibaba Qwen (Qwen3 TR, Table 11)	arxiv_2505.09388 p.14	m2025.qwen3.qwen3-235b-think.mmlu-redux
gpt-4.1-2025-04-14	HumanEval	0-shot CoT, assistant system prompt	94.5	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	m2025.simple-evals.gpt-4.1.humaneval
o3	SWE-bench Verified	launch chart	69.1	OpenAI (o3 and o4-mini launch post, Apr 16 2025, chart data)	openai_o3_o4mini_blog html	m2025.o3-blog.o3.swebench
o3 (no tools)	AIME 2024	no tools	91.6	OpenAI (o3 and o4-mini launch post, Apr 16 2025, chart data)	openai_o3_o4mini_blog html	m2025.o3-blog.o3.aime24.notools
o3 (no tools)	AIME 2025	no tools	88.9	OpenAI (o3 and o4-mini launch post, Apr 16 2025, chart data)	openai_o3_o4mini_blog html	m2025.o3-blog.o3.aime25.notools
o3 (no tools)	GPQA Diamond	no tools	83.3	OpenAI (o3 and o4-mini launch post, Apr 16 2025, chart data)	openai_o3_o4mini_blog html	m2025.o3-blog.o3.gpqa.notools
o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	MMLU	0-shot CoT, no tools	92.9	OpenAI (simple-evals README, commit 6ead091, 2025-07-09)	openai_simple_evals_readme md	m2025.simple-evals.o3.mmlu
o4-mini	SWE-bench Verified	launch chart	68.1	OpenAI (o3 and o4-mini launch post, Apr 16 2025, chart data)	openai_o3_o4mini_blog html	m2025.o3-blog.o4-mini.swebench

o4-mini (no tools)	AIME 2024	no tools	93.4	OpenAI (o3 and o4-mini launch post, Apr 16 2025, chart data)	openai_o3_o4mini_blog html	m2025.o3-blog.o4-mini.aime24.notools
o4-mini (no tools)	AIME 2025	no tools	92.7	OpenAI (o3 and o4-mini launch post, Apr 16 2025, chart data)	openai_o3_o4mini_blog html	m2025.o3-blog.o4-mini.aime25.notools
o4-mini (no tools)	GPQA Diamond	no tools	81.4	OpenAI (o3 and o4-mini launch post, Apr 16 2025, chart data)	openai_o3_o4mini_blog html	m2025.o3-blog.o4-mini.gpqa.notools
o4-mini (with python only)	AIME 2025	with Python interpreter	99.5	OpenAI (o3 and o4-mini launch post, Apr 16 2025, chart data)	openai_o3_o4mini_blog html	m2025.o3-blog.o4-mini.aime25.python

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.3	o1 (Sept 2024 pre-release)	pass@1	2024-09	m2024.o1blog.o1.mmlu.pass1
MMLU-Pro	84.0	DeepSeek-R1	EM	2025-01	m2025.r1v1.deepseek-r1.mmlu-pro.em
HumanEval	93.7	Claude 3.5 Sonnet (New)	0-shot	2024-10	m2024.c35oct.claude-3.5-sonnet-new.humaneval.0-shot
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	m2024.qwen25.qwen2.5-72b-instruct.math.instruct
MATH-500	97.3	DeepSeek-R1	Pass@1	2025-01	m2025.r1v1.deepseek-r1.math500.pass1
GPQA Diamond	83.3	o3 (no tools)	no tools	2025-04	m2025.o3-blog.o3.gpqa.notools
SWE-bench Verified	69.1	o3	launch chart	2025-04	m2025.o3-blog.o3.swebench
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	m2025.o3-blog.o4-mini.aime24.notools
AIME 2025	99.5	o4-mini (with python only)	with Python interpreter	2025-04	m2025.o3-blog.o4-mini.aime25.python

2025-05

May 2025. 19 verified scores for 5 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Claude Opus 4	GPQA Diamond	without extended thinking	74.9	Anthropic (Claude 4 launch post, May 22 2025)	anthropic_claude4_blog html	m2025.claude4-blog.claude-opus-4.gpqa.no-et
Claude Opus 4	MMMLU	without extended thinking	87.4	Anthropic (Claude 4 launch post, May 22 2025)	anthropic_claude4_blog html	m2025.claude4-blog.claude-opus-4.mmmlu.no-et
Claude Opus 4	SWE-bench Verified (full 500)	no extended thinking, bash + edit-tool scaffold	72.5	Anthropic (Claude 4 launch post, May 22 2025)	anthropic_claude4_blog html	m2025.claude4-blog.claude-opus-4.swebench
Claude Opus 4	SWE-bench Verified (full 500)	high compute: parallel attempts + scoring model	79.4	Anthropic (Claude 4 launch post, May 22 2025)	anthropic_claude4_blog html	m2025.claude4-blog.claude-opus-4.swebench.parallel
Claude Sonnet 4	MMMLU	without extended thinking	85.4	Anthropic (Claude 4 launch post, May 22 2025)	anthropic_claude4_blog html	m2025.claude4-blog.claude-sonnet-4.mmmlu.no-et
Claude Sonnet 4	SWE-bench Verified (full 500)	no extended thinking, bash + edit-tool scaffold	72.7	Anthropic (Claude 4 launch post, May 22 2025)	anthropic_claude4_blog html	m2025.claude4-blog.claude-sonnet-4.swebench
Claude Sonnet 4	SWE-bench Verified (full 500)	high compute: parallel attempts + scoring model	80.2	Anthropic (Claude 4 launch post, May 22 2025)	anthropic_claude4_blog html	m2025.claude4-blog.claude-sonnet-4.swebench.parallel

DeepSeek-R1-0528	AIME 2024	Pass@1 (16 samples)	91.4	DeepSeek (DeepSeek-R1-0528 Hugging Face model card)	hf_deepseek_r1_0528_jun html	m2025.hf-r1-0528.deepseek-r1-0528.aime24
DeepSeek-R1-0528	AIME 2025	Pass@1 (16 samples)	87.5	DeepSeek (DeepSeek-R1-0528 Hugging Face model card)	hf_deepseek_r1_0528_jun html	m2025.hf-r1-0528.deepseek-r1-0528.aime25
DeepSeek-R1-0528	GPQA Diamond	Pass@1 (16 samples)	81.0	DeepSeek (DeepSeek-R1-0528 Hugging Face model card)	hf_deepseek_r1_0528_jun html	m2025.hf-r1-0528.deepseek-r1-0528.gpqa
DeepSeek-R1-0528	LiveCodeBench (2408-2505)	Pass@1	73.3	DeepSeek (DeepSeek-R1-0528 Hugging Face model card)	hf_deepseek_r1_0528_jun html	m2025.hf-r1-0528.deepseek-r1-0528.lcb
DeepSeek-R1-0528	MMLU-Pro	EM	85.0	DeepSeek (DeepSeek-R1-0528 Hugging Face model card)	hf_deepseek_r1_0528_jun html	m2025.hf-r1-0528.deepseek-r1-0528.mmlu-pro
DeepSeek-R1-0528	MMLU-Redux	EM	93.4	DeepSeek (DeepSeek-R1-0528 Hugging Face model card)	hf_deepseek_r1_0528_jun html	m2025.hf-r1-0528.deepseek-r1-0528.mmlu-redux
DeepSeek-R1-0528	SWE-bench Verified	Resolved, Agentless framework (card note: 'We use Agentless framework to evaluate model performance on SWE-Verified')	57.6	DeepSeek (DeepSeek-R1-0528 Hugging Face model card)	hf_deepseek_r1_0528_jun html	m2025.hf-r1-0528.deepseek-r1-0528.swebench
Vicuna-13b	HumanEval	pass@1	12.5	Microsoft (WizardLM v3, May 2025)	arxiv_2304.12244v3 p.7	m2023.xu2023v3.vicuna-13b.humaneval
Vicuna-13b	MMLU	OpenLLM harness (shots not stated)	50.84	Microsoft (WizardLM v3, May 2025)	arxiv_2304.12244v3 p.7	m2023.xu2023v3.vicuna-13b.mmlu
WizardLM-13b	GSM8K	4-shot, pass@1	37.15	Microsoft (WizardLM v3, May 2025)	arxiv_2304.12244v3 p.7	m2023.xu2023v3.wizardlm-13b.gsm8k
WizardLM-13b	HumanEval	pass@1	24.0	Microsoft (WizardLM v3, May 2025)	arxiv_2304.12244v3 p.7	m2023.xu2023v3.wizardlm-13b.humaneval
WizardLM-13b	MMLU	OpenLLM harness (shots not stated)	52.92	Microsoft (WizardLM v3, May 2025)	arxiv_2304.12244v3 p.7	m2023.xu2023v3.wizardlm-13b.mmlu

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.3	o1 (Sept 2024 pre-release)	pass@1	2024-09	m2024.o1blog.o1.mmlu.pass1
MMLU-Pro	85.0	DeepSeek-R1-0528	EM	2025-05	m2025.hf-r1-0528.deepseek-r1-0528.mmlu-pro
HumanEval	93.7	Claude 3.5 Sonnet (New)	0-shot	2024-10	m2024.c35oct.claude-3.5-sonnet-new.humaneval.0-shot
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	m2024.qwen25.qwen2.5-72b-instruct.math.instruct
MATH-500	98.0	Qwen3-235B-A22B (Thinking)	thinking mode	2025-05	m2025.qwen3.qwen3-235b-think.math500
GPQA Diamond	84.0	Gemini2.5-Pro	as reported by Qwen	2025-05	m2025.qwen3.gemini-2.5-pro.gpqa
SWE-bench Verified	80.2	Claude Sonnet 4	high compute: parallel attempts + scoring model	2025-05	m2025.claude4-blog.claude-sonnet-4.swebench.parallel
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	m2025.o3-blog.o4-mini.aime24.notools
AIME 2025	99.5	o4-mini (with python only)	with Python interpreter	2025-04	m2025.o3-blog.o4-mini.aime25.python

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Qwen3 Technical Report, v1, PDF page 14: Table 11: Comparison among Qwen3-235B-A22B (Thinking) and other reasoning baselines.; Table 12: Comparison among Qwen3-235B-A22B (Non-thinking) and other non-reasoning baselines.

2025-06

June 2025. 12 verified scores for 3 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Gemini 2.5 Pro	AIME 2025	single attempt	88.0	Google (Gemini 2.5 TR v1, Table 4)	arxiv_2507.06261 p.14	m2025.gemini25-tr.gemini-2.5-pro.aime25
Gemini 2.5 Pro	GPQA Diamond	single attempt	86.4	Google (Gemini 2.5 TR v1, Table 4)	arxiv_2507.06261 p.14	m2025.gemini25-tr.gemini-2.5-pro.gpqa
Gemini 2.5 Pro	Global MMLU (Lite)	Table 3	89.2	Google (Gemini 2.5 TR v1, Table 3)	arxiv_2507.06261 p.13	m2025.gemini25-tr.gemini-2.5-pro.global-mmlu-lite
Gemini 2.5 Pro	LiveCodeBench	Table 4	74.2	Google (Gemini 2.5 TR v1, Table 4)	arxiv_2507.06261 p.14	m2025.gemini25-tr.gemini-2.5-pro.lcb
Gemini 2.5 Pro	SWE-bench Verified	multiple attempts	67.2	Google (Gemini 2.5 TR v1, Table 4)	arxiv_2507.06261 p.14	m2025.gemini25-tr.gemini-2.5-pro.swebench.multi
Gemini 2.5 Pro	SWE-bench Verified	single attempt	59.6	Google (Gemini 2.5 TR v1, Table 4)	arxiv_2507.06261 p.14	m2025.gemini25-tr.gemini-2.5-pro.swebench.single
Magistral Medium	LiveCodeBench v5	average over 16 runs	59.4	Mistral AI (Magistral TR, Table 2)	arxiv_2506.10910 p.9	m2025.magistral.magistral-medium.lcb-v5
MiniMax-M1-80k	AIME 2024	80K thinking budget	86.0	MiniMax (MiniMax-M1 TR v1, Table 2)	arxiv_2506.13585 p.13	m2025.minimax-m1.minimax-m1-80k.aime24
MiniMax-M1-80k	GPQA Diamond	80K thinking budget	70.0	MiniMax (MiniMax-M1 TR v1, Table 2)	arxiv_2506.13585 p.13	m2025.minimax-m1.minimax-m1-80k.gpqa
MiniMax-M1-80k	LiveCodeBench (24/8-25/5)	80K thinking budget	65.0	MiniMax (MiniMax-M1 TR v1, Table 2)	arxiv_2506.13585 p.13	m2025.minimax-m1.minimax-m1-80k.lcb
MiniMax-M1-80k	MATH-500	80K thinking budget	96.8	MiniMax (MiniMax-M1 TR v1, Table 2)	arxiv_2506.13585 p.13	m2025.minimax-m1.minimax-m1-80k.math500
MiniMax-M1-80k	MMLU-Pro	80K thinking budget	81.1	MiniMax (MiniMax-M1 TR v1, Table 2)	arxiv_2506.13585 p.13	m2025.minimax-m1.minimax-m1-80k.mmlu-pro

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.3	o1 (Sept 2024 pre-release)	pass@1	2024-09	m2024.o1blog.o1.mmlu.pass1
MMLU-Pro	85.0	DeepSeek-R1-0528	EM	2025-05	m2025.hf-r1-0528.deepseek-r1-0528.mmlu-pro
HumanEval	93.7	Claude 3.5 Sonnet (New)	0-shot	2024-10	m2024.c35oct.claude-3.5-sonnet-new.humaneval.0-shot
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	m2024.qwen25.qwen2.5-72b-instruct.math.instruct
MATH-500	98.0	Qwen3-235B-A22B (Thinking)	thinking mode	2025-05	m2025.qwen3.qwen3-235b-think.math500
GPQA Diamond	84.0	Gemini2.5-Pro	as reported by Qwen	2025-05	m2025.qwen3.gemini-2.5-pro.gpqa
SWE-bench Verified	80.2	Claude Sonnet 4	high compute: parallel attempts + scoring model	2025-05	m2025.claude4-blog.claude-sonnet-4.swebench.parallel
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	m2025.o3-blog.o4-mini.aime24.notools
AIME 2025	99.5	o4-mini (with python only)	with Python interpreter	2025-04	m2025.o3-blog.o4-mini.aime25.python

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Magistral, v1, PDF page 10: Table 3: Performance of Magistral Small compared with different training setups across various bench-.; Table 4: Magistral Medium's pass@1 performance on multilingual versions of the AIME 2024 benchmark.; Table 5: Cross-domain generalization during math-only and code-only RL for a 24B model.

2025-07

July 2025. 17 verified scores for 6 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
GLM-4.5	LiveCodeBench (2407-2501)	GLM evaluation	72.9	Zhipu AI (GLM-4.5 TR, Table 4)	arxiv_2508.06471 p.16	m2025.glm45.glm-4.5.lcb
GLM-4.5	MATH-500	GLM evaluation	98.2	Zhipu AI (GLM-4.5 TR, Table 4)	arxiv_2508.06471 p.16	m2025.glm45.glm-4.5.math500
GLM-4.5	MMLU-Pro	GLM evaluation	84.6	Zhipu AI (GLM-4.5 TR, Table 4)	arxiv_2508.06471 p.16	m2025.glm45.glm-4.5.mmlu-pro
Grok 4	MMLU-Pro	as evaluated/reported by Zhipu	86.6	Zhipu AI (GLM-4.5 TR, Table 4)	arxiv_2508.06471 p.16	m2025.glm45.grok-4.mmlu-pro
Grok 4 (no tools) [chart code GROK_4_NT]	AIME 2025	no tools	91.7	xAI (Grok 4 launch post, Jul 9 2025, chart data)	xai_grok4_blog html	m2025.grok4-blog.grok-4.aime25.notools
Grok 4 (no tools) [chart code GROK_4_NT]	GPQA	no tools	87.5	xAI (Grok 4 launch post, Jul 9 2025, chart data)	xai_grok4_blog html	m2025.grok4-blog.grok-4.gpqa.notools
Grok 4 (no tools) [chart code GROK_4_NT]	LiveCodeBench (Jan-May)	no tools	79.0	xAI (Grok 4 launch post, Jul 9 2025, chart data)	xai_grok4_blog html	m2025.grok4-blog.grok-4.lcb.notools
Grok 4 Heavy [chart code GROK_4_HEAVY]	AIME 2025	with tools (Python)	100.0	xAI (Grok 4 launch post, Jul 9 2025, chart data)	xai_grok4_blog html	m2025.grok4-blog.grok-4-heavy.aime25.tools
Grok 4 Heavy [chart code GROK_4_HEAVY]	GPQA	with tools	88.4	xAI (Grok 4 launch post, Jul 9 2025, chart data)	xai_grok4_blog html	m2025.grok4-blog.grok-4-heavy.gpqa.tools
Grok 4 Heavy [chart code GROK_4_HEAVY]	LiveCodeBench (Jan-May)	with tools	79.4	xAI (Grok 4 launch post, Jul 9 2025, chart data)	xai_grok4_blog html	m2025.grok4-blog.grok-4-heavy.lcb.tools
Kimi-K2-Instruct	MATH-500	Acc.	97.4	Moonshot AI (Kimi K2 TR, Table 3)	arxiv_2507.20534 p.16	m2025.kimi-k2.kimi-k2-instruct.math500
Kimi-K2-Instruct	MMLU	EM	89.5	Moonshot AI (Kimi K2 TR, Table 3)	arxiv_2507.20534 p.16	m2025.kimi-k2.kimi-k2-instruct.mmlu
Kimi-K2-Instruct	MMLU-Pro	EM	81.1	Moonshot AI (Kimi K2 TR, Table 3)	arxiv_2507.20534 p.16	m2025.kimi-k2.kimi-k2-instruct.mmlu-pro
Qwen3-235B-A22B-Thinking-2507	AIME 2025	model card table	92.3	Alibaba Qwen (Qwen3-235B-A22B-Thinking-2507 Hugging Face model card)	hf_qwen3_235b_thinking_2507 html	m2025.hf-qwen3-2507.qwen3-235b-thinking-2507.aime25
Qwen3-235B-A22B-Thinking-2507	GPQA	model card table	81.1	Alibaba Qwen (Qwen3-235B-A22B-Thinking-2507 Hugging Face model card)	hf_qwen3_235b_thinking_2507 html	m2025.hf-qwen3-2507.qwen3-235b-thinking-2507.gpqa
Qwen3-235B-A22B-Thinking-2507	LiveCodeBench v6 (25.02-25.05)	model card table	74.1	Alibaba Qwen (Qwen3-235B-A22B-Thinking-2507 Hugging Face model card)	hf_qwen3_235b_thinking_2507 html	m2025.hf-qwen3-2507.qwen3-235b-thinking-2507.lcb-v6
Qwen3-235B-A22B-Thinking-2507	MMLU-Pro	model card table	84.4	Alibaba Qwen (Qwen3-235B-A22B-Thinking-2507 Hugging Face model card)	hf_qwen3_235b_thinking_2507 html	m2025.hf-qwen3-2507.qwen3-235b-thinking-2507.mmlu-pro

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.9	o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	0-shot CoT, no tools	2025-07	m2025.simple-evals.o3.mmlu
MMLU-Pro	85.0	DeepSeek-R1-0528	EM	2025-05	m2025.hf-r1-0528.deepseek-r1-0528.mmlu-pro
HumanEval	97.6	o3-mini-high	0-shot CoT (simple-evals)	2025-07	m2025.simple-evals.o3-mini-high.humaneval
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	m2024.qwen25.qwen2.5-72b-instruct.math.instruct
MATH-500	98.0	Qwen3-235B-A22B (Thinking)	thinking mode	2025-05	m2025.qwen3.qwen3-235b-think.math500
GPQA Diamond	86.4	Gemini 2.5 Pro	single attempt	2025-07	m2025.gemini25-tr.gemini-2.5-pro.gpqa

SWE-bench Verified	80.2	Claude Sonnet 4	high compute: parallel attempts + scoring model	2025-05	`m2025.claude4-blog.claude-sonnet-4.swebench.parallel`
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	`m2025.o3-blog.o4-mini.aime24.notools`
AIME 2025	100.0	Grok 4 Heavy [chart code GROK_4_HEAVY]	with tools (Python)	2025-07	`m2025.grok4-blog.grok-4-heavy.aime25.tools`

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Kimi K2: Open Agentic Intelligence, v2, PDF page 1: Figure 1: Kimi K2 main results.2.

2025-08

August 2025. 13 verified scores for 8 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Claude Opus 4.1	SWE-bench Verified	no extended thinking; simple two-tool scaffold (bash + string-replace file editor); full 500 problems	74.5	Anthropic (Claude Opus 4.1 post, Aug 5 2025)	anthropic_claude_opus41_blog html	`m2025.opus41-blog.claude-opus-4.1.swebench`
DeepSeek V3.1-NonThinking	SWE-bench Verified	agent mode	66.0	DeepSeek (DeepSeek-V3.1 Hugging Face model card)	hf_deepseek_v31 html	`m2025.hf-v31.deepseek-v3.1-nonthinking.swebench`
DeepSeek V3.1-Thinking	AIME 2025	Pass@1	88.4	DeepSeek (DeepSeek-V3.1 Hugging Face model card)	hf_deepseek_v31 html	`m2025.hf-v31.deepseek-v3.1-thinking.aime25`
DeepSeek V3.1-Thinking	GPQA Diamond	Pass@1	80.1	DeepSeek (DeepSeek-V3.1 Hugging Face model card)	hf_deepseek_v31 html	`m2025.hf-v31.deepseek-v3.1-thinking.gpqa`
DeepSeek V3.1-Thinking	LiveCodeBench (2408-2505)	Pass@1	74.8	DeepSeek (DeepSeek-V3.1 Hugging Face model card)	hf_deepseek_v31 html	`m2025.hf-v31.deepseek-v3.1-thinking.lcb`
DeepSeek V3.1-Thinking	MMLU-Pro	EM	84.8	DeepSeek (DeepSeek-V3.1 Hugging Face model card)	hf_deepseek_v31 html	`m2025.hf-v31.deepseek-v3.1-thinking.mmlu-pro`
GPT-5 (thinking)	AIME 2025	without tools	94.6	OpenAI (GPT-5 launch post, Aug 7 2025)	openai_gpt5_blog html	`m2025.gpt5-blog.gpt-5.aime25.notools`
GPT-5 (thinking)	SWE-bench Verified	launch prose	74.9	OpenAI (GPT-5 launch post, Aug 7 2025)	openai_gpt5_blog html	`m2025.gpt5-blog.gpt-5.swebench`
GPT-5 pro	GPQA Diamond	without tools, extended reasoning	88.4	OpenAI (GPT-5 launch post, Aug 7 2025)	openai_gpt5_blog html	`m2025.gpt5-blog.gpt-5-pro.gpqa.notools`
GPT-5 pro (no tools)	AIME 2025	no tools	96.7	OpenAI (GPT-5 launch post, chart data)	openai_gpt5_blog html	`m2025.gpt5-blog.gpt-5-pro.aime25.notools`
GPT-5 pro (python)	AIME 2025	with Python tool	100.0	OpenAI (GPT-5 launch post, Aug 7 2025) chart data	openai_gpt5_blog html	`m2025.gpt5-blog.gpt-5-pro.aime25.python`
gpt-oss-120b (high)	AIME 2025	no tools, reasoning effort high	92.5	OpenAI (gpt-oss-120b & gpt-oss-20b Model Card, Table 3)	arxiv_2508.10925 p.11	`m2025.gpt-oss.gpt-oss-120b-high.aime25.notools`
gpt-oss-120b (high)	MMLU	reasoning effort high	90.0	OpenAI (gpt-oss-120b & gpt-oss-20b Model Card, Table 3)	arxiv_2508.10925 p.11	`m2025.gpt-oss.gpt-oss-120b-high.mmlu`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.9	o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	0-shot CoT, no tools	2025-07	`m2025.simple-evals.o3.mmlu`

MMLU-Pro	86.6	Grok 4	as evaluated/reported by Zhipu	2025-08	`m2025.glm45.grok-4.mmlu-pro`
HumanEval	97.6	o3-mini-high	0-shot CoT (simple-evals)	2025-07	`m2025.simple-evals.o3-mini-high.humaneval`
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	`m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot`
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	`m2024.qwen25.qwen2.5-72b-instruct.math.instruct`
MATH-500	98.2	GLM-4.5	GLM evaluation	2025-08	`m2025.glm45.glm-4.5.math500`
GPQA Diamond	88.4	GPT-5 pro	without tools, extended reasoning	2025-08	`m2025.gpt5-blog.gpt-5-pro.gpqa.notools`
SWE-bench Verified	80.2	Claude Sonnet 4	high compute: parallel attempts + scoring model	2025-05	`m2025.claude4-blog.claude-sonnet-4.swebench.parallel`
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	`m2025.o3-blog.o4-mini.aime24.notools`
AIME 2025	100.0	Grok 4 Heavy [chart code GROK_4_HEAVY]	with tools (Python)	2025-07	`m2025.grok4-blog.grok-4-heavy.aime25.tools`

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- GLM-4.5: Agentic, Reasoning, and Coding (ARC) Foundation Models, v1, PDF page 16: Figure 11: One example of user prompt we used for TAU-bench.; Table 4: Results on Reasoning Benchmarks.; Table 5: Results on SWE-bench Verified and Terminal-Bench.
- gpt-oss-120b & gpt-oss-20b Model Card, v1, PDF page 6: Figure 1: Main capabilities evaluations.

2025-09

September 2025. 12 verified scores for 4 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Claude Sonnet 4.5	SWE-bench Verified (full 500)	10-trial average, no test-time compute, 200K thinking budget, two-tool scaffold	77.2	Anthropic (Claude Sonnet 4.5 launch post, Sep 29 2025, methodology)	anthropic_claude_sonnet45_blog html	`m2025.sonnet45-blog.claude-sonnet-4.5.swebench`
Claude Sonnet 4.5	SWE-bench Verified (full 500)	high compute: parallel attempts + scoring model	82.0	Anthropic (Claude Sonnet 4.5 launch post, Sep 29 2025, methodology)	anthropic_claude_sonnet45_blog html	`m2025.sonnet45-blog.claude-sonnet-4.5.swebench.parallel`
Claude Sonnet 4.5 (Thinking)	AIME 2025	no tools; non-asterisked baseline 'quoted from' the list of public sources in footnote 3.1 (GPT-5 post, GPT-5 for Developers, GPT-5 system card, claude-sonnet-4-5 post, grok-4 post, deepseek-v3.2 post, Terminal-Bench and Vals AI leaderboards, artificialanalysis); per-cell source not stated	87.0	Moonshot AI (Kimi-K2-Thinking Hugging Face model card)	hf_kimi_k2_thinking html	`m2025.hf-k2-thinking.claude-sonnet-4.5.aime25.notools`
Claude Sonnet 4.5 (Thinking)	AIME 2025	with Python; non-asterisked baseline quoted from the public sources listed in footnote 3.1; per-cell source not stated	100.0	Moonshot AI (Kimi-K2-Thinking Hugging Face model card)	hf_kimi_k2_thinking html	`m2025.hf-k2-thinking.claude-sonnet-4.5.aime25.python`
DeepSeek-V3.2-Exp	AIME 2025	reasoning mode, without tool use	89.3	DeepSeek (DeepSeek-V3.2-Exp Hugging Face model card)	hf_deepseek_v32_exp html	`m2025.hf-v32exp.deepseek-v3.2-exp.aime25`
DeepSeek-V3.2-Exp	GPQA Diamond	reasoning mode, without tool use	79.9	DeepSeek (DeepSeek-V3.2-Exp Hugging Face model card)	hf_deepseek_v32_exp html	`m2025.hf-v32exp.deepseek-v3.2-exp.gpqa`
DeepSeek-V3.2-Exp	LiveCodeBench	reasoning mode, without tool use	74.1	DeepSeek (DeepSeek-V3.2-Exp Hugging Face model card)	hf_deepseek_v32_exp html	`m2025.hf-v32exp.deepseek-v3.2-exp.lcb`

DeepSeek-V3.2-Exp	MMLU-Pro	reasoning mode, without tool use	85.0	DeepSeek (DeepSeek-V3.2-Exp Hugging Face model card)	hf_deepseek_v32_exp html	m2025.hf-v32exp.deepseek-v3.2-exp.mmlu-pro
GLM-4.6	AIME 2025	as reported by Z.ai	93.9	Zhipu AI / Z.ai (GLM-4.7 launch post, table)	zai_glm47_blog html	m2025.glm47-blog.glm-4.6.aime25
GLM-4.6	LiveCodeBench v6	as reported by Z.ai	82.8	Zhipu AI / Z.ai (GLM-4.7 launch post, table)	zai_glm47_blog html	m2025.glm47-blog.glm-4.6.lcb-v6
GLM-4.6	MMLU-Pro	as reported by Z.ai	83.2	Zhipu AI / Z.ai (GLM-4.7 launch post, table)	zai_glm47_blog html	m2025.glm47-blog.glm-4.6.mmlu-pro
GLM-4.6	SWE-bench Verified	as reported by Z.ai	68.0	Zhipu AI / Z.ai (GLM-4.7 launch post, table)	zai_glm47_blog html	m2025.glm47-blog.glm-4.6.swebench

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.9	o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	0-shot CoT, no tools	2025-07	m2025.simple-evals.o3.mmlu
MMLU-Pro	86.6	Grok 4	as evaluated/reported by Zhipu	2025-08	m2025.glm45.grok-4.mmlu-pro
HumanEval	97.6	o3-mini-high	0-shot CoT (simple-evals)	2025-07	m2025.simple-evals.o3-mini-high.humaneval
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	m2024.qwen25.qwen2.5-72b-instruct.math.instruct
MATH-500	98.2	GLM-4.5	GLM evaluation	2025-08	m2025.glm45.glm-4.5.math500
GPQA Diamond	88.4	GPT-5 pro	without tools, extended reasoning	2025-08	m2025.gpt5-blog.gpt-5-pro.gpqa.notools
SWE-bench Verified	82.0	Claude Sonnet 4.5	high compute: parallel attempts + scoring model	2025-09	m2025.sonnet45-blog.claude-sonnet-4.5.swebench.parallel
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	m2025.o3-blog.o4-mini.aime24.notools
AIME 2025	100.0	Grok 4 Heavy [chart code GROK_4_HEAVY]	with tools (Python)	2025-07	m2025.grok4-blog.grok-4-heavy.aime25.tools

2025-10

October 2025. 1 verified scores for 1 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Claude Haiku 4.5	SWE-bench Verified (full 500)	50-trial average, no test-time compute, 128K thinking budget, default sampling, full 500 problems, two-tool scaffold, WITH a prompt addition ('You should use tools as much as possible, ideally more than 100 times...')	73.3	Anthropic (Claude Haiku 4.5 post, Oct 15 2025, methodology)	anthropic_claude_haiku45_blog html	m2025.haiku45-blog.claude-haiku-4.5.swebench

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.9	o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	0-shot CoT, no tools	2025-07	m2025.simple-evals.o3.mmlu
MMLU-Pro	86.6	Grok 4	as evaluated/reported by Zhipu	2025-08	m2025.glm45.grok-4.mmlu-pro
HumanEval	97.6	o3-mini-high	0-shot CoT (simple-evals)	2025-07	m2025.simple-evals.o3-mini-high.humaneval

GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	`m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot`
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	`m2024.qwen25.qwen2.5-72b-instruct.math.instruct`
MATH-500	98.2	GLM-4.5	GLM evaluation	2025-08	`m2025.glm45.glm-4.5.math500`
GPQA Diamond	88.4	GPT-5 pro	without tools, extended reasoning	2025-08	`m2025.gpt5-blog.gpt-5-pro.gpqa.notools`
SWE-bench Verified	82.0	Claude Sonnet 4.5	high compute: parallel attempts + scoring model	2025-09	`m2025.sonnet45-blog.claude-sonnet-4.5.swebench.parallel`
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	`m2025.o3-blog.o4-mini.aime24.notools`
AIME 2025	100.0	Grok 4 Heavy [chart code GROK_4_HEAVY]	with tools (Python)	2025-07	`m2025.grok4-blog.grok-4-heavy.aime25.tools`

2025-11

November 2025. 19 verified scores for 6 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Claude Opus 4.5	AIME 2025	extended thinking	92.8	Moonshot AI (Kimi K2.5 TR, Table 4)	arxiv_2602.02276 p.12	`m2025.kimi-k25.claude-opus-4.5.aime25`
Claude Opus 4.5	GPQA Diamond	extended thinking	87.0	Moonshot AI (Kimi K2.5 TR, Table 4)	arxiv_2602.02276 p.12	`m2025.kimi-k25.claude-opus-4.5.gpqa`
Claude Opus 4.5	MMLU-Pro	extended thinking; starred = Moonshot internal evaluation	89.3	Moonshot AI (Kimi K2.5 TR, Table 4)	arxiv_2602.02276 p.12	`m2025.kimi-k25.claude-opus-4.5.mmlu-pro`
Claude Opus 4.5	SWE-bench Verified	extended thinking	80.9	Moonshot AI (Kimi K2.5 TR, Table 4)	arxiv_2602.02276 p.12	`m2025.kimi-k25.claude-opus-4.5.swebench`
GPT-5.1 Thinking	AIME 2025	no tools, high effort	94.0	OpenAI (GPT-5.2 launch post, Dec 11 2025)	openai_gpt52_blog html	`m2025.gpt52-blog.gpt-5.1-thinking.aime25.notools`
GPT-5.1 Thinking	GPQA Diamond	no tools, high effort	88.1	OpenAI (GPT-5.2 launch post, Dec 11 2025)	openai_gpt52_blog html	`m2025.gpt52-blog.gpt-5.1-thinking.gpqa.notools`
GPT-5.1 Thinking	MMMLU	high effort	89.5	OpenAI (GPT-5.2 launch post, Dec 11 2025)	openai_gpt52_blog html	`m2025.gpt52-blog.gpt-5.1-thinking.mmmlu`
GPT-5.1 Thinking	SWE-bench Verified	high effort	76.3	OpenAI (GPT-5.2 launch post, Dec 11 2025)	openai_gpt52_blog html	`m2025.gpt52-blog.gpt-5.1-thinking.swebench`
Gemini 3 Deep Think	GPQA Diamond	launch prose (Deep Think mode)	93.8	Google (Gemini 3 launch post, Nov 18 2025)	google_gemini3_blog html	`m2025.gemini3-blog.gemini-3-deep-think.gpqa`
Gemini 3 Pro	GPQA Diamond	launch prose	91.9	Google (Gemini 3 launch post, Nov 18 2025)	google_gemini3_blog html	`m2025.gemini3-blog.gemini-3-pro.gpqa`
Gemini 3 Pro	SWE-bench Verified	launch prose	76.2	Google (Gemini 3 launch post, Nov 18 2025)	google_gemini3_blog html	`m2025.gemini3-blog.gemini-3-pro.swebench`
Gemini-3.0 Pro	AIME 2025	Pass@1	95.0	DeepSeek (DeepSeek-V3.2 TR, Table 2)	arxiv_2512.02556 p.13	`m2025.dsv32.gemini-3-pro.aime25`
Gemini-3.0 Pro	LiveCodeBench	Pass@1-COT	90.7	DeepSeek (DeepSeek-V3.2 TR, Table 2)	arxiv_2512.02556 p.13	`m2025.dsv32.gemini-3-pro.lcb`
Gemini-3.0 Pro	MMLU-Pro	EM	90.1	DeepSeek (DeepSeek-V3.2 TR, Table 2)	arxiv_2512.02556 p.13	`m2025.dsv32.gemini-3-pro.mmlu-pro`

Kimi K2 Thinking	AIME 2025	no tools, avg@32	94.5	Moonshot AI (Kimi-K2-Thinking Hugging Face model card)	hf_kimi_k2_thinking html	m2025.hf-k2-thinking.kimi-k2-thinking.aime25.notools
Kimi K2 Thinking	AIME 2025	with Python, avg@16	99.1	Moonshot AI (Kimi-K2-Thinking Hugging Face model card)	hf_kimi_k2_thinking html	m2025.hf-k2-thinking.kimi-k2-thinking.aime25.python
Kimi K2 Thinking	LiveCodeBench v6	no tools; 128k thinking-token budget (2.2); in-house harness with rewritten system prompt, averaged over 5 runs (5.2-5.3)	83.1	Moonshot AI (Kimi-K2-Thinking Hugging Face model card)	hf_kimi_k2_thinking html	m2025.hf-k2-thinking.kimi-k2-thinking.lcb-v6
Kimi K2 Thinking	MMLU-Pro	no tools	84.6	Moonshot AI (Kimi-K2-Thinking Hugging Face model card)	hf_kimi_k2_thinking html	m2025.hf-k2-thinking.kimi-k2-thinking.mmlu-pro
Kimi K2 Thinking	SWE-bench Verified	with tools; in-house harness derived from SWE-agent with clamped Bash/Edit context and REWRITTEN system prompt; averaged over 5 runs (footnotes 5.2-5.3); INT4	71.3	Moonshot AI (Kimi-K2-Thinking Hugging Face model card)	hf_kimi_k2_thinking html	m2025.hf-k2-thinking.kimi-k2-thinking.swebench

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.9	o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	0-shot CoT, no tools	2025-07	m2025.simple-evals.o3.mmlu
MMLU-Pro	86.6	Grok 4	as evaluated/reported by Zhipu	2025-08	m2025.glm45.grok-4.mmlu-pro
HumanEval	97.6	o3-mini-high	0-shot CoT (simple-evals)	2025-07	m2025.simple-evals.o3-mini-high.humaneval
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	m2024.qwen25.qwen2.5-72b-instruct.math.instruct
MATH-500	98.2	GLM-4.5	GLM evaluation	2025-08	m2025.glm45.glm-4.5.math500
GPQA Diamond	93.8	Gemini 3 Deep Think	launch prose (Deep Think mode)	2025-11	m2025.gemini3-blog.gemini-3-deep-think.gpqa
SWE-bench Verified	82.0	Claude Sonnet 4.5	high compute: parallel attempts + scoring model	2025-09	m2025.sonnet45-blog.claude-sonnet-4.5.swebench.parallel
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	m2025.o3-blog.o4-mini.aime24.notools
AIME 2025	100.0	Grok 4 Heavy [chart code GROK_4_HEAVY]	with tools (Python)	2025-07	m2025.grok4-blog.grok-4-heavy.aime25.tools

2025-12

December 2025. 21 verified scores for 6 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
DeepSeek-V3.2 Thinking	AIME 2025	Pass@1	93.1	DeepSeek (DeepSeek-V3.2 TR, Table 2)	arxiv_2512.02556 p.13	m2025.dsv32.deepseek-v3.2.aime25
DeepSeek-V3.2 Thinking	LiveCodeBench	Pass@1-COT	83.3	DeepSeek (DeepSeek-V3.2 TR, Table 2)	arxiv_2512.02556 p.13	m2025.dsv32.deepseek-v3.2.lcb
DeepSeek-V3.2 Thinking	MMLU-Pro	EM	85.0	DeepSeek (DeepSeek-V3.2 TR, Table 2)	arxiv_2512.02556 p.13	m2025.dsv32.deepseek-v3.2.mmlu-pro
DeepSeek-V3.2 Thinking	SWE-bench Verified	Resolved	73.1	DeepSeek (DeepSeek-V3.2 TR, Table 2)	arxiv_2512.02556 p.13	m2025.dsv32.deepseek-v3.2.swebench
DeepSeek-V3.2-Speciale	AIME 2025	Pass@1	96.0	DeepSeek (DeepSeek-V3.2 TR, Table 3)	arxiv_2512.02556 p.15	m2025.dsv32.deepseek-v3.2-speciale.aime25
DeepSeek-V3.2-Speciale	GPQA Diamond	Pass@1	85.7	DeepSeek (DeepSeek-V3.2 TR, Table 3)	arxiv_2512.02556 p.15	m2025.dsv32.deepseek-v3.2-speciale.gpqa
DeepSeek-V3.2-Speciale	LiveCodeBench	Pass@1-COT	88.7	DeepSeek (DeepSeek-V3.2 TR, Table 3)	arxiv_2512.02556 p.15	m2025.dsv32.deepseek-v3.2-speciale.lcb
GLM-4.7	AIME 2025	as reported by Z.ai	95.7	Zhipu AI / Z.ai (GLM-4.7 launch post, table)	zai_glm47_blog html	m2025.glm47-blog.glm-4.7.aime25
GLM-4.7	GPQA Diamond	as reported by Z.ai	85.7	Zhipu AI / Z.ai (GLM-4.7 launch post, table)	zai_glm47_blog html	m2025.glm47-blog.glm-4.7.gpqa

GLM-4.7	LiveCodeBench v6	as reported by Z.ai	84.9	Zhipu AI / Z.ai (GLM-4.7 launch post, table)	zai_glm47_blog html	m2025.glm47-blog.glm-4.7.lcb-v6
GLM-4.7	MMLU-Pro	as reported by Z.ai	84.3	Zhipu AI / Z.ai (GLM-4.7 launch post, table)	zai_glm47_blog html	m2025.glm47-blog.glm-4.7.mmlu-pro
GLM-4.7	SWE-bench Verified	as reported by Z.ai	73.8	Zhipu AI / Z.ai (GLM-4.7 launch post, table)	zai_glm47_blog html	m2025.glm47-blog.glm-4.7.swebench
GPT-5.2 Pro	AIME 2025	no tools, xhigh effort	100.0	OpenAI (GPT-5.2 launch post, Dec 11 2025)	openai_gpt52_blog html	m2025.gpt52-blog.gpt-5.2-pro.aime25.notools
GPT-5.2 Pro	GPQA Diamond	no tools, xhigh effort	93.2	OpenAI (GPT-5.2 launch post, Dec 11 2025)	openai_gpt52_blog html	m2025.gpt52-blog.gpt-5.2-pro.gpqa.notools
GPT-5.2 Thinking	AIME 2025	no tools, xhigh effort	100.0	OpenAI (GPT-5.2 launch post, Dec 11 2025)	openai_gpt52_blog html	m2025.gpt52-blog.gpt-5.2-thinking.aime25.notools
GPT-5.2 Thinking	FrontierMath Tier 1-3	with Python, xhigh effort	40.3	OpenAI (GPT-5.2 launch post, Dec 11 2025)	openai_gpt52_blog html	m2025.gpt52-blog.gpt-5.2-thinking.frontiermath-t1-3
GPT-5.2 Thinking	GPQA Diamond	no tools, xhigh effort	92.4	OpenAI (GPT-5.2 launch post, Dec 11 2025)	openai_gpt52_blog html	m2025.gpt52-blog.gpt-5.2-thinking.gpqa.notools
GPT-5.2 Thinking	MMMLU	xhigh effort	89.6	OpenAI (GPT-5.2 launch post, Dec 11 2025)	openai_gpt52_blog html	m2025.gpt52-blog.gpt-5.2-thinking.mmmlu
GPT-5.2 Thinking	SWE-bench Verified	xhigh effort	80.0	OpenAI (GPT-5.2 launch post, Dec 11 2025)	openai_gpt52_blog html	m2025.gpt52-blog.gpt-5.2-thinking.swebench
Gemini 3 Flash	GPQA Diamond	launch prose	90.4	Google (Gemini 3 Flash launch post, Dec 17 2025)	google_gemini3_flash_blog html	m2025.gemini3flash-blog.gemini-3-flash.gpqa
Gemini 3 Flash	SWE-bench Verified	launch prose	78.0	Google (Gemini 3 Flash launch post, Dec 17 2025)	google_gemini3_flash_blog html	m2025.gemini3flash-blog.gemini-3-flash.swebench

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.9	o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	0-shot CoT, no tools	2025-07	m2025.simple-evals.o3.mmlu
MMLU-Pro	90.1	Gemini-3.0 Pro	EM	2025-12	m2025.dsv32.gemini-3-pro.mmlu-pro
HumanEval	97.6	o3-mini-high	0-shot CoT (simple-evals)	2025-07	m2025.simple-evals.o3-mini-high.humaneval
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	m2024.qwen25.qwen2.5-72b-instruct.math.instruct
MATH-500	98.2	GLM-4.5	GLM evaluation	2025-08	m2025.glm45.glm-4.5.math500
GPQA Diamond	93.8	Gemini 3 Deep Think	launch prose (Deep Think mode)	2025-11	m2025.gemini3-blog.gemini-3-deep-think.gpqa
SWE-bench Verified	82.0	Claude Sonnet 4.5	high compute: parallel attempts + scoring model	2025-09	m2025.sonnet45-blog.claude-sonnet-4.5.swebench.parallel
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	m2025.o3-blog.o4-mini.aime24.notools
AIME 2025	100.0	Grok 4 Heavy [chart code GROK_4_HEAVY]	with tools (Python)	2025-07	m2025.grok4-blog.grok-4-heavy.aime25.tools

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- DeepSeek-V3.2: Pushing the Frontier of Open Large Language Models, v1, PDF page 1: Figure 1: Benchmark of DeepSeek-V3.2 and its counterparts.

2026-01

January 2026. 6 verified scores for 1 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Kimi K2.5	AIME 2025	Avg@64, max 96k completion tokens (appendix E)	96.1	Moonshot AI, Kimi K2.5 technical report (arXiv:2602.02276v2) Table 4	arxiv_2602.02276 p.12	m2026.kimi.kimi-k2-5.aime-2025.table4

Kimi K2.5	GPQA-Diamond	avg of 8 runs (appendix E)	87.6	Moonshot AI, Kimi K2.5 technical report (arXiv:2602.02276v2) Table 4	arxiv_2602.02276 p.12	m2026.kimi.kimi-k2-5.gpqa-diamond.table4
Kimi K2.5	LiveCodeBench v6	Table 4; coding task averaged over 5 runs (Avg@5, appendix E.5); temperature 1.0, top-p 0.95; mode NOT stated	85.0	Moonshot AI, Kimi K2.5 technical report (arXiv:2602.02276v2) Table 4	arxiv_2602.02276 p.12	m2026.kimi.kimi-k2-5.livecodebench-v6.table4
Kimi K2.5	MMLU-Pro	Table 4, Kimi K2.5 general config (temperature 1.0, top-p 0.95, 256k context); thinking/instant mode NOT stated	87.1	Moonshot AI, Kimi K2.5 technical report (arXiv:2602.02276v2) Table 4	arxiv_2602.02276 p.12	m2026.kimi.kimi-k2-5.mmlu-pro.table4
Kimi K2.5	SWE-bench Verified	internal minimal-tool framework (bash, create_file, insert, view, str_replace, submit) with system prompts specifically tailored for repository-level code manipulation; non-thinking mode; Avg@5	76.8	Moonshot AI, Kimi K2.5 technical report (arXiv:2602.02276v2) Table 4	arxiv_2602.02276 p.12	m2026.kimi.kimi-k2-5.swe-bench-verified.table4
Kimi K2.5	Terminal-Bench 2.0	Terminus-2 framework with the provided JSON parser, NON-THINKING mode, Avg@5	50.8	Moonshot AI, Kimi K2.5 technical report (arXiv:2602.02276v2) Table 4	arxiv_2602.02276 p.12	m2026.kimi.kimi-k2-5.terminal-bench-2-0.table4

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.9	o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	0-shot CoT, no tools	2025-07	m2025.simple-evals.o3.mmlu
MMLU-Pro	90.1	Gemini-3.0 Pro	EM	2025-12	m2025.dsv32.gemini-3-pro.mmlu-pro
HumanEval	97.6	o3-mini-high	0-shot CoT (simple-evals)	2025-07	m2025.simple-evals.o3-mini-high.humaneval
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	m2024.qwen25.qwen2.5-72b-instruct.math.instruct
MATH-500	98.2	GLM-4.5	GLM evaluation	2025-08	m2025.glm45.glm-4.5.math500
GPQA Diamond	93.8	Gemini 3 Deep Think	launch prose (Deep Think mode)	2025-11	m2025.gemini3-blog.gemini-3-deep-think.gpqa
SWE-bench Verified	82.0	Claude Sonnet 4.5	high compute: parallel attempts + scoring model	2025-09	m2025.sonnet45-blog.claude-sonnet-4.5.swebench.parallel
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	m2025.o3-blog.o4-mini.aime24.notools
AIME 2025	100.0	Grok 4 Heavy [chart code GROK_4_HEAVY]	with tools (Python)	2025-07	m2025.grok4-blog.grok-4-heavy.aime25.tools

2026-02

February 2026. 23 verified scores for 6 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Claude Opus 4.6	AIME 2025	5 trials, adaptive thinking, max effort, no tools; Anthropic flags possible contamination	99.79	Anthropic, Claude Opus 4.6 System Card (Feb 2026)	anthropic_claude_opus46_system_card p.25	m2026.opus46card.claude-opus-4-6.aime-2025
Claude Opus 4.6	GPQA Diamond	5 trials, adaptive thinking, max effort	91.31	Anthropic, Claude Opus 4.6 System Card (Feb 2026)	anthropic_claude_opus46_system_card p.25	m2026.opus46card.claude-opus-4-6.gpqa-diamond
Claude Opus 4.6	MMMLU	5 trials, all non-English pairings, adaptive thinking, max effort	91.05	Anthropic, Claude Opus 4.6 System Card (Feb 2026)	anthropic_claude_opus46_system_card p.25	m2026.opus46card.claude-opus-4-6.mmmlu
Claude Opus 4.6	SWE-bench Verified	prompt-modified harness (non-standard)	81.4	Anthropic, Claude Opus 4.6 System Card (Feb 2026)	anthropic_claude_opus46_system_card p.19	m2026.opus46card.claude-opus-4-6.swe-bench-verified.prompt-modified
Claude Opus 4.6	SWE-bench Verified	averaged over 25 trials, adaptive thinking, max effort, standard prompt	80.84	Anthropic, Claude Opus 4.6 System Card (Feb 2026)	anthropic_claude_opus46_system_card p.19	m2026.opus46card.claude-opus-4-6.swe-bench-verified.standard

Claude Sonnet 4.6	GPQA Diamond	10 trials	89.9	Anthropic, Claude Sonnet 4.6 System Card (Feb 17, 2026)	anthropic_claude_sonnet46_system_card p.23	m2026.sonnet46card.claude-sonnet-4-6.gpqa-diamond
Claude Sonnet 4.6	MMMLU	10 trials, all non-English pairings	89.3	Anthropic, Claude Sonnet 4.6 System Card (Feb 17, 2026)	anthropic_claude_sonnet46_system_card p.23	m2026.sonnet46card.claude-sonnet-4-6.mmmlu
Claude Sonnet 4.6	SWE-bench Verified	prompt-modified harness (non-standard)	80.2	Anthropic, Claude Sonnet 4.6 System Card (Feb 17, 2026)	anthropic_claude_sonnet46_system_card p.16	m2026.sonnet46card.claude-sonnet-4-6.swe-bench-verified.prompt-modified
Claude Sonnet 4.6	SWE-bench Verified	standard prompt	79.6	Anthropic, Claude Sonnet 4.6 System Card (Feb 17, 2026)	anthropic_claude_sonnet46_system_card p.16	m2026.sonnet46card.claude-sonnet-4-6.swe-bench-verified.standard
GLM-5	AIME 2026 I	max generation length 131,072 tokens	92.7	Zhipu AI, GLM-5 technical report (arXiv:2602.15763v2) Table 7	arxiv_2602.15763 p.23	m2026.glm5.glm-5.aime-2026-i.table7
GLM-5	GPQA-Diamond	max generation length 131,072 tokens	86.0	Zhipu AI, GLM-5 technical report (arXiv:2602.15763v2) Table 7	arxiv_2602.15763 p.23	m2026.glm5.glm-5.gpqa-diamond.table7
GLM-5	SWE-bench Verified	OpenHands with tailored instruction prompt	77.8	Zhipu AI, GLM-5 technical report (arXiv:2602.15763v2) Table 7	arxiv_2602.15763 p.23	m2026.glm5.glm-5.swe-bench-verified.table7
GPT-5.3-Codex (xhigh)	SWE-Bench Pro (Public)	xhigh	56.8	OpenAI, Introducing GPT-5.3-Codex	openai_gpt53codex_blog html	m2026.gpt53codex.gpt-5-3-codex.swe-bench-pro
GPT-5.3-Codex (xhigh)	Terminal-Bench 2.0	xhigh	77.3	OpenAI, Introducing GPT-5.3-Codex (Feb 5, 2026)	openai_gpt53codex_blog html	m2026.gpt53codex.gpt-5-3-codex.terminal-bench-2-0
Gemini 3.1 Pro	GPQA Diamond	Thinking (High), no tools	94.3	Google DeepMind, Gemini 3.1 Pro model card (published 19 Feb 2026)	google_gemini31pro_model_card html	m2026.g31card.gemini-3-1-pro.gpqa-diamond
Gemini 3.1 Pro	MMMLU	Thinking (High)	92.6	Google DeepMind, Gemini 3.1 Pro model card (published 19 Feb 2026)	google_gemini31pro_model_card html	m2026.g31card.gemini-3-1-pro.mmmlu
Gemini 3.1 Pro	SWE-bench Verified	Thinking (High), single attempt	80.6	Google DeepMind, Gemini 3.1 Pro model card (published 19 Feb 2026)	google_gemini31pro_model_card html	m2026.g31card.gemini-3-1-pro.swe-bench-verified
Gemini 3.1 Pro	Terminal-Bench 2.0	Thinking (High), Terminus-2 harness	68.5	Google DeepMind, Gemini 3.1 Pro model card (published 19 Feb 2026)	google_gemini31pro_model_card html	m2026.g31card.gemini-3-1-pro.terminal-bench-2-0
Qwen3.5-397B-A17B	AIME 2026	as in model card table	91.3	Alibaba Qwen, Hugging Face model card Qwen/Qwen3.5-397B-A17B (capture 2026-02-16)	hf_qwen35_397b_model_card html	m2026.qwen35card.qwen3-5-397b-a17b.aime26
Qwen3.5-397B-A17B	GPQA (subset not stated)	as in model card table	88.4	Alibaba Qwen, Hugging Face model card Qwen/Qwen3.5-397B-A17B (capture 2026-02-16)	hf_qwen35_397b_model_card html	m2026.qwen35card.qwen3-5-397b-a17b.gpqa
Qwen3.5-397B-A17B	LiveCodeBench v6	as in model card table	83.6	Alibaba Qwen, Hugging Face model card Qwen/Qwen3.5-397B-A17B (capture 2026-02-16)	hf_qwen35_397b_model_card html	m2026.qwen35card.qwen3-5-397b-a17b.livecodebench-v6
Qwen3.5-397B-A17B	MMLU-Pro	thinking, as in model card table	87.8	Alibaba Qwen, Hugging Face model card Qwen/Qwen3.5-397B-A17B (capture 2026-02-16)	hf_qwen35_397b_model_card html	m2026.qwen35card.qwen3-5-397b-a17b.mmlu-pro
Qwen3.5-397B-A17B	SWE-bench Verified	as in model card table	76.4	Alibaba Qwen, Hugging Face model card Qwen/Qwen3.5-397B-A17B (capture 2026-02-16)	hf_qwen35_397b_model_card html	m2026.qwen35card.qwen3-5-397b-a17b.swe-bench-verified

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim

MMLU	92.9	o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	0-shot CoT, no tools	2025-07	`m2025.simple-evals.o3.mmlu`
MMLU-Pro	90.1	Gemini-3.0 Pro	EM	2025-12	`m2025.dsv32.gemini-3-pro.mmlu-pro`
HumanEval	97.6	o3-mini-high	0-shot CoT (simple-evals)	2025-07	`m2025.simple-evals.o3-mini-high.humaneval`
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	`m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot`
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	`m2024.qwen25.qwen2.5-72b-instruct.math.instruct`
MATH-500	98.2	GLM-4.5	GLM evaluation	2025-08	`m2025.glm45.glm-4.5.math500`
GPQA Diamond	94.3	Gemini 3.1 Pro	Thinking (High), no tools	2026-02	`m2026.g31card.gemini-3-1-pro.gpqa-diamond`
SWE-bench Verified	82.0	Claude Sonnet 4.5	high compute: parallel attempts + scoring model	2025-09	`m2025.sonnet45-blog.claude-sonnet-4.5.swebench.parallel`
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	`m2025.o3-blog.o4-mini.aime24.notools`
AIME 2025	100.0	Grok 4 Heavy [chart code GROK_4_HEAVY]	with tools (Python)	2025-07	`m2025.grok4-blog.grok-4-heavy.aime25.tools`

Primary-source pages from documents first published this month (not redistributed; links open the pinned version at the cited page):

- Kimi K2.5: Visual Agentic Intelligence, v2, PDF page 12: Table 4: Performance comparison of Kimi K2.5 against open-source and proprietary models.
- GLM-5: from Vibe Coding to Agentic Engineering, v2, PDF page 23: Table 7: Comparison between GLM-5 and open-source/proprietary models.

2026-03

March 2026. 5 verified scores for 2 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
GPT-5.4	FrontierMath Tier 1-3	reasoning effort xhigh	47.6	OpenAI, Introducing GPT-5.4 (Mar 5, 2026)	openai_gpt54_blog html	`m2026.gpt54.gpt-5-4.frontiermath-tier-1-3`
GPT-5.4	GPQA Diamond	reasoning effort xhigh	92.8	OpenAI, Introducing GPT-5.4 (Mar 5, 2026)	openai_gpt54_blog html	`m2026.gpt54.gpt-5-4.gpqa-diamond`
GPT-5.4	SWE-Bench Pro (Public)	reasoning effort xhigh	57.7	OpenAI, Introducing GPT-5.4 (Mar 5, 2026)	openai_gpt54_blog html	`m2026.gpt54.gpt-5-4.swe-bench-pro`
GPT-5.4	Terminal-Bench 2.0	reasoning effort xhigh	75.1	OpenAI, Introducing GPT-5.4 (Mar 5, 2026)	openai_gpt54_blog html	`m2026.gpt54.gpt-5-4.terminal-bench-2-0`
GPT-5.4 Pro	GPQA Diamond	reasoning effort xhigh	94.4	OpenAI, Introducing GPT-5.4 (Mar 5, 2026)	openai_gpt54_blog html	`m2026.gpt54.gpt-5-4-pro.gpqa-diamond`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.9	o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	0-shot CoT, no tools	2025-07	`m2025.simple-evals.o3.mmlu`
MMLU-Pro	90.1	Gemini-3.0 Pro	EM	2025-12	`m2025.dsv32.gemini-3-pro.mmlu-pro`
HumanEval	97.6	o3-mini-high	0-shot CoT (simple-evals)	2025-07	`m2025.simple-evals.o3-mini-high.humaneval`
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	`m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot`
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	`m2024.qwen25.qwen2.5-72b-instruct.math.instruct`
MATH-500	98.2	GLM-4.5	GLM evaluation	2025-08	`m2025.glm45.glm-4.5.math500`
GPQA Diamond	94.4	GPT-5.4 Pro	reasoning effort xhigh	2026-03	`m2026.gpt54.gpt-5-4-pro.gpqa-diamond`

SWE-bench Verified	82.0	Claude Sonnet 4.5	high compute: parallel attempts + scoring model	2025-09	m2025.sonnet45-blog.claude-sonnet-4.5.swebench.parallel
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	m2025.o3-blog.o4-mini.aime24.notools
AIME 2025	100.0	Grok 4 Heavy [chart code GROK_4_HEAVY]	with tools (Python)	2025-07	m2025.grok4-blog.grok-4-heavy.aime25.tools

2026-04

April 2026. 12 verified scores for 3 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Claude Opus 4.7	GPQA Diamond	10 trials	94.2	Anthropic, Claude Opus 4.7 System Card	anthropic_claude_opus47_system_card p.193	m2026.opus47card.claude-opus-4-7.gpqa-diamond
Claude Opus 4.7	MMMLU	3 trials, adaptive thinking, max effort	91.5	Anthropic, Claude Opus 4.7 System Card	anthropic_claude_opus47_system_card p.193	m2026.opus47card.claude-opus-4-7.mmmlu
Claude Opus 4.7	SWE-bench Pro	adaptive thinking, max effort, 5 trials	64.3	Anthropic, Claude Opus 4.7 System Card	anthropic_claude_opus47_system_card p.192	m2026.opus47card.claude-opus-4-7.swe-bench-pro
Claude Opus 4.7	SWE-bench Verified	adaptive thinking, max effort, 5 trials	87.6	Anthropic, Claude Opus 4.7 System Card	anthropic_claude_opus47_system_card p.192	m2026.opus47card.claude-opus-4-7.swe-bench-verified
Claude Opus 4.7	Terminal-Bench 2.0	Terminus-2 harness, thinking disabled	69.4	Anthropic, Claude Opus 4.7 System Card	anthropic_claude_opus47_system_card p.193	m2026.opus47card.claude-opus-4-7.terminal-bench-2-0
DeepSeek-V4-Pro (Think Max)	GPQA Diamond	Think Max, Pass@1	90.1	DeepSeek, Hugging Face model card deepseek-ai/DeepSeek-V4-Pro (capture 2026-04-24)	hf_deepseek_v4_pro_model_card html	m2026.dsv4card.deepseek-v4-pro-max.gpqa-diamond
DeepSeek-V4-Pro (Think Max)	LiveCodeBench (version not stated)	Think Max, Pass@1	93.5	DeepSeek, Hugging Face model card deepseek-ai/DeepSeek-V4-Pro (capture 2026-04-24)	hf_deepseek_v4_pro_model_card html	m2026.dsv4card.deepseek-v4-pro-max.livecodebench
DeepSeek-V4-Pro (Think Max)	MMLU-Pro	Think Max, EM	87.5	DeepSeek, Hugging Face model card deepseek-ai/DeepSeek-V4-Pro (capture 2026-04-24)	hf_deepseek_v4_pro_model_card html	m2026.dsv4card.deepseek-v4-pro-max.mmlu-pro
DeepSeek-V4-Pro (Think Max)	SWE-bench Verified	Think Max, resolved rate	80.6	DeepSeek, Hugging Face model card deepseek-ai/DeepSeek-V4-Pro (capture 2026-04-24)	hf_deepseek_v4_pro_model_card html	m2026.dsv4card.deepseek-v4-pro-max.swe-bench-verified
GPT-5.5	GPQA Diamond	reasoning effort xhigh	93.6	OpenAI, Introducing GPT-5.5 (Apr 23, 2026)	openai_gpt55_blog html	m2026.gpt55.gpt-5-5.gpqa-diamond
GPT-5.5	SWE-Bench Pro (Public)	reasoning effort xhigh	58.6	OpenAI, Introducing GPT-5.5 (Apr 23, 2026)	openai_gpt55_blog html	m2026.gpt55.gpt-5-5.swe-bench-pro
GPT-5.5	Terminal-Bench 2.0	reasoning effort xhigh	82.7	OpenAI, Introducing GPT-5.5 (Apr 23, 2026)	openai_gpt55_blog html	m2026.gpt55.gpt-5-5.terminal-bench-2-0

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.9	o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	0-shot CoT, no tools	2025-07	m2025.simple-evals.o3.mmlu
MMLU-Pro	90.1	Gemini-3.0 Pro	EM	2025-12	m2025.dsv32.gemini-3-pro.mmlu-pro
HumanEval	97.6	o3-mini-high	0-shot CoT (simple-evals)	2025-07	m2025.simple-evals.o3-mini-high.humaneval
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	m2024.qwen25.qwen2.5-72b-instruct.math.instruct
MATH-500	98.2	GLM-4.5	GLM evaluation	2025-08	m2025.glm45.glm-4.5.math500

GPQA Diamond	94.4	GPT-5.4 Pro	reasoning effort xhigh	2026-03	`m2026.gpt54.gpt-5-4-pro.gpqa-diamond`
SWE-bench Verified	87.6	Claude Opus 4.7	adaptive thinking, max effort, 5 trials	2026-04	`m2026.opus47card.claude-opus-4-7.swe-bench-verified`
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	`m2025.o3-blog.o4-mini.aime24.notools`
AIME 2025	100.0	Grok 4 Heavy [chart code GROK_4_HEAVY]	with tools (Python)	2025-07	`m2025.grok4-blog.grok-4-heavy.aime25.tools`

2026-05

May 2026. 2 verified scores for 1 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Gemini 3.5 Flash	SWE-Bench Pro (Public)	single attempt	55.1	Google DeepMind, Gemini 3.5 Flash model card (published 19 May 2026)	google_gemini35flash_model_card html	`m2026.g35flash.gemini-3-5-flash.swe-bench-pro`
Gemini 3.5 Flash	Terminal-Bench 2.1	Terminus-2 harness	76.2	Google DeepMind, Gemini 3.5 Flash model card (published 19 May 2026)	google_gemini35flash_model_card html	`m2026.g35flash.gemini-3-5-flash.terminal-bench-2-1`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.9	o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	0-shot CoT, no tools	2025-07	`m2025.simple-evals.o3.mmlu`
MMLU-Pro	90.1	Gemini-3.0 Pro	EM	2025-12	`m2025.dsv32.gemini-3-pro.mmlu-pro`
HumanEval	97.6	o3-mini-high	0-shot CoT (simple-evals)	2025-07	`m2025.simple-evals.o3-mini-high.humaneval`
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	`m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot`
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	`m2024.qwen25.qwen2.5-72b-instruct.math.instruct`
MATH-500	98.2	GLM-4.5	GLM evaluation	2025-08	`m2025.glm45.glm-4.5.math500`
GPQA Diamond	94.4	GPT-5.4 Pro	reasoning effort xhigh	2026-03	`m2026.gpt54.gpt-5-4-pro.gpqa-diamond`
SWE-bench Verified	87.6	Claude Opus 4.7	adaptive thinking, max effort, 5 trials	2026-04	`m2026.opus47card.claude-opus-4-7.swe-bench-verified`
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	`m2025.o3-blog.o4-mini.aime24.notools`
AIME 2025	100.0	Grok 4 Heavy [chart code GROK_4_HEAVY]	with tools (Python)	2025-07	`m2025.grok4-blog.grok-4-heavy.aime25.tools`

2026-06

June 2026. 5 verified scores for 2 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Claude Sonnet 5	SWE-bench Pro	standard configuration, 5 trials	63.2	Anthropic, Claude Sonnet 5 System Card	anthropic_claude_sonnet5_system_card p.116	`m2026.sonnet5card.claude-sonnet-5.swe-bench-pro`
Claude Sonnet 5	SWE-bench Verified	adaptive thinking, max effort, 5 trials (standard configuration)	85.2	Anthropic, Claude Sonnet 5 System Card	anthropic_claude_sonnet5_system_card p.116	`m2026.sonnet5card.claude-sonnet-5.swe-bench-verified`

GLM-5.2	AIME 2026	as in model card table	99.2	Zhipu AI (Z.ai), Hugging Face model card zai-org/GLM-5.2 (capture 2026-06-16)	hf_glm52_model_card html	m2026.glm52card.glm-5-2.aime-2026
GLM-5.2	GPQA-Diamond	as in model card table	91.2	Zhipu AI (Z.ai), Hugging Face model card zai-org/GLM-5.2 (capture 2026-06-16)	hf_glm52_model_card html	m2026.glm52card.glm-5-2.gpqa-diamond
GLM-5.2	SWE-bench Pro	as in model card table	62.1	Zhipu AI (Z.ai), Hugging Face model card zai-org/GLM-5.2 (capture 2026-06-16)	hf_glm52_model_card html	m2026.glm52card.glm-5-2.swe-bench-pro

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.9	o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	0-shot CoT, no tools	2025-07	m2025.simple-evals.o3.mmlu
MMLU-Pro	90.1	Gemini-3.0 Pro	EM	2025-12	m2025.dsv32.gemini-3-pro.mmlu-pro
HumanEval	97.6	o3-mini-high	0-shot CoT (simple-evals)	2025-07	m2025.simple-evals.o3-mini-high.humaneval
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	m2024.qwen25.qwen2.5-72b-instruct.math.instruct
MATH-500	98.2	GLM-4.5	GLM evaluation	2025-08	m2025.glm45.glm-4.5.math500
GPQA Diamond	94.4	GPT-5.4 Pro	reasoning effort xhigh	2026-03	m2026.gpt54.gpt-5-4-pro.gpqa-diamond
SWE-bench Verified	87.6	Claude Opus 4.7	adaptive thinking, max effort, 5 trials	2026-04	m2026.opus47card.claude-opus-4-7.swe-bench-verified
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	m2025.o3-blog.o4-mini.aime24.notools
AIME 2025	100.0	Grok 4 Heavy [chart code GROK_4_HEAVY]	with tools (Python)	2025-07	m2025.grok4-blog.grok-4-heavy.aime25.tools

2026-07

July 2026. 5 verified scores for 2 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
GPT-5.6 Sol	GPQA Diamond	effort not stated in the table	94.6	OpenAI, GPT-5.6 launch post (Jul 9, 2026)	openai_gpt56_blog html	m2026.gpt56.gpt-5-6-sol.gpqa-diamond
GPT-5.6 Sol	SWE-Bench Pro	effort not stated in the table	64.6	OpenAI, GPT-5.6 launch post (Jul 9, 2026)	openai_gpt56_blog html	m2026.gpt56.gpt-5-6-sol.swe-bench-pro
GPT-5.6 Sol	Terminal-Bench 2.1	effort not stated in the table	88.8	OpenAI, GPT-5.6 launch post (Jul 9, 2026)	openai_gpt56_blog html	m2026.gpt56.gpt-5-6-sol.terminal-bench-2-1
Grok 4.5	SWE-Bench Pro	resolve rate	64.7	xAI (page branded SpaceXAI), Introducing Grok 4.5 (Jul 8, 2026)	xai_grok45_blog html	m2026.grok45.grok-4-5.swe-bench-pro
Grok 4.5	Terminal-Bench 2.1	effort not stated for Grok 4.5	83.3	xAI (page branded SpaceXAI), Introducing Grok 4.5 (Jul 8, 2026)	xai_grok45_blog html	m2026.grok45.grok-4-5.terminal-bench-2-1

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.9	o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	0-shot CoT, no tools	2025-07	m2025.simple-evals.o3.mmlu
MMLU-Pro	90.1	Gemini-3.0 Pro	EM	2025-12	m2025.dsv32.gemini-3-pro.mmlu-pro
HumanEval	97.6	o3-mini-high	0-shot CoT (simple-evals)	2025-07	m2025.simple-evals.o3-mini-high.humaneval
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot

MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	`m2024.qwen25.qwen2.5-72b-instruct.math.instruct`
MATH-500	98.2	GLM-4.5	GLM evaluation	2025-08	`m2025.glm45.glm-4.5.math500`
GPQA Diamond	94.6	GPT-5.6 Sol	effort not stated in the table	2026-07	`m2026.gpt56.gpt-5-6-sol.gpqa-diamond`
SWE-bench Verified	87.6	Claude Opus 4.7	adaptive thinking, max effort, 5 trials	2026-04	`m2026.opus47card.claude-opus-4-7.swe-bench-verified`
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	`m2025.o3-blog.o4-mini.aime24.notools`
AIME 2025	100.0	Grok 4 Heavy [chart code GROK_4_HEAVY]	with tools (Python)	2025-07	`m2025.grok4-blog.grok-4-heavy.aime25.tools`

2026-08

August 2026. 3 verified scores for 1 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Qwen3.8-Max	GPQA Diamond	as in model card table	92.6	Alibaba Qwen, Hugging Face model card Qwen/Qwen3.8-2.4T-A95B (capture 2026-08-12)	hf_qwen38_2p4t_model_card html	`m2026.qwen38card.qwen3-8-max.gpqa-diamond`
Qwen3.8-Max	SWE-bench Pro, Qwen-refined version ('Problematic tasks corrected and all baselines evaluated on the refined benchmark')	Claude Code harness, temp 1.0, top_p 0.95, 256K context (footnote 3)	67.7	Alibaba Qwen, Hugging Face model card Qwen/Qwen3.8-2.4T-A95B (capture 2026-08-12)	hf_qwen38_2p4t_model_card html	`m2026.qwen38card.qwen3-8-max.swe-bench-pro`
Qwen3.8-Max	Terminal-Bench 2.1	Claude Code harness, avg@10, 5-hour timeout, max_tokens=131,072 (footnote 2); baselines are best published scores across harnesses	86.6	Alibaba Qwen, Hugging Face model card Qwen/Qwen3.8-2.4T-A95B (capture 2026-08-12)	hf_qwen38_2p4t_model_card html	`m2026.qwen38card.qwen3-8-max.terminal-bench-2-1`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.9	o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	0-shot CoT, no tools	2025-07	`m2025.simple-evals.o3.mmlu`
MMLU-Pro	90.1	Gemini-3.0 Pro	EM	2025-12	`m2025.dsv32.gemini-3-pro.mmlu-pro`
HumanEval	97.6	o3-mini-high	0-shot CoT (simple-evals)	2025-07	`m2025.simple-evals.o3-mini-high.humaneval`
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	`m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot`
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	`m2024.qwen25.qwen2.5-72b-instruct.math.instruct`
MATH-500	98.2	GLM-4.5	GLM evaluation	2025-08	`m2025.glm45.glm-4.5.math500`
GPQA Diamond	94.6	GPT-5.6 Sol	effort not stated in the table	2026-07	`m2026.gpt56.gpt-5-6-sol.gpqa-diamond`
SWE-bench Verified	87.6	Claude Opus 4.7	adaptive thinking, max effort, 5 trials	2026-04	`m2026.opus47card.claude-opus-4-7.swe-bench-verified`
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	`m2025.o3-blog.o4-mini.aime24.notools`
AIME 2025	100.0	Grok 4 Heavy [chart code GROK_4_HEAVY]	with tools (Python)	2025-07	`m2025.grok4-blog.grok-4-heavy.aime25.tools`

2026-09

September 2026. 8 verified scores for 5 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Claude Opus 5.5	Humanity's Last Exam	with tools, adaptive thinking, max effort	67.7	Anthropic, Introducing Claude Opus 5.5 (Sep 22, 2026)	anthropic_claude_opus55_page html	`m2026.opus55page.claude-opus-5-5.hle.with-tools`
Claude Opus 5.5	Terminal-Bench 4.0	xhigh effort (Anthropic setup; harness not named for the Opus 5.5 run)	66.4	Anthropic, Introducing Claude Opus 5.5 (Sep 22, 2026)	anthropic_claude_opus55_page html	`m2026.opus55page.claude-opus-5-5.terminal-bench-4-0.xhigh`
Claude Opus 5.5 (Artificial Analysis run: Claude Opus 5.5 (Max, Default Fallback))	Terminal-Bench 4.0	Artificial Analysis harness, effort 'max'	59.6	Artificial Analysis model page (Wayback)	aa_claude_opus55 html	`m2026.aa.claude-opus-5-5.terminal-bench-4-0`
Claude Sonnet 5.5	Humanity's Last Exam	with tools	64.5	Anthropic, Introducing Claude Sonnet 5.5 (Sep 28, 2026)	anthropic_claude_sonnet55_page html	`m2026.sonnet55page.claude-sonnet-5-5.hle.with-tools`
Claude Sonnet 5.5	Terminal-Bench 4.0	effort not stated for Sonnet 5.5 (footnote 1 states xhigh only for Opus 5.5)	70.6	Anthropic, Introducing Claude Sonnet 5.5 (Sep 28, 2026)	anthropic_claude_sonnet55_page html	`m2026.sonnet55page.claude-sonnet-5-5.terminal-bench-4-0`
Claude Sonnet 5.5 (Artificial Analysis run: Claude Sonnet 5.5 (max with fallback))	Terminal-Bench 4.0	Artificial Analysis harness, effort 'max'	63.6	Artificial Analysis model page (Wayback)	aa_claude_sonnet55 html	`m2026.aa.claude-sonnet-5-5.terminal-bench-4-0`
GPT-6 Astra	GPQA Diamond	maximum at any effort	96.0	OpenAI, GPT-6 Astra launch post (Wayback capture 2026-09-03)	openai_gpt6_astra_blog html	`m2026.astra.gpt-6-astra.gpqa-diamond`
GPT-6 Astra	Terminal-Bench 4.0	maximum at any effort (eval table)	57.7	OpenAI, GPT-6 Astra launch post (Wayback capture 2026-09-03)	openai_gpt6_astra_blog html	`m2026.astra.gpt-6-astra.terminal-bench-4-0.table`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.9	o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	0-shot CoT, no tools	2025-07	`m2025.simple-evals.o3.mmlu`
MMLU-Pro	90.1	Gemini-3.0 Pro	EM	2025-12	`m2025.dsv32.gemini-3-pro.mmlu-pro`
HumanEval	97.6	o3-mini-high	0-shot CoT (simple-evals)	2025-07	`m2025.simple-evals.o3-mini-high.humaneval`
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	`m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot`
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	`m2024.qwen25.qwen2.5-72b-instruct.math.instruct`
MATH-500	98.2	GLM-4.5	GLM evaluation	2025-08	`m2025.glm45.glm-4.5.math500`
GPQA Diamond	96.0	GPT-6 Astra	maximum at any effort	2026-09	`m2026.astra.gpt-6-astra.gpqa-diamond`
SWE-bench Verified	87.6	Claude Opus 4.7	adaptive thinking, max effort, 5 trials	2026-04	`m2026.opus47card.claude-opus-4-7.swe-bench-verified`
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	`m2025.o3-blog.o4-mini.aime24.notools`
AIME 2025	100.0	Grok 4 Heavy [chart code GROK_4_HEAVY]	with tools (Python)	2025-07	`m2025.grok4-blog.grok-4-heavy.aime25.tools`

2026-10

October 2026. 2 verified scores for 2 model(s).

Model	Benchmark	Protocol (as stated)	Value	Reported by	Source	Claim
Claude Haiku 5.5	Terminal-Bench 4.0	as in launch table (effort not stated)	39.2	Anthropic, Introducing Claude Haiku 5.5 (Oct 7, 2026)	anthropic_claude_haiku55_page html	`m2026.haiku55page.claude-haiku-5-5.terminal-bench-4-0`
Gemini 4 Argon	Terminal-Bench 4.0	as in model card table	57.4	Google DeepMind, Gemini 4 Argon model card (published 13 Oct 2026)	google_gemini4argon_model_card html	`m2026.argon.gemini-4-argon.terminal-bench-4-0`

Highest value published by the end of this month:

Benchmark	Value	Model	Protocol	Published	Claim
MMLU	92.9	o3 (row label 'o3' with footnote [^9], whose text reads 'The default reasoning level for o3-mini is "medium"')	0-shot CoT, no tools	2025-07	`m2025.simple-evals.o3.mmlu`
MMLU-Pro	90.1	Gemini-3.0 Pro	EM	2025-12	`m2025.dsv32.gemini-3-pro.mmlu-pro`
HumanEval	97.6	o3-mini-high	0-shot CoT (simple-evals)	2025-07	`m2025.simple-evals.o3-mini-high.humaneval`
GSM8K	96.8	Llama 3 405B (Llama 3.1 405B Instruct)	8-shot CoT	2024-07	`m2024.llama3herd.llama-3.1-405b.gsm8k.8-shot-cot`
MATH	83.1	Qwen2.5-72B-Instruct	not stated in table	2024-12	`m2024.qwen25.qwen2.5-72b-instruct.math.instruct`
MATH-500	98.2	GLM-4.5	GLM evaluation	2025-08	`m2025.glm45.glm-4.5.math500`
GPQA Diamond	96.0	GPT-6 Astra	maximum at any effort	2026-09	`m2026.astra.gpt-6-astra.gpqa-diamond`
SWE-bench Verified	87.6	Claude Opus 4.7	adaptive thinking, max effort, 5 trials	2026-04	`m2026.opus47card.claude-opus-4-7.swe-bench-verified`
AIME 2024	93.4	o4-mini (no tools)	no tools	2025-04	`m2025.o3-blog.o4-mini.aime24.notools`
AIME 2025	100.0	Grok 4 Heavy [chart code GROK_4_HEAVY]	with tools (Python)	2025-07	`m2025.grok4-blog.grok-4-heavy.aime25.tools`